

Workbook

Chapters 0–7

Minimal theory, taken from the dictaat's own definitions, theorems and strategies, followed immediately by the dictaat's own exercises: every Oefening of Hoofdstuk 0–7 (1–30, 101–123, 201–220, 301–330, 401–420, 501–524, 601–635, 701–730) is either worked in the text or solved in Appendix S.

- **Source.** S. Wepster, *Infinitesimaalrekening*, editie 2026, Mathematisch Instituut, Universiteit Utrecht: Blok 1, Hoofdstuk 0–7 (pages 17–113). Definition, Stelling, Voorbeeld and Oefening numbers are the dictaat's; nothing is used that the dictaat does not contain or take for granted (vwo algebra and, in the marked exercises, basic calculus).
- **Which exercises are worked where.** Roughly half of the exercises, chosen to show each technique once, are worked in full in the chapter text. All others appear in NOW YOU boxes and are solved in Appendix S. Exercises the dictaat marks (*Extra*), and the (Extra) sections 3.5, 6.6 and 7.1, are outside the exam material but included.
- **How to use it.** Read one theory box, cover the solution of the worked exercise that follows, attempt it, then compare. Do the NOW YOU exercises before opening Appendix S. Reflect after every exercise, as the dictaat asks: could you have seen the answer coming, would another approach work, what if the question were changed slightly?

How this workbook works

Every section is a sequence of short cycles: **one piece of theory, then the exercises that use it.**

Definitie n.m

The dictaat's definitions and notation, condensed; the things it lists under "hoofdpunten: uit je hoofd weten".

Stelling / Voorbeeld

A result you may use, with the idea behind it, or a worked example from the dictaat that serves as the model for the exercises.

Strategy

The dictaat's methods (separating variables, integrating factor, substitution, partial fractions, function investigation) as step-by-step recipes.

Exercise n badge

An Oefening worked in full, with the checks the dictaat insists on (differentiate your primitive, substitute your solution, sanity-check a value). Cover the solution and try first.

NOW YOU

The remaining exercises for that piece of theory, all solved in Appendix S.

Watch out

Where marks are lost: the dictaat's own warnings and the errors its author sees most in exams.

Conventions of the dictaat

- $\log$ is the **natural** logarithm (base e); $\ln$ is never used. $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The inverse function of f is written f_{inv} , not f^{-1} (to avoid confusion with $1/f$); likewise $\arcsin, \arccos, \arctan$ and $\operatorname{arsinh}, \operatorname{arcosh}, \operatorname{artanh}$.
- "The function $f(x)$ " is sloppy but used; $f : A \rightarrow B, x \mapsto f(x)$ is correct. Intervals (a, b) open, $[a, b]$ closed; ∞ is never a number.

- Exercises are numbered per chapter: Oefening 203 is the third exercise of Hoofdstuk 2. The dictaat marks easy theory-checks and “switching” problems with icons that are not reproduced here; (Extra) marks exercises outside the exam material.
- Exam: no calculator, no formula sheet; sum and double-angle formulas (Oefening 26) and the standard primitives (§6.1) must be known by heart.

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Chapter 0

Voorkennis: prior knowledge refreshed

Contents

- A** Algebraic skills (§0.1)
- B** Functions and inverse functions (§0.2)
- C** Polynomials and factorising (§0.3)
- D** Trigonometry (§0.4)
- E** The logarithm (§0.5)

Planning

Exercises 1–30 (all worked or solved).

The dictaat's own "hoofdpunten": when an inverse function exists and how to find it; $|x|$ and $\sqrt{x^2} = |x|$; composition $f \circ g$; even and odd; the factor theorem; the unit circle, exact values at "nice" angles, symmetries, sum formulas (needed hard in Chapter 6); basic properties of log and exp.

A ALGEBRAIC SKILLS

§0.1 · the point of these exercises

These exercises refresh vwo algebra. The dictaat insists that you **find your own way to check an answer**: substitute back, multiply out again, try a special value ("sanity check"). The one identity to recognise instantly: $a^2 - b^2 = (a - b)(a + b)$.

Exercise 2

from the dictaat

Simplify: (a) $(2 - \sqrt{3})^2$; (b) $(2 - \sqrt{3})(2 + \sqrt{3})$; (c) $\frac{2025^2 - 1}{2024}$; (d) $\frac{(2 + \sqrt{\pi} - y)^2 - (2 - \sqrt{\pi} + y)^2}{\sqrt{\pi} - y}$
(choose a handy approach!).

(a) $(2 - \sqrt{3})^2 = 4 - 4\sqrt{3} + 3 = 7 - 4\sqrt{3}$. (b) $(2 - \sqrt{3})(2 + \sqrt{3}) = 4 - 3 = 1$.

(c) $2025^2 - 1 = (2025 - 1)(2025 + 1) = 2024 \cdot 2026$, so the quotient is 2026.

(d) Put $b = \sqrt{\pi} - y$; then the numerator is $(2 + b)^2 - (2 - b)^2 = ((2 + b) - (2 - b))((2 + b) + (2 - b)) = 2b \cdot 4 = 8b$, and dividing by b gives 8 (for $y \neq \sqrt{\pi}$). Expanding both squares would give the same, with ten times the work. $\square$

Exercise 4

from the dictaat

(a) For which r does $2x^2 + rx + 6 = 0$ have exactly one solution x ? Answer in the form $r = n\sqrt{m}$, $n \in \mathbb{Z}$, $m \in \mathbb{N}$ minimal. (b) $\frac{3t}{t + \frac{1}{1+1/t}} = 1$ has exactly one solution; which? (c) Find all u with $\frac{1}{1 + \frac{1}{1+u}} < 2$. (d)

Solve $\sqrt{v} + 1 = \sqrt{v+1}$ for $v \in \mathbb{R}$.

(a) Exactly one solution iff the discriminant is 0: $r^2 - 4 \cdot 2 \cdot 6 = r^2 - 48 = 0$, so $r = \pm\sqrt{48} = \pm 4\sqrt{3}$. Both $r = 4\sqrt{3}$ and $r = -4\sqrt{3}$ work ($n = \pm 4$, $m = 3$).

(b) The fraction only makes sense for $t \neq 0$ and $t \neq -1$. Simplify from the inside: $\frac{1}{1 + \frac{1}{1+u}} = \frac{t}{t+1}$, so the denominator is $t + \frac{t}{t+1} = \frac{t(t+1) + t}{t+1} = \frac{t(t+2)}{t+1}$. The equation becomes $\frac{3t(t+1)}{t(t+2)} = 1$, i.e. (as $t \neq 0$)

$3(t+1) = t+2$, so $t = -\frac{1}{2}$. Check: $1 + 1/t = -1$, $\frac{1}{-1} = -1$, $t + (-1) = -\frac{3}{2}$, and $\frac{-3/2}{-3/2} = 1$. ✓

(c) Need $u \neq -1$ and $1 + \frac{1}{1+u} \neq 0$, i.e. $u \neq -2$. Then $1 + \frac{1}{1+u} = \frac{u+2}{u+1}$ and the inequality is $\frac{u+1}{u+2} < 2$.

Never multiply by $u+2$ without knowing its sign; instead bring everything to one side: $\frac{u+1}{u+2} - 2 = \frac{u+1-2u-4}{u+2} = \frac{-(u+3)}{u+2} < 0$, i.e. $\frac{u+3}{u+2} > 0$: both factors positive ($u > -2$) or both negative ($u < -3$).

Removing $u = -1$: $u \in (-\infty, -3) \cup (-2, -1) \cup (-1, \infty)$.

(d) Both sides are defined for $v \geq 0$ and non-negative, so squaring is reversible: $v + 2\sqrt{v} + 1 = v + 1$, so $\sqrt{v} = 0$, $v = 0$. Check: $\sqrt{0} + 1 = 1 = \sqrt{1}$. ✓ $\square$

Exercise 5

from the dictaat

Find all c with: (a) $(3+c)(c^2+2) = (2c+2)(3+c)$; (b) $1-c = \sqrt{1-2c+c^2}$; (c) $2^{c^2} \cdot 2^{2c+1} = 16$. (d) Check that $c = -3$ is a solution of all three and $c = 3$ of none.

(a) Do not divide by $3+c$ (you would lose a solution). Bring to one side and factor: $(3+c)(c^2+2-2c-2) = (3+c)(c^2-2c) = c(c-2)(c+3) = 0$, so $c \in \{-3, 0, 2\}$.

(b) $1-2c+c^2 = (1-c)^2$, so the right side is $\sqrt{(1-c)^2} = |1-c|$. The equation $1-c = |1-c|$ holds exactly when $1-c \geq 0$: all $c \leq 1$.

(c) $2^{c^2+2c+1} = 2^4$, so $c^2+2c+1 = (c+1)^2 = 4$, $c+1 = \pm 2$: $c = 1$ or $c = -3$.

(d) -3 appears in (a), satisfies $-3 \leq 1$ in (b), and appears in (c); 3 is not in (a), fails $3 \leq 1$, and $3^2+6+1 = 16 \neq 4$. ✓ $\square$

NOW YOU 1, 3 · solutions in Appendix S

1 All positive divisors of $60, 61, \dots, 65$; which are squares? 3 Check $\frac{1}{6} - \frac{1}{7} = \frac{1}{6 \cdot 7}$; for which p, q is $\frac{1}{p} - \frac{1}{q} = \frac{1}{pq}$?

B FUNCTIONS AND INVERSE FUNCTIONS

Definition 0.1 · function, domain, codomain, range

A **function** $f: A \rightarrow B$ is a rule assigning to every $x \in A$ exactly one $y = f(x) \in B$, written $x \mapsto f(x)$; y is the **image** (beeld) of x and x an **original** of y . A is the **domain**, B the **codomain**, and the **range** (bereik) is the set of $y \in B$ that actually occur as images. If no domain is given, take the largest subset of $\mathbb{R}$ on which the formula makes sense. Voorbeeld 0.2: $x \mapsto x^2$ on $\mathbb{R}$ has range $[0, \infty)$; 5 has the two originals $\pm\sqrt{5}$. Voorbeeld 0.3: “ $f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \sqrt{x}$ ” is wrong (negative x); $f: [0, \infty) \rightarrow \mathbb{R}$ is right. Strictly, “the function $f(x)$ ” is wrong ($f(x)$ is a number), but the dictaat uses it anyway. **Piecewise** definitions (Voorbeeld 0.4): $\text{sgn } x = -1, 0, +1$ for $x < 0, x = 0, x > 0$; in graphs a closed dot is on the graph, an open dot is not.

Definition 0.5 and Voorbeeld 0.7 · inverse function

If $f: A \rightarrow B$ is **bijective** (a one-to-one pairing of A with B), its **inverse** $f_{\text{inv}}: B \rightarrow A$ is the unique function with $f_{\text{inv}}(f(x)) = x$. The dictaat writes f_{inv} , not f^{-1} , to avoid confusion with $1/f$. $y = f(x) \Leftrightarrow x = f_{\text{inv}}(y)$; the graphs are mirror images in the line $y = x$. Voorbeeld 0.6: $x \mapsto (x-1)^2$ is not bijective on $\mathbb{R} \rightarrow \mathbb{R}$ ($f(-1) = f(3) = 4$), nor on $[1, \infty) \rightarrow \mathbb{R}$ (negative numbers are never images), but it is bijective $[1, \infty) \rightarrow [0, \infty)$ with inverse $y \mapsto 1 + \sqrt{y}$. **To find the inverse:** write $y = f(x)$ and solve for x ; then swap the letters.

Exercise 9

from the dictaat

Find the inverse of the bijective functions (a) $p : x \mapsto 1 + \sqrt[3]{2x}$; (b) $q : x \mapsto \frac{x^2 - 4}{x + 2}$; (c) $r : x \mapsto \frac{3x}{4 + 5x}$, and state on which domain and range each is bijective.

(a) $y = 1 + \sqrt[3]{2x} \Leftrightarrow (y - 1)^3 = 2x \Leftrightarrow x = \frac{1}{2}(y - 1)^3$. So $p_{\text{inv}}(x) = \frac{1}{2}(x - 1)^3$; the cube root is defined for all reals, so $p : \mathbb{R} \rightarrow \mathbb{R}$ is bijective (domain $\mathbb{R}$, range $\mathbb{R}$).

(b) For $x \neq -2$: $q(x) = \frac{(x-2)(x+2)}{x+2} = x - 2$. Domain $\mathbb{R} \setminus \{-2\}$; the value $-2 - 2 = -4$ is never taken, so the range is $\mathbb{R} \setminus \{-4\}$. From $y = x - 2$: $q_{\text{inv}}(x) = x + 2$ on $\mathbb{R} \setminus \{-4\}$, with range $\mathbb{R} \setminus \{-2\}$.

(c) Domain $x \neq -\frac{4}{5}$. $y = \frac{3x}{4+5x} \Leftrightarrow 4y + 5xy = 3x \Leftrightarrow x(3 - 5y) = 4y \Leftrightarrow x = \frac{4y}{3 - 5y}$, provided $y \neq \frac{3}{5}$ (and $y = \frac{3}{5}$ would need $3x = \frac{3}{5}(4 + 5x)$, i.e. $0 = \frac{12}{5}$: impossible, so $\frac{3}{5}$ is not in the range). Hence $r : \mathbb{R} \setminus \{-\frac{4}{5}\} \rightarrow \mathbb{R} \setminus \{\frac{3}{5}\}$ is bijective with $r_{\text{inv}}(x) = \frac{4x}{3 - 5x}$. $\square$

Exercise 10

from the dictaat

Let f be bijective with inverse f_{inv} and let $s(x) = 1 - 2f(3 - 4x)$. Show how to compute s_{inv} from f_{inv} .

Put $y = s(x) = 1 - 2f(3 - 4x)$ and peel the layers off from the outside: $f(3 - 4x) = \frac{1 - y}{2}$, so $3 - 4x = f_{\text{inv}}\left(\frac{1 - y}{2}\right)$, so $x = \frac{3 - f_{\text{inv}}\left(\frac{1 - y}{2}\right)}{4}$. Hence $s_{\text{inv}}(y) = \frac{1}{4}\left(3 - f_{\text{inv}}\left(\frac{1 - y}{2}\right)\right)$: the inverse undoes the steps $x \mapsto 3 - 4x \mapsto f(\cdot) \mapsto 1 - 2(\cdot)$ in reverse order. $\square$

NOW YOU 6, 7, 8 · solutions in Appendix S

6 Sketch the graphs of $x \mapsto x^2$ and $x \mapsto \sqrt{x}$ by hand. **7** Give a piecewise formula for the Heaviside function. **8** Fill in the steps of Voorbeeld 0.7 (inverse of $f(x) = 3 - \frac{1}{2+x}$).

Absolute value
§0.2.1

$|x| = x$ for $x \geq 0$ and $-x$ for $x < 0$; equivalently $|x| = \sqrt{x^2}$. **Distance:** $|x - y|$ is the distance between x and y on the number line, so $|x - 1| < 3$ means “less than 3 away from 1”. Recognise $|x|$ when it is hidden, as in $\sqrt{x^2}$ or $\sqrt{1 - \cos^2 x}$.

Exercise 11

from the dictaat

Solve $|x - 1| \leq |x - 3|$.

Distance: the points no further from 1 than from 3 are those on the side of the midpoint 2 nearer to 1: $x \leq 2$.

Algebra: both sides are ≥ 0 , so squaring is allowed: $x^2 - 2x + 1 \leq x^2 - 6x + 9 \Leftrightarrow 4x \leq 8 \Leftrightarrow x \leq 2$. $\square$

NOW YOU 12 · solutions in Appendix S

12 Simplify $\sqrt{1 - \cos^2 x}$ as far as possible; sanity check at $x = -\frac{\pi}{6}$.

Compositions, even and odd functions

§0.2.2 and Definition 0.8

$(f \circ g)(x) = f(g(x))$ (“ f after g ”: apply g first). f is **even** if $f(x) = f(-x)$ for all x in its domain, **odd** if $f(x) = -f(-x)$. Even: symmetric in the y -axis (x^2 , $\cos$, $|x|$); odd: symmetric about the origin (x , x^3 , $\sin$).

Exercise 15

from the dictaat

$f(x) = \sqrt{x}$, $g(x) = x^2$. Give domain and range of each; simplify $f \circ g$ and $g \circ f$; what is the difference?

f : domain $[0, \infty)$, range $[0, \infty)$. g : domain $\mathbb{R}$, range $[0, \infty)$.

$(f \circ g)(x) = \sqrt{x^2} = |x|$, defined for all $x \in \mathbb{R}$ (since $x^2 \geq 0$ is always in the domain of f).

$(g \circ f)(x) = (\sqrt{x})^2 = x$, but only for $x \geq 0$ (the domain of f).

Difference: different domains ($\mathbb{R}$ versus $[0, \infty)$) and different formulas on the negative axis: $f \circ g$ is $|x|$, not x . On $[0, \infty)$ they agree. □

Exercise 17

from the dictaat

Are there values of p for which $f: x \mapsto |x+p| + |x-4|$ is even? Odd?

Even. For x larger than both 4 and $|p|$ all absolute values open with a plus sign: $f(x) = (x+p) + (x-4) = 2x+p-4$ and $f(-x) = |-x+p| + |-x-4| = (x-p) + (x+4) = 2x-p+4$. Evenness forces $p-4 = -p+4$, so $p=4$. Conversely for $p=4$: $f(-x) = |-x+4| + |-x-4| = |x-4| + |x+4| = f(x)$ for all x . So f is even exactly for $p=4$.

Odd. An odd function defined at 0 has $f(0) = -f(0)$, so $f(0) = 0$. Here $f(0) = |p| + 4 \geq 4 > 0$. So f is odd for no p . □

Exercise 19

from the dictaat

v even and w odd, both defined on all of $\mathbb{R}$. Decide whether vw , w^2 , $v \circ w$, $w \circ v$, $w \circ w$ are even, odd, or neither.

Replace x by $-x$ and use $v(-x) = v(x)$, $w(-x) = -w(x)$:

$(vw)(-x) = v(x) \cdot (-w(x)) = -(vw)(x)$: **odd**. $w^2(-x) = (-w(x))^2 = w(x)^2$: **even**.

$(v \circ w)(-x) = v(-w(x)) = v(w(x))$: **even**. $(w \circ v)(-x) = w(v(x))$: **even**.

$(w \circ w)(-x) = w(-w(x)) = -w(w(x))$: **odd**. □

NOW YOU 13, 14, 16, 18 · solutions in Appendix S

13 $p(x) = 1+x$, $q(x) = x^2$: find $p \circ p$, $p \circ q$, $q \circ p$, $q \circ q$. **14** $m(x) = -x$ satisfies $(m \circ m)(x) = x$; find more functions f with $(f \circ f)(x) = x$. **16** Examples of even, odd, neither, both. **18** “Every odd function has $f(0) = 0$ ”: true? Counterexample? Something similar for even functions?

C POLYNOMIALS, DIVISIBILITY AND FACTORISING
§0.3 · polynomials, long division, the factor theorem

A **polynomial** (veelterm) of degree n is $a_0 + a_1x + \dots + a_nx^n$ with $a_n \neq 0$. **Factorising** is the reverse of expanding: $4 - 16x^2 = 4(1 - 2x)(1 + 2x)$, $1 - x^4 = (1 - x)(1 + x)(1 + x^2)$. **Long division (staartdeling)** of a polynomial by one of equal or lower degree: divide the highest-power terms, subtract, repeat; the

dictaat's example gives

$$\frac{8x^3 - 12x^2 + 6x + 3}{2x^2 - x} = 4x - 4 + \frac{2x + 3}{2x^2 - x}, \quad \text{remainder } 2x + 3 \text{ (lower degree than the divisor).}$$

Stelling 0.9 (factorstelling): $p(x_0) = 0$ iff $x - x_0$ is a factor of p . (Proof: divide, $p(x) = (x - x_0)q(x) + c$; substituting x_0 gives $c = p(x_0)$.) **Strategy:** to factorise, try $x = 0, \pm 1, \pm 2, \dots$ to find a zero, divide out the factor, repeat; a polynomial of degree n has at most n zeros. **Vocabulary:** factors are multiplied, terms are added.

Exercise 20(a)

from the dictaat

Divide $\frac{6x^4 - 4x^3 - x^2 + 6x - 12}{2x^2 - 3}$ and check by multiplying back.

$6x^4 \div 2x^2 = 3x^2$; subtract $3x^2(2x^2 - 3) = 6x^4 - 9x^2$: remainder $-4x^3 + 8x^2 + 6x - 12$.

$-4x^3 \div 2x^2 = -2x$; subtract $-2x(2x^2 - 3) = -4x^3 + 6x$: remainder $8x^2 - 12$.

$8x^2 \div 2x^2 = 4$; subtract $4(2x^2 - 3) = 8x^2 - 12$: remainder 0.

So the quotient is $3x^2 - 2x + 4$ with remainder 0: $2x^2 - 3$ is a factor. Check: $(2x^2 - 3)(3x^2 - 2x + 4) = 6x^4 - 4x^3 + 8x^2 - 9x^2 + 6x - 12 = 6x^4 - 4x^3 - x^2 + 6x - 12$. ✓ $\square$

Exercise 21(a)

from the dictaat

Factorise $x^3 + 2x^2 - 5x - 6$ and check by expanding.

Try small integers: $x = 1$: $1 + 2 - 5 - 6 \neq 0$; $x = -1$: $-1 + 2 + 5 - 6 = 0$. So $x + 1$ is a factor (Stelling 0.9). Dividing: $x^3 + 2x^2 - 5x - 6 = (x + 1)(x^2 + x - 6)$, and $x^2 + x - 6 = (x + 3)(x - 2)$. Result: $(x + 1)(x + 3)(x - 2)$. Check: $(x + 1)(x^2 + x - 6) = x^3 + x^2 - 6x + x^2 + x - 6 = x^3 + 2x^2 - 5x - 6$. ✓ $\square$

NOW YOU 20(b), 21(b), 21(c) · solutions in Appendix S

20(b) $\frac{6x^4 - 4x^3 - x^2 + 6x - 12}{x + 2x^3}$. **21(b)** $45x^4(x - 14) - 50x^3(x - 14) + 30x^2(x - 14)$. **21(c)** $x - x^3 + x^5 - x^7$.

D TRIGONOMETRY

§0.4 · the unit circle

Angles in radians. The point of the unit circle at angle t (counterclockwise from the positive x -axis) is $(\cos t, \sin t)$; $\tan t = \sin t / \cos t$ is the length cut off on the vertical tangent line at $(1, 0)$ (similar triangles). Symmetries read off from the circle: $\sin(-t) = -\sin t$, $\cos(-t) = \cos t$, $\sin(\frac{\pi}{2} - t) = \cos t$, $\sin^2 t + \cos^2 t = 1$ (Pythagoras). The sum rule $\sin(x + y) = \sin x \cos y + \cos x \sin y$ is given; everything else in Oefening 26 follows from it. Exact values come from the square with diagonal 1 (angles $\frac{\pi}{4}$, sides $\frac{1}{2}\sqrt{2}$) and the equilateral triangle with side 1 (angles $\frac{\pi}{3}$; half of it has angles $\frac{\pi}{6}, \frac{\pi}{3}$ and sides $\frac{1}{2}, \frac{1}{2}\sqrt{3}$).

Exercise 23

from the dictaat

 Fill in the table of $\sin t$, $\cos t$, $\tan t$ for $t = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$.

t	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin t$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1
$\cos t$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0
$\tan t$	0	$\frac{1}{3}\sqrt{3}$	1	$\sqrt{3}$	undefined

 Memory aid: the sines are $\frac{1}{2}\sqrt{0}, \frac{1}{2}\sqrt{1}, \frac{1}{2}\sqrt{2}, \frac{1}{2}\sqrt{3}, \frac{1}{2}\sqrt{4}$; the cosines are the same row reversed. $\square$
Exercise 25

from the dictaat

 Use the table and the symmetries of the circle to find: $\sin \frac{2\pi}{3}, \cos \frac{2\pi}{3}, \tan \frac{2\pi}{3}, \sin \frac{5\pi}{6}, \cos \frac{5\pi}{6}, \tan \frac{5\pi}{6}, \sin \pi, \cos \pi, \tan \pi, \sin(-\frac{\pi}{4}), \cos(-\frac{3\pi}{4}), \tan \frac{3\pi}{4}, \sin \frac{4\pi}{3}, \cos \frac{5\pi}{3}, \tan \frac{6\pi}{3}, \sin(-\frac{\pi}{6}), \cos(-\frac{2\pi}{3}), \tan(-\frac{5\pi}{6})$.

Reduce each angle to a “nice” angle in the first quadrant and fix the sign from the quadrant (sin positive above the axis, cos positive to the right, tan positive in quadrants I and III).

 $\frac{2\pi}{3} = \pi - \frac{\pi}{3}$ (II): $\sin = \frac{1}{2}\sqrt{3}, \cos = -\frac{1}{2}, \tan = -\sqrt{3}$. $\frac{5\pi}{6} = \pi - \frac{\pi}{6}$ (II): $\sin = \frac{1}{2}, \cos = -\frac{1}{2}\sqrt{3}, \tan = -\frac{1}{3}\sqrt{3}$.

 π : $\sin \pi = 0, \cos \pi = -1, \tan \pi = 0$. $\sin(-\frac{\pi}{4}) = -\frac{1}{2}\sqrt{2}; \cos(-\frac{3\pi}{4}) = \cos \frac{3\pi}{4} = -\frac{1}{2}\sqrt{2}; \tan \frac{3\pi}{4} = -1$.

 $\frac{4\pi}{3} = \pi + \frac{\pi}{3}$ (III): $\sin = -\frac{1}{2}\sqrt{3}$. $\frac{5\pi}{3} = 2\pi - \frac{\pi}{3}$ (IV): $\cos = \frac{1}{2}$. $\tan \frac{6\pi}{3} = \tan 2\pi = 0$.

 $\sin(-\frac{\pi}{6}) = -\frac{1}{2}; \cos(-\frac{2\pi}{3}) = \cos \frac{2\pi}{3} = -\frac{1}{2}; \tan(-\frac{5\pi}{6}) = -\tan \frac{5\pi}{6} = \frac{1}{3}\sqrt{3}$. $\square$
Exercise 26

from the dictaat

 From $\sin(x+y) = \sin x \cos y + \cos x \sin y$ derive: (a) $\sin(x-y)$; (b) $\cos(x+y)$ using $\cos(x+y) = \sin((\frac{\pi}{2} - x) - y)$; (c) $\tan(x+y)$ in terms of $\tan$ only; (d) the double-angle formulas.

 (a) Replace y by $-y$ and use $\cos(-y) = \cos y, \sin(-y) = -\sin y$: $\sin(x-y) = \sin x \cos y - \cos x \sin y$.

 (b) $\cos(x+y) = \sin((\frac{\pi}{2} - x) - y) = \sin(\frac{\pi}{2} - x) \cos y - \cos(\frac{\pi}{2} - x) \sin y = \cos x \cos y - \sin x \sin y$, using $\sin(\frac{\pi}{2} - x) = \cos x$ and $\cos(\frac{\pi}{2} - x) = \sin x$.

 (c) $\tan(x+y) = \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y}$; divide numerator and denominator by $\cos x \cos y$: $\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$.

 (d) Put $y = x$: $\sin 2x = 2 \sin x \cos x$; $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$ (using $\sin^2 x + \cos^2 x = 1$); $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$. $\square$
NOW YOU 22, 24, 27 (Extra) · solutions in Appendix S

22 Angles (radians) and sides of the square with diagonal 1 and the equilateral triangle with side 1. **24** (a) Why are the labels $\sin, \cos, \tan$ placed where they are in the unit circle? (b) Signs in $\sin(\pi \pm t), \cos(\pi \pm t), \tan(\pi \pm t)$. (c) Check the four basic identities in the figure. **27 (Extra)** A point moves along the unit circle at speed 1; conclude $\sin' = \cos, \cos' = -\sin$.

E THE LOGARITHM

Definitions 0.10, 0.11 · logarithm, base, change of base

A **logarithm** is a function $\log : (0, \infty) \rightarrow \mathbb{R}$ with $\log(ab) = \log a + \log b$ for all $a, b > 0$ (it turns multiplication into addition; that is why Napier and Briggs invented it). Its **base** (grondtal) is the number g with $\log g = 1$; notation ${}^g\log x$ (or $\log_g x$). All rules follow from the definition (Oefening 28). **Change of base:** ${}^h\log x = \frac{{}^g\log x}{{}^g\log h}$ (the right side satisfies the functional equation and takes the value 1 at h). **From now on** $\log$ means the **natural logarithm**, base $e = 2.71828\dots$ (the primitive of $1/x$, the inverse of $x \mapsto e^x$).

Exercise 28

from the dictaat

For $a, b > 0$ derive from the definition, one by one: (a) $\log 1 = 0$; (b) $\log a^2 = 2 \log a$; (c) $\log \frac{1}{a} = -\log a$; (d) $\log a^n = n \log a$ for $n \in \mathbb{Z}$; (e) $\log a^{p/q} = \frac{p}{q} \log a$ for $\frac{p}{q} \in \mathbb{Q}$.

(a) $\log 1 = \log(1 \cdot 1) = \log 1 + \log 1$, so $\log 1 = 0$.

(b) $\log a^2 = \log(a \cdot a) = \log a + \log a = 2 \log a$.

(c) $0 = \log 1 = \log(a \cdot \frac{1}{a}) = \log a + \log \frac{1}{a}$, so $\log \frac{1}{a} = -\log a$.

(d) For $n > 0$ repeat (b): $\log a^n = \log(a \cdots a) = n \log a$. For $n = 0$: $\log a^0 = \log 1 = 0 = 0 \cdot \log a$. For $n < 0$: $\log a^n = \log \frac{1}{a^{-n}} = -\log a^{-n} = -(-n) \log a = n \log a$ by (c).

(e) First $p = 1$: $a = (a^{1/q})^q$, so by (d) $\log a = q \log a^{1/q}$, i.e. $\log a^{1/q} = \frac{1}{q} \log a$. Then $\log a^{p/q} = \log(a^{1/q})^p = p \log a^{1/q} = \frac{p}{q} \log a$.

(The dictaat remarks that $\log a^c = c \log a$ also holds for irrational c , but that needs §6.6.) $\square$

Exercise 30

from the dictaat

Without a calculator: is $(\log 3)^2 - \log(3^2)$ negative, zero or positive?

$(\log 3)^2 - \log(3^2) = (\log 3)^2 - 2 \log 3 = \log 3 (\log 3 - 2)$. Now $\log 3 > \log 1 = 0$, and $\log 3 < 2$ because $3 < e^2$ (as $e > 1.7$ gives $e^2 > 2.89$; more crudely $e^2 > 2^2 = 4 > 3$) and $\log$ is increasing. So the product is **negative**. (Indeed $\log 3 \approx 1.10$, giving ≈ -0.99 .) $\square$

NOW YOU 29 · solutions in Appendix S

29 (a) Why do logarithms with bases g and h differ everywhere by a constant factor? (b) Which factor? (c) What can you say about ${}^g\log h \cdot {}^h\log g$?

Watch out

- $\sqrt{x^2} = |x|$, never x ; $\sqrt{1 - \cos^2 x} = |\sin x|$ (Oefening 12).
- f_{inv} is the inverse under composition, not $1/f$. Before inverting, check bijectivity and state domain and range.
- Do not divide an equation by an expression that can be 0 (Oefening 5a); do not multiply an inequality by an expression of unknown sign (Oefening 4c).
- $\log$ is the natural logarithm in this course; “log” on a calculator is often base 10.

Chapter 1

Complex numbers

Contents

- A Why complex numbers (§1.1)
- B The set $\mathbb{C}$ (§1.2)
- C Modulus, argument, conjugate (§1.3)
- D Calculations in $\mathbb{C}$ (§1.4)
- E The complex exponential (§1.5)

Planning

Exercises 101–123 (all worked or solved).
 Hoofdpunten: use the complex plane; real and imaginary part, modulus, argument, conjugate, $e^{i\varphi}$; elementary calculations in $\mathbb{C}$; solving simple polynomial equations in $\mathbb{C}$.

A WHY COMPLEX NUMBERS ARE UNAVOIDABLE

§1.1 · Cardano and Bombelli

Cardano (1545) solved $x^3 + px = q$ by the formula $x = \sqrt[3]{\frac{q}{2} + \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}} + \sqrt[3]{\frac{q}{2} - \sqrt{(\frac{q}{2})^2 + (\frac{p}{3})^3}}$. It works for $x^3 + 6x = 20$ ($x = 2$), but for $x^3 - 15x = 4$ it needs $\sqrt{-121}$, although $x = 4$ is visibly a solution. Bombelli (1572) just computed on with $\sqrt{-121}$: since $(2 \pm \sqrt{-1})^3 = 2 \pm \sqrt{-121}$, the formula gives $x = (2 + \sqrt{-1}) + (2 - \sqrt{-1}) = 4$. Moral: the complex numbers forced themselves upon us. (No exercises; not exam material.)

B THE SET OF COMPLEX NUMBERS

Definition 1.1 · $\mathbb{C}$, real and imaginary part

i is a new number with $i^2 = -1$ ($i = \sqrt{-1}$, and $-i$ is the other square root of -1). $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$; for $z = a + bi$, $\operatorname{Re} z = a$ and $\operatorname{Im} z = b$ (both real). $\mathbb{R} \subseteq \mathbb{C}$ via $x = x + 0i$. **Fundamental theorem of algebra:** every non-constant polynomial has a zero in $\mathbb{C}$, so all zeros of all polynomials lie in $\mathbb{C}$ (no further extension needed). For $b \geq 0$: $\sqrt{-b} = i\sqrt{b}$.

Exercise 101

from the dictaat

Compute the complex zeros of $P(x) = x^2 - 4x + 13$ with the abc-formula, in the form $a + bi$.

$$x = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i. \text{ Check: } (2 + 3i)^2 - 4(2 + 3i) + 13 = 4 + 12i - 9 - 8 - 12i + 13 = 0. \checkmark \quad \square$$

C THE COMPLEX PLANE: MODULUS, ARGUMENT, CONJUGATE

Definitions 1.2, 1.3 · modulus, argument, polar form, conjugate

Plot $a + bi$ as the point (a, b) : real axis horizontal, imaginary axis vertical (the Argand plane). The **modulus** $|z| = \sqrt{a^2 + b^2}$ is the distance to 0; the **argument** $\arg z$ is the angle from the positive real axis, counterclockwise, with the convention $-\pi < \arg z \leq \pi$ ($\arg 0$ is undefined). $\tan(\arg z) = \frac{b}{a}$ when $a \neq 0$, but two angles in $(-\pi, \pi]$ have the same tangent: **always sketch** z to pick the right one. Positive reals have argument 0, negative reals π , bi has argument $\pm \frac{\pi}{2}$. With $r = |z|$, $\varphi = \arg z$: $z = r(\cos \varphi + i \sin \varphi)$ (**polar coordinates**). The **conjugate** $\bar{z} = a - bi$ is the mirror image of z in the real axis.

Exercise 103

from the dictaat

Modulus and argument of $1 + i$, $3i$, -20 , $-1 - i$; draw them.

$1 + i$: $|1 + i| = \sqrt{1+1} = \sqrt{2}$, in quadrant I on the diagonal: $\arg = \frac{\pi}{4}$. $3i$: $|3i| = 3$, on the positive imaginary axis: $\arg = \frac{\pi}{2}$.

-20 : $|-20| = 20$, negative real: $\arg = \pi$. $-1 - i$: $|-1 - i| = \sqrt{2}$, quadrant III on the diagonal: $\arg = -\frac{3\pi}{4}$ (not $\frac{5\pi}{4}$: the argument must lie in $(-\pi, \pi]$). □

Exercise 104

from the dictaat

$z = a + bi$. Give $\arg z$ in terms of $\arctan$ for: (a) $a > 0, b \geq 0$; (b) $a > 0, b < 0$; (c) $a < 0, b \geq 0$; (d) $a < 0, b < 0$; (e) $a = 0, b > 0$; (f) $a = 0, b < 0$.

$\arctan$ takes values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, i.e. it returns the angle of a point in the *right* half-plane.

(a) and (b): z is in the right half-plane, so $\arg z = \arctan \frac{b}{a}$ (in $[0, \frac{\pi}{2})$ for (a), in $(-\frac{\pi}{2}, 0)$ for (b)).

(c) z in quadrant II: $\arctan \frac{b}{a}$ is the angle of $-z$ (which is in quadrant IV), so $\arg z = \arctan \frac{b}{a} + \pi$.

(d) z in quadrant III: $\arg z = \arctan \frac{b}{a} - \pi$ (adding π would leave $(-\pi, \pi]$).

(e) $\arg z = \frac{\pi}{2}$; (f) $\arg z = -\frac{\pi}{2}$ ($\tan$ undefined). □

Exercise 105

from the dictaat

Determine $\arg z$ for $z = -\frac{1}{2}\sqrt{3} - \frac{1}{2}i$.

$|z| = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$ and z is in quadrant III (both parts negative). The reference angle has $\tan = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$, i.e. $\frac{\pi}{6}$. So $\arg z = -\pi + \frac{\pi}{6} = -\frac{5\pi}{6}$ (case (d) of 104: $\arctan \frac{-1/2}{-\sqrt{3}/2} - \pi = \frac{\pi}{6} - \pi$). Indeed $z = \cos(-\frac{5\pi}{6}) + i \sin(-\frac{5\pi}{6})$. □

NOW YOU 102, 106, 107, 108, 109, 110 · solutions in Appendix S

102 Draw $1 + 2i$, $3i$, -1 , $-4 - i$, $2 - i$, 4 . **106** Why is $\arg z$ undefined for $z = 0$? **107** Check $\operatorname{Re} \bar{z} = \operatorname{Re} z$, $\operatorname{Im} \bar{z} = -\operatorname{Im} z$, $|\bar{z}| = |z|$, $\arg \bar{z} = -\arg z$, $\bar{\bar{z}} = z$. **108** $\bar{z}$ for $1 + i$, $3i$, -20 , $-1 - i$. **109** Solve $z^2 + 2z + 2 = 0$. **110** If z solves $az^2 + bz + c = 0$ with $a, b, c \in \mathbb{R}$, so does $\bar{z}$.

D CALCULATIONS IN $\mathbb{C}$

Definition 1.4 and Stelling 1.5 · sum, product, conjugation rules, division

$(a+bi) + (c+di) = (a+c) + (b+d)i$ and $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$: this is ordinary expanding of brackets with $i^2 = -1$. **Stelling 1.5:** $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, $z\bar{z} = |z|^2$, $z + \bar{z} = 2\operatorname{Re} z$.

Division: for $w \neq 0$, $\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}$: multiply numerator and denominator by the conjugate of the denominator, which makes the denominator real.

Exercise 111(a)

from the dictaat

$z = 3 - 2i$, $w = 1 + i$: compute $z + w$ and zw .

$z + w = (3+1) + (-2+1)i = 4 - i$. $zw = (3-2i)(1+i) = 3 + 3i - 2i - 2i^2 = 3 + i + 2 = 5 + i$. (With the formula: $ac - bd = 3 \cdot 1 - (-2) \cdot 1 = 5$, $ad + bc = 3 \cdot 1 + (-2) \cdot 1 = 1$.) $\square$

Exercise 114

from the dictaat

Write $\frac{z}{w}$ in the form $a + bi$: (a) $z = 1$, $w = 2 + 3i$; (b) $z = 1 + i$, $w = i$; (c) $z = -2 - i$, $w = 3 - 4i$.

$$(a) \frac{1}{2+3i} = \frac{2-3i}{(2+3i)(2-3i)} = \frac{2-3i}{4+9} = \frac{2}{13} - \frac{3}{13}i.$$

$$(b) \frac{1+i}{i} = \frac{(1+i)(-i)}{i(-i)} = \frac{-i-i^2}{1} = 1-i. \text{ (Check: } i(1-i) = i+1. \checkmark)$$

$$(c) \frac{-2-i}{3-4i} = \frac{(-2-i)(3+4i)}{9+16} = \frac{-6-8i-3i-4i^2}{25} = \frac{-2-11i}{25} = -\frac{2}{25} - \frac{11}{25}i. \quad \square$$

NOW YOU 111(b),(c), 112, 113, 115 · solutions in Appendix S

111 (b) $z = i$, $w = 1 - i$; (c) $z = w = -4 + 6i$. **112** Check that Definition 1.4 is “expanding brackets with $i^2 = -1$ ”. **113** Prove parts (a), (c), (d) of Stelling 1.5 by writing out. **115** Commutativity, associativity and distributivity in $\mathbb{C}$ by writing out.

E THE COMPLEX EXPONENTIAL FUNCTION

Stelling 1.6, Definition 1.8, Stelling 1.11 · multiply moduli, add arguments

$|zw| = |z||w|$ and $\arg(zw) = \arg z + \arg w$ (modulo 2π); proof: multiply the polar forms and use the sum formulas. So powers are easy in polar form: Voorbeeld 1.7, $(\sqrt{3} + i)^3$ has modulus $2^3 = 8$ and argument $3 \cdot \frac{\pi}{6} = \frac{\pi}{2}$, hence equals $8i$. **Definition 1.8:** $e^{it} = \cos t + i \sin t$ for real t (consistent: $e^0 = 1$, $\frac{d}{dt} e^{it} = ie^{it}$, $e^{is} e^{it} = e^{i(s+t)}$, Voorbeeld 1.9). Hence $z = re^{i\varphi}$ with $r = |z|$, $\varphi = \arg z$, and $re^{i\varphi} \cdot se^{i\theta} = rse^{i(\varphi+\theta)}$. **De Moivre:** $(\cos \varphi + i \sin \varphi)^n = \cos n\varphi + i \sin n\varphi$. For $z = a + ib$: $e^z = e^a(\cos b + i \sin b)$.

Voorbeeld 1.10 · solving $z^n = w$

Write $z = re^{i\varphi}$ and $w = se^{i\theta}$. Then $z^n = r^n e^{in\varphi}$, so $r^n = s$ gives $r = s^{1/n}$ (moduli are ≥ 0), and $n\varphi = \theta + 2k\pi$ gives $\varphi = \frac{\theta + 2k\pi}{n}$, $k = 0, 1, \dots, n-1$ (n different solutions, equally spaced on a circle; other k repeat them). Example: $z^6 = 1$ has $r = 1$, $\varphi = \frac{k\pi}{3}$: $1, \frac{1}{2} \pm \frac{1}{2}\sqrt{3}i, -\frac{1}{2} \pm \frac{1}{2}\sqrt{3}i, -1$. Convert back to $a + bi$ with

cos, sin of nice angles.

Exercise 117

from the dictaat

Write $(1+i)^7$ in the form $a+bi$.

$1+i = \sqrt{2}e^{i\pi/4}$, so $(1+i)^7 = (\sqrt{2})^7 e^{7i\pi/4} = 8\sqrt{2}e^{-i\pi/4}$ (since $\frac{7\pi}{4} = 2\pi - \frac{\pi}{4}$). Hence $(1+i)^7 = 8\sqrt{2}(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4}) = 8\sqrt{2}(\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i) = 8 - 8i$. (Check: $(1+i)^2 = 2i$, $(1+i)^4 = -4$, $(1+i)^7 = (-4)(2i)(1+i) = -8i + 8$. ✓) $\square$

Exercise 120

from the dictaat

Write i and $1+i$ in the form $re^{i\varphi}$.

$|i| = 1$, $\arg i = \frac{\pi}{2}$: $i = e^{i\pi/2}$. $|1+i| = \sqrt{2}$, $\arg(1+i) = \frac{\pi}{4}$: $1+i = \sqrt{2}e^{i\pi/4}$. $\square$

Exercise 123

from the dictaat

Solve: (a) $z^4 = 16$; (b) $z^2 = i$; (c) $z^6 - 2z^3 + 1 = 49$.

(a) $z = re^{i\varphi}$: $r^4 = 16$ gives $r = 2$; $4\varphi = 0 + 2k\pi$ gives $\varphi = \frac{k\pi}{2}$, $k = 0, 1, 2, 3$: $z = 2, 2i, -2, -2i$.

(b) $|i| = 1$, $\arg i = \frac{\pi}{2}$: $r = 1$ and $2\varphi = \frac{\pi}{2} + 2k\pi$, $\varphi = \frac{\pi}{4}$ or $\frac{5\pi}{4} \equiv -\frac{3\pi}{4}$: $z = \pm e^{i\pi/4} = \pm(\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i)$. So yes, $\sqrt{i}$ is a complex number: $(\frac{1+i}{\sqrt{2}})^2 = \frac{2i}{2} = i$.

(c) Substitute $w = z^3$: $w^2 - 2w - 48 = 0$, $w = \frac{2 \pm \sqrt{4+192}}{2} = \frac{2 \pm 14}{2}$, so $w = 8$ or $w = -6$.

$z^3 = 8$: $r = 2$, $\varphi = \frac{2k\pi}{3}$: $z = 2, 2e^{2\pi i/3} = -1 + \sqrt{3}i, 2e^{-2\pi i/3} = -1 - \sqrt{3}i$.

$z^3 = -6 = 6e^{i\pi}$: $r = \sqrt[3]{6}$, $3\varphi = \pi + 2k\pi$, $\varphi = \frac{\pi}{3}, \pi, -\frac{\pi}{3}$: $z = \sqrt[3]{6}(\frac{1}{2} + \frac{1}{2}\sqrt{3}i), -\sqrt[3]{6}, \sqrt[3]{6}(\frac{1}{2} - \frac{1}{2}\sqrt{3}i)$.

Six solutions in total, as expected for degree 6. $\square$

NOW YOU 116, 118, 119, 121, 122 · solutions in Appendix S

116 Adding $a+bi$ and $c+di$ is adding the vectors (a,b) and (c,d) . **118** Geometric meaning of multiplication by i . **119** $e^{i\pi}$ and $4e^{i\pi/3}$ in the form $a+bi$. **121** $|e^{i\varphi}| = 1$: all $e^{i\varphi}$ lie on the unit circle. **122** Draw the six solutions of $z^6 = 1$; the conjugate of each solution is again a solution.

Watch out

- $\arg z \in (-\pi, \pi]$: $-1-i$ has argument $-\frac{3\pi}{4}$, not $\frac{5\pi}{4}$. tan alone does not determine the quadrant: sketch.
- $|z|^2 = z\bar{z}$, not z^2 . $\overline{zw} = \bar{z}\bar{w}$ is the conjugate of the product; do not confuse with $\bar{z}\bar{w}$ written out.
- An equation $z^n = w$ has n solutions; after finding the modulus, list $k = 0, \dots, n-1$ and then stop ($k = n$ repeats $k = 0$).

Chapter 2

Limits, differentiation and integration

Contents

- A Limits (§2.1)
- B Continuity (§2.2)
- C Differentiation (§2.3)
- D Integration (§2.4)
- E Integration with common sense (§2.5)

Planning

Exercises 201–220 (all worked or solved).

Hoofdpunten: the limit definition of the derivative and its picture; the rules for limits; the rules for differentiating, above all the chain rule; integration as “sum of strips”, and the elementary rules that follow from that picture; some definite integrals without (much) computation.

A LIMITS

Definition 2.1 (informal), one-sided limits, Stelling 2.3 · limits and their rules

$\lim_{x \rightarrow a} f(x) = b$ means: $f(x)$ can be made as close to b as we like by taking x close enough to a . Two things to keep sharp: the limit says nothing about $x = a$ itself ($f(a)$ need not equal b , f need not even be defined at a); and it does not claim that $f(x) = b$ for some x . **Left limit** $\lim_{x \rightarrow a^-} f(x)$ ($x \uparrow a$), **right limit** $\lim_{x \rightarrow a^+} f(x)$ ($x \downarrow a$): if they differ, the limit does not exist; if they agree, that common value is the limit. $\lim_{x \rightarrow a} f(x) = +\infty$ means f takes arbitrarily large values near a (strictly the limit then does not exist, ∞ is not a number, but we distinguish this from the case of different one-sided limits).

Rules (Stelling 2.3): if $f(x) \rightarrow \varphi$ and $g(x) \rightarrow \gamma$ as $x \rightarrow a$, then $f \pm g \rightarrow \varphi \pm \gamma$, $fg \rightarrow \varphi\gamma$, $f/g \rightarrow \varphi/\gamma$ if $\gamma \neq 0$, and $\lim_{x \rightarrow a} f(g(x)) = \lim_{x \rightarrow \gamma} f(x)$ (substitution).

Stelling 2.4 · squeeze theorem (insluitstelling)

If $g(x) \leq f(x) \leq h(x)$ on an open interval around a (except possibly at a) and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = \lambda$, then $\lim_{x \rightarrow a} f(x) = \lambda$. The creative step is choosing g and h . Voorbeeld 2.5: for $x \sin \frac{1}{x}$ use $-1 \leq \sin \frac{1}{x} \leq 1$, but since x can be negative, squeeze between $g(x) = -|x|$ and $h(x) = |x|$ (Oefening 204). Standard use: “bounded times something tending to 0” tends to 0.

Exercise 201

from the dictaat

Determine the limits, if they exist: (a) $\lim_{k \rightarrow 1} \frac{k^5 - k^2}{k - 1}$; (b) $\lim_{p \rightarrow \infty} e^p$; (c) $\lim_{q \rightarrow 0^-} e^{1/q}$; (d) $\lim_{t \rightarrow \infty} \frac{\sin t}{t}$; (e) $\lim_{z \rightarrow 0^+} (x - \frac{1}{z})$; (f) $\lim_{x \rightarrow \infty} \log \frac{1}{1 + x^2}$.

(a) Numerator and denominator both tend to 0; factor: $k^5 - k^2 = k^2(k^3 - 1) = k^2(k - 1)(k^2 + k + 1)$, so for $k \neq 1$ the quotient equals $k^2(k^2 + k + 1)$, a polynomial, with limit $1 \cdot 3 = 3$ (Oefening 203b).

(b) e^p grows without bound: $+\infty$ (no real limit).

(c) As $q \rightarrow 0^-$, $1/q \rightarrow -\infty$ and $e^{1/q} \rightarrow 0$. (From the right, $1/q \rightarrow +\infty$ and $e^{1/q} \rightarrow +\infty$: the two-sided limit does not exist.)

(d) $-\frac{1}{t} \leq \frac{\sin t}{t} \leq \frac{1}{t}$ for $t > 0$, and $\pm \frac{1}{t} \rightarrow 0$: by the squeeze theorem the limit is 0.

- (e) $\frac{1}{z} \rightarrow +\infty$ as $z \downarrow 0$, so $x - \frac{1}{z} \rightarrow -\infty$ for any fixed x : no real limit.
 (f) $\frac{1}{1+x^2} \rightarrow 0^+$ and $\log y \rightarrow -\infty$ as $y \downarrow 0$, so the limit is $-\infty$. □

Exercise 205

from the dictaat

Determine: (a) $\lim_{x \rightarrow 0} \sin(2x) \cos(x^{-2})$; (b) $\lim_{x \rightarrow \infty} \sin(5x)e^{-5x}$; (c) $\lim_{x \rightarrow \infty} \frac{4x - 3 \cos(2\sqrt{x} + 1)}{5 - 6x}$.

(a) $|\sin(2x) \cos(x^{-2})| \leq |\sin 2x|$ because $|\cos| \leq 1$, i.e. $-|\sin 2x| \leq f(x) \leq |\sin 2x|$, and $|\sin 2x| \rightarrow 0$. Squeeze: the limit is 0. (The factor $\cos(x^{-2})$ oscillates wildly, but it is bounded.)

(b) $-e^{-5x} \leq \sin(5x)e^{-5x} \leq e^{-5x}$ and $e^{-5x} \rightarrow 0$: limit 0.

(c) Divide numerator and denominator by x : $\frac{4 - 3 \cos(2\sqrt{x} + 1)/x}{5/x - 6}$. Here $|3 \cos(\cdot)/x| \leq 3/x \rightarrow 0$ (squeeze) and $5/x \rightarrow 0$, so by the quotient rule the limit is $\frac{4}{-6} = -\frac{2}{3}$. □

NOW YOU 202, 203, 204 · solutions in Appendix S

202 (a) $\lim_{x \rightarrow 0} \frac{e^x}{1+x}$ and $\lim_{x \rightarrow 1} \frac{e^{x-1}}{x}$. (b) Why is $\lim_{x \rightarrow a-b} f(x) = \lim_{x \rightarrow a} f(x-b)$, and how does that apply to (a)? **203** From the rules: $\lim k f(x) = k\varphi$; $\lim_{x \rightarrow a} P(x) = P(a)$ for polynomials; $\lim f(x)^k = \varphi^k$ for $k \in \mathbb{Q}$, $k > 0$. **204** Finish Voorbeeld 2.5: $\lim_{x \rightarrow 0} |x| = 0$, $\lim_{x \rightarrow 0} (-|x|) = 0$, squeeze $x \sin \frac{1}{x}$, sketch.

B CONTINUITY
Definition 2.6 and Voorbeeld 2.7

f is **continuous at a** if a is in the domain and $\lim_{x \rightarrow a} f(x) = f(a)$; continuous if this holds at every point of the domain (no jumps). Watch fractions: $f(x) = \frac{x^2-9}{x-3}$ is not defined at 3, but for $x \neq 3$ it equals $x+3$, so $\lim_{x \rightarrow 3} f(x) = 6$ (cancel the factor $x-3$ only for $x \neq 3$). Continuity plays a small role in this course; differentiability is what matters.

C DIFFERENTIATION
Definition 2.8 and Oefening 206 · the derivative as a limit

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

provided the limit exists as a real number: the slope of the chord through $(a, f(a))$ and $(x, f(x))$ “just before” $x - a$ becomes 0 (figure 2.3: the small right triangle with legs $x - a$ and $f(x) - f(a)$). Remember the formula together with the picture. Notations: $f' = \frac{df}{dx} = \frac{d}{dx}f$, and $\dot{x}$ for $\frac{dx}{dt}$ in physics.

Stelling 2.9 · rules

$\frac{d}{dx}x^r = rx^{r-1}$, $\sin' = \cos$, $\cos' = -\sin$, $\log' x = \frac{1}{x}$, $(e^x)' = e^x$; sum rule; product rule $(fg)' = f'g + fg'$; quotient rule $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$; **chain rule** $f(g(x))' = f'(g(x))g'(x)$, i.e. $(f \circ g)' = (f' \circ g) \cdot g'$ (Oefening 208). The quotient rule follows from the product and chain rules applied to $f \cdot g^{-1}$ (Oefening 207).

Exercise 209

from the dictaat

Differentiate with the rules: (a) $f(x) = e^{x^3}(1-x^2)$; (b) $g(x) = \sin(x^2 + \sqrt{3x})$; (c) $h(x) = \frac{\log(4x+1)}{4x}$; (d) $j(x) = 2^x$ (rewrite as an e -power); (e) $k(x) = \log(\log(\pi x))$; (f) $m(x) = \sqrt[3]{x^3+1} e^{\sin x}$.

(a) Product and chain rule: $f'(x) = 3x^2 e^{x^3}(1-x^2) + e^{x^3}(-2x) = e^{x^3}(3x^2 - 3x^4 - 2x)$. (If the exponent is meant to be $x^3(1-x^2) = x^3 - x^5$, then $f'(x) = (3x^2 - 5x^4)e^{x^3-x^5}$.)

(b) Chain rule, inner derivative $2x + \frac{3}{2\sqrt{3x}}$: $g'(x) = \cos(x^2 + \sqrt{3x}) \left(2x + \frac{3}{2\sqrt{3x}}\right)$. (If the inner function is $x^2 + \sqrt{3}x$, the inner derivative is $2x + \sqrt{3}$.)

(c) Quotient rule with $(\log(4x+1))' = \frac{4}{4x+1}$: $h'(x) = \frac{\frac{4}{4x+1} \cdot 4x - 4\log(4x+1)}{16x^2} = \frac{\frac{4x}{4x+1} - \log(4x+1)}{4x^2}$.

(d) $2^x = e^{x \log 2}$, so $j'(x) = \log 2 \cdot e^{x \log 2} = 2^x \log 2$.

(e) Chain rule twice: $k'(x) = \frac{1}{\log(\pi x)} \cdot \frac{1}{\pi x} \cdot \pi = \frac{1}{x \log(\pi x)}$.

(f) $((x^3+1)^{1/3})' = \frac{1}{3}(x^3+1)^{-2/3} \cdot 3x^2 = x^2(x^3+1)^{-2/3}$ and $(e^{\sin x})' = \cos x e^{\sin x}$, so $m'(x) = e^{\sin x} \left(\frac{x^2}{(x^3+1)^{2/3}} + \cos x \sqrt[3]{x^3+1} \right)$. $\square$

Exercise 210

from the dictaat

(a) Show $1 + \tan^2 x = \frac{1}{\cos^2 x}$. (b) Differentiate $\tan x$. (c) Conclude that $\tan'$ has two very different-looking forms.

(a) $1 + \tan^2 x = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$. (b) Quotient rule: $\tan' x = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x}$.

(c) $\tan' x = \frac{1}{\cos^2 x} = 1 + \tan^2 x$; the second form is what makes Oefening 317 work. $\square$

§2.3.1 · recognise a limit as a derivative

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\sin x - \sin 0}{x - 0} = \sin'(0) = \cos 0 = 1$: a **standard limit**. Whenever a limit has the shape $\frac{f(x)-f(a)}{x-a}$ with $x \rightarrow a$, it equals $f'(a)$.

Exercise 211

from the dictaat

Compute in the same way: (a) $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x}$; (b) $\lim_{x \rightarrow 1} \frac{e^x - e}{x - 1}$.

(a) $\cos 0 = 1$, so this is $\lim_{x \rightarrow 0} \frac{\cos x - \cos 0}{x - 0} = \cos'(0) = -\sin 0 = 0$. (b) $e = e^1$, so this is $\frac{d}{dx} e^x \Big|_{x=1} = e^1 = e$. $\square$

Exercise 212

from the dictaat

Which is true: (A) every continuous function is differentiable; (B) every differentiable function is continuous? Why?

(B) is true. If $f'(a)$ exists then $f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a) \rightarrow f'(a) \cdot 0 = 0$ as $x \rightarrow a$ (product rule for limits), i.e. $\lim_{x \rightarrow a} f(x) = f(a)$.

(A) is false. $f(x) = |x|$ is continuous at 0, but $\frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x}$ is -1 for $x < 0$ and $+1$ for $x > 0$: the one-sided limits differ, so $f'(0)$ does not exist (the graph has a corner). $\square$

NOW YOU 206, 207, 208 · solutions in Appendix S

206 Why can Definition 2.8 be written with $x \rightarrow a$ instead of $h \rightarrow 0$? **207** Derive the quotient rule from $f \cdot g^{-1}$. **208** Complete $(\dots)' = \dots g'$ in composition notation.

D INTEGRATION

Definitions 2.10 and Stelling 2.11 · primitives, the fundamental theorem

g is a **primitive** (antiderivative) of f if $g' = f$. Primitives are unique up to a constant: the **indefinite integral** $\int f(x) dx = g(x) + c$ is the whole set of primitives (c : integration constant; easily forgotten, and the dictaat lets it pop up unexpectedly). The **definite integral** is a number: $\int_a^b f(x) dx = g(b) - g(a)$ (the constant cancels, Oefening 213). **Picture (figure 2.4)**: $\int_a^b f(x) dx$ is the sum ($\int$ is a long s) of strips of height $f(x)$ and infinitely small width dx between a and b ; heights can be negative. **Hoofdstelling 2.11**: for continuous f on $[a, b]$, $F(x) = \int_a^x f(t) dt$ is a primitive of f , and $\int_a^b f(x) dx = F(b) - F(a)$ for any primitive F . The integration variable is a dummy: $\int f(x) dx = \int f(t) dt$; never reuse it outside the integral ($x^2 + \int_0^x e^t dt$ is right, $x^2 + \int_0^x e^x dx$ is wrong).

NOW YOU 213, 214 (Extra) · solutions in Appendix S

213 If $\tilde{g} = g + c$ then $\tilde{g}(b) - \tilde{g}(a) = g(b) - g(a)$: no integration constant in definite integrals. **214 (Extra)** With $g(x) = \int_0^x f(t) dt$: write $g(x + dx)$ and $dg = g(x + dx) - g(x)$ as integrals and argue $\frac{dg}{dx} = f(x)$.

E INTEGRATION WITH COMMON SENSE

Stelling 2.12 · rules for definite integrals

(a) for $f \geq 0$ continuous, $\int_a^b f$ is the area under the graph; (b) $\int_a^a f = 0$; (c) $\int_a^b f = -\int_b^a f$; (d) $\int_a^b f = \int_a^c f + \int_c^b f$ (cut up); (e) $\int_a^b f(x) dx = \int_{a+c}^{b+c} f(x-c) dx$ (translate); (f) linearity; (g) f odd $\Rightarrow \int_{-a}^a f = 0$; (h) f even $\Rightarrow \int_{-a}^a f = 2 \int_0^a f$; (i) changing f at a single point does not change the integral; (j) $f \leq g$ on $(a, b) \Rightarrow \int_a^b f \leq \int_a^b g$; (k) $|\int_a^b f| \leq \int_a^b |f|$. Do not memorise; recognise when they help. Voorbeeld 2.13: $\int_{-1}^1 (x-4) dx = 0 - 4 \cdot 2 = -8$ (odd part vanishes, rectangle 2×1 four times); or translate to $\int_{-5}^{-3} x dx = -8$ (signed area of a trapezium); $\int_{-2}^2 \sqrt{4-x^2} dx = 2\pi$ (half disc of radius 2).

Exercise 215

from the dictaat

Compare, with a sketch: (a) the area enclosed by $y = x - 1$, $x = 0$, $y = 0$; (b) the integral $\int_0^1 (x - 1) dx$.

(a) The region is the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, -1)$: area $\frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$.

(b) $\int_0^1 (x - 1) dx = [\frac{1}{2}x^2 - x]_0^1 = -\frac{1}{2}$. The integral is minus the area, because the strips have negative height: the triangle lies below the x -axis. Area is non-negative; a definite integral is signed. $\square$

Exercise 218

from the dictaat

Compute with the basic rules but without primitives: (a) $\int_0^{2\pi} (\pi x + e^\pi \sin x) dx$; (b) $\int_{-2}^0 (x+1) \cos(x+1) dx$; (c) $\int_0^6 \sqrt{9 - (x-3)^2} dx$; (d) $\int_{-1}^3 (2(x-1)^3 - 4 \sin(x-1) + \operatorname{sgn} x + 5|x|) dx$; (e) $\int_0^{\pi/2} \sin^2 t dt$.

(a) Linearity: $\pi \int_0^{2\pi} x dx + e^\pi \int_0^{2\pi} \sin x dx$. The first integral is the area of the triangle with base 2π and height 2π : $2\pi^2$. The second is 0: translating by π , $\int_0^{2\pi} \sin x dx = \int_{-\pi}^\pi \sin(x+\pi) dx = -\int_{-\pi}^\pi \sin x dx = 0$ by oddness (or: the positive hump on $[0, \pi]$ cancels the negative one on $[\pi, 2\pi]$). Total: $2\pi^3$.

(b) Translate by 1: $\int_{-2}^0 (x+1) \cos(x+1) dx = \int_{-1}^1 x \cos x dx = 0$, since $x \cos x$ is odd (odd times even).

(c) $y = \sqrt{9 - (x-3)^2}$ is the upper half of the circle with centre $(3, 0)$ and radius 3: the integral is the area of a half disc, $\frac{1}{2} \pi \cdot 3^2 = \frac{9}{2} \pi$.

(d) Split by linearity. Translating by 1: $\int_{-1}^3 (2(x-1)^3 - 4 \sin(x-1)) dx = \int_{-2}^2 (2x^3 - 4 \sin x) dx = 0$ (odd integrand). $\int_{-1}^3 \operatorname{sgn} x dx = \int_{-1}^0 (-1) dx + \int_0^3 1 dx = -1 + 3 = 2$ (the value at 0 is irrelevant, rule (i)). $\int_{-1}^3 5|x| dx = 5(\frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 3 \cdot 3) = 25$ (two triangles). Total: 27.

(e) The substitution $t \mapsto \frac{\pi}{2} - t$ reflects $[0, \frac{\pi}{2}]$ onto itself and turns $\sin$ into $\cos$, so $\int_0^{\pi/2} \sin^2 t dt = \int_0^{\pi/2} \cos^2 t dt$. Their sum is $\int_0^{\pi/2} (\sin^2 t + \cos^2 t) dt = \int_0^{\pi/2} 1 dt = \frac{\pi}{2}$, so each equals $\frac{\pi}{4}$. $\square$

Exercise 219

from the dictaat

Show with Stelling 2.12, without a primitive, that $\int_a^b x dx = \frac{1}{2}(b^2 - a^2)$.

First $\int_0^c x dx = \frac{1}{2}c^2$ for every c : for $c \geq 0$ it is the area of the triangle with legs c and c ; for $c < 0$, by the reversal rule $\int_0^c x dx = -\int_c^0 x dx$, and $\int_c^0 x dx$ is minus the area $\frac{1}{2}c^2$ of the triangle below the axis, so again $\frac{1}{2}c^2$. Now cut up: $\int_a^b x dx = \int_a^0 x dx + \int_0^b x dx = -\int_0^a x dx + \int_0^b x dx = \frac{1}{2}b^2 - \frac{1}{2}a^2$. $\square$

NOW YOU 216, 217, 220 · solutions in Appendix S

216 Sketch the rules of Stelling 2.12 that surprise you. **217** Match the graphs of f with those of $\int_a^x f(t) dt$.

220 The Gini index $G = 2 \int_0^1 (x - v(x)) dx$: sketch v for a socialist and a capitalist society, interpret G graphically, find its extreme values.

Watch out

- A limit at a ignores $f(a)$; one-sided limits that differ mean “does not exist”; $\pm\infty$ is a description of behaviour, not a number.
- The chain rule: differentiate the outer function at the inner one, then multiply by the inner derivative. “Kettingregel vergeten?” is the dictaat’s first question when you are stuck.
- A definite integral is a signed quantity; area is not. Odd/even symmetry and translation remove most of the work in §2.5 problems.
- Never write the integration variable outside its integral.

Chapter 3

Approximating functions

Contents

- A Linearisation (§3.1)
- B Second-order approximation (§3.2)
- C Taylor approximation of order n (§3.3)
- D l'Hôpital, standard limits (§3.4)
- E (Extra) Infinite sums (§3.5)

Planning

Exercises 301–330 (all worked or solved).

Hoofdpunten: notation $f^{(n)}$ and $n!$; the Taylor formula (with steunpunt and order); using Taylor approximations for values and for limits; l'Hôpital; the standard limits $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow \infty} \frac{\log x}{x} = 0$, $\lim_{t \rightarrow \infty} \frac{t^x}{e^t} = 0$.

A LINEARISATION

§3.1 · the tangent line as approximation

The **linearisation** of f at the **steunpunt** (base point) a is the tangent line, written in powers of $x - a$:

$$P_1(x) = f(a) + f'(a)(x - a), \quad P_1(a) = f(a), \quad P_1'(a) = f'(a).$$

Voorbeeld 3.1: $\sqrt{5}$ with $f(x) = \sqrt{x}$, $a = 4$: $f(4) = 2$, $f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$, so $\sqrt{5} \approx 2 + \frac{1}{4}(5 - 4) = 2\frac{1}{4}$; self-check: $(2\frac{1}{4})^2 = 5\frac{1}{16}$, close to 5.

Exercise 301

from the dictaat

Linear approximations of: (a) $\sqrt{e}$ with $f(x) = e^x$, $a = 0$; (b) $\log 3$ with $f(x) = \log(1 + x)$, $a = 0$; (c) $\sqrt[3]{4}$ with $f(x) = \sqrt[3]{x}$, $a = 1$.

(a) $f(0) = 1$, $f'(0) = 1$: $P_1(x) = 1 + x$, and $\sqrt{e} = e^{1/2} \approx 1 + \frac{1}{2} = 1.5$.

(b) $f(0) = 0$, $f'(x) = \frac{1}{1+x}$, $f'(0) = 1$: $P_1(x) = x$, and $\log 3 = f(2) \approx 2$.

(c) $f(1) = 1$, $f'(x) = \frac{1}{3}x^{-2/3}$, $f'(1) = \frac{1}{3}$: $P_1(x) = 1 + \frac{1}{3}(x - 1)$, and $\sqrt[3]{4} \approx 1 + \frac{1}{3} \cdot 3 = 2$.

(True values: 1.6487, 1.0986, 1.5874: (b) and (c) are far from the steunpunt, so the approximations are poor.) □

Exercise 302

from the dictaat

The approximation of $\sqrt{5}$ was slightly too large. Predict, without a calculator, whether the answers in 301 are too large or too small. Hint: the second derivative.

The tangent line lies below the graph where $f'' > 0$ (convex) and above it where $f'' < 0$ (concave), so P_1 is too small for convex f and too large for concave f (near a ; the second-order term $\frac{1}{2}f''(a)(x - a)^2$ has the sign of f'').

(a) $(e^x)'' = e^x > 0$: 1.5 is too small. (b) $\log(1 + x)'' = -\frac{1}{(1+x)^2} < 0$: 2 is too large. (c) $(x^{1/3})'' = -\frac{2}{9}x^{-5/3} < 0$ for $x > 0$: 2 is too large. ($\sqrt{x}$ is concave, which explains Voorbeeld 3.1.) □

B SECOND-ORDER APPROXIMATION

§3.2 · matching the curvature

$P_2(x) = f(a) + f'(a)(x-a) + c_2(x-a)^2$ with $c_2 = \frac{1}{2}f''(a)$ satisfies $P_2(a) = f(a)$, $P'_2(a) = f'(a)$, $P''_2(a) = f''(a)$ (Oefening 303). More work does not always give a better approximation, especially far from the steunpunt; the cure is even more work (next section).

Exercise 303

from the dictaat

(a) Show that $P_2(x) = f(a) + f'(a)(x-a) + c_2(x-a)^2$ satisfies $P_2(a) = f(a)$, $P'_2(a) = f'(a)$ for every c_2 . (b) Show that $c_2 = \frac{1}{2}f''(a)$ gives $P''_2(a) = f''(a)$. (c) Redo Voorbeeld 3.1 with a second-order approximation; check that it is sometimes better and sometimes not.

(a) $P_2(a) = f(a) + 0 + 0$. $P'_2(x) = f'(a) + 2c_2(x-a)$, so $P'_2(a) = f'(a)$.

(b) $P''_2(x) = 2c_2$, so $P''_2(a) = f''(a)$ iff $c_2 = \frac{1}{2}f''(a)$.

(c) $f(x) = \sqrt{x}$, $a = 4$: $f''(x) = -\frac{1}{4}x^{-3/2}$, $f''(4) = -\frac{1}{32}$, $c_2 = -\frac{1}{64}$: $P_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$. At $x = 5$: $P_2(5) = 2.25 - 0.015625 = 2.234375$ versus $\sqrt{5} = 2.23607$: error 0.0017 instead of 0.014 for P_1 . Better. At $x = 20$: $P_1(20) = 6$, $P_2(20) = 6 - \frac{256}{64} = 2$, while $\sqrt{20} = 4.47$: P_2 is now worse than P_1 (far from the steunpunt the quadratic term dominates in the wrong direction). $\square$

NOW YOU 304 · solutions in Appendix S

304 Look at 302 again, now that you know P_2 : the correction term $\frac{1}{2}f''(a)(x-a)^2$ has the sign of $f''(a)$.

C TAYLOR APPROXIMATION OF ORDER n

Definitions 3.2–3.4 · $f^{(k)}$, $k!$, the Taylor polynomial

$f^{(k)}$ is the k -th derivative, $f^{(0)} = f$; $k! = k(k-1) \cdots 2 \cdot 1$, $0! = 1$. The **Taylor approximation of order n with steunpunt a** is

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k,$$

the unique polynomial of degree $\leq n$ with $P_n^{(k)}(a) = f^{(k)}(a)$ for $k = 0, \dots, n$ (Oefening 306). Raising the order adds terms without redoing earlier work. **The standard ones at $a = 0$** (Oefening 307): $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots$; $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$; $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots$; $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$; also $\frac{1}{1-x} = 1 + x + x^2 + \dots$ and $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$ (Oefening 308).

Voorbeelden 3.5, 3.6 and Oefening 316 · three shortcuts

Substitution: if $u \rightarrow 0$ when $x \rightarrow 0$, substitute u into a known expansion: $\log(1+x^2) = x^2 - \frac{1}{2}x^4 + \dots$; $\log(\cos x) = \log(1+u)$ with $u = \cos x - 1 = -\frac{1}{2}x^2 + \frac{1}{24}x^4 + \text{h.o.t.}$, so $\log \cos x = u - \frac{1}{2}u^2 + \dots = -\frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{2} \cdot \frac{1}{4}x^4 + \dots = -\frac{1}{2}x^2 - \frac{1}{12}x^4 + \text{h.o.t.}$ Keep track of which terms are higher order than you need. **Algebra:** $\log \frac{1+x}{1-x} = \log(1+x) - \log(1-x)$ (Oefening 314). **A differential equation:** if $f' = (2x-2)f$ (Oefening 316) or $\tan' = 1 + \tan^2$ (317), differentiate the relation repeatedly and substitute $x = 0$ instead of computing messy quotients.

Exercise 308

from the dictaat

Fourth-order Taylor approximations at 0 of: (a) $\frac{1}{x-1}$; (b) $\sqrt{1+x}$; (c) $\sqrt{1+x^2}$.

(a) $f(x) = (x-1)^{-1}$: $f' = -(x-1)^{-2}$, $f'' = 2(x-1)^{-3}$, $f''' = -6(x-1)^{-4}$, $f'''' = 24(x-1)^{-5}$; at 0: $-1, -1, -2, -6, -24$, so $c_k = f^{(k)}(0)/k! = -1$: $P_4(x) = -1 - x - x^2 - x^3 - x^4$. (Check: $\frac{1}{x-1} = -\frac{1}{1-x} = -(1+x+x^2+\dots)$.)

(b) $f = (1+x)^{1/2}$: $f' = \frac{1}{2}(1+x)^{-1/2}$, $f'' = -\frac{1}{4}(1+x)^{-3/2}$, $f''' = \frac{3}{8}(1+x)^{-5/2}$, $f'''' = -\frac{15}{16}(1+x)^{-7/2}$; at 0: $1, \frac{1}{2}, -\frac{1}{4}, \frac{3}{8}, -\frac{15}{16}$. Divide by $k!$: $P_4(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4$.

(c) Substitute $u = x^2$ in (b): $\sqrt{1+x^2} = 1 + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \text{h.o.t.}$, so $P_4(x) = 1 + \frac{1}{2}x^2 - \frac{1}{8}x^4$ (the x^3 term of (b) would give x^6). $\square$

Exercise 309

from the dictaat

Compute $\sin 37^\circ$ to three decimals with a Taylor approximation, and check.

Radians first: $37^\circ = \frac{37\pi}{180} \approx 0.64577$. With $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$: $x^3 \approx 0.26930$, $x^5 \approx 0.11230$, so $\sin 37^\circ \approx 0.64577 - 0.04488 + 0.00094 = 0.60182$. Three decimals: 0.602. Check: $\sin 37^\circ \approx 0.6018$ (and $\sin 30^\circ = 0.5 < 0.602 < 0.707 = \sin 45^\circ$, plausible). Using 37 instead of 0.6458 in the polynomial gives nonsense: degrees must be converted. $\square$

Exercise 314

from the dictaat

Find a fifth-degree Taylor approximation of $\log \frac{1+x}{1-x}$ at 0, the smart way.

$\log \frac{1+x}{1-x} = \log(1+x) - \log(1-x)$. From Oefening 307/313: $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$ and $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5}$ (substitute $-x$). Subtracting, the even powers cancel and the odd ones double:

$$\log \frac{1+x}{1-x} \approx 2x + \frac{2}{3}x^3 + \frac{2}{5}x^5.$$

(The function is odd, so only odd powers can occur; Oefening 311.) $\square$

Exercise 316

from the dictaat

Taylor approximation of $f(x) = e^{x^2-2x}$ at 0, reusing results: (a) $f(0)$; (b) show $f'(x) = (2x-2)f(x)$, find $f'(0)$; (c) differentiate again, find $f''(0)$; (d) as many $f^{(n)}(0)$ as you can.

(a) $f(0) = e^0 = 1$. (b) Chain rule: $f'(x) = (2x-2)e^{x^2-2x} = (2x-2)f(x)$, so $f'(0) = -2 \cdot 1 = -2$.

(c) Product rule on $f' = (2x-2)f$: $f'' = 2f + (2x-2)f'$, so $f''(0) = 2 \cdot 1 + (-2)(-2) = 6$.

(d) Keep differentiating the relation: $f''' = 2f' + 2f' + (2x-2)f'' = 4f' + (2x-2)f''$, so $f'''(0) = 4(-2) - 2 \cdot 6 = -20$; $f'''' = 4f'' + 2f'' + (2x-2)f''' = 6f'' + (2x-2)f'''$, so $f''''(0) = 36 + 40 = 76$; in general $f^{(n+1)} = 2nf^{(n-1)} + (2x-2)f^{(n)}$, giving $f^{(n+1)}(0) = 2nf^{(n-1)}(0) - 2f^{(n)}(0)$: $f^{(5)}(0) = 8 \cdot 6 - 2 \cdot 76 = -104$, $f^{(6)}(0) = 10(-20) - 2(-104) = 8$, ... Hence $P_4(x) = 1 - 2x + 3x^2 - \frac{10}{3}x^3 + \frac{19}{6}x^4$. $\square$

Exercise 317

from the dictaat

Redo Oefening 312 (P_5 of $\tan x$ at 0) efficiently, using $\tan' x = 1 + \tan^2 x$.

Write $T = \tan x$, so $T' = 1 + T^2$ and $T(0) = 0$. Differentiate the relation repeatedly, always substituting $T' = 1 + T^2$: $T'' = 2TT' = 2T + 2T^3$; $T''' = (2 + 6T^2)T' = (2 + 6T^2)(1 + T^2) = 2 + 8T^2 + 6T^4$; $T'''' =$

$$(16T + 24T^3)(1 + T^2) = 16T + 40T^3 + 24T^5; T^{(5)} = (16 + 120T^2 + 120T^4)(1 + T^2).$$

At $x = 0$ ($T = 0$): $T = 0$, $T' = 1$, $T'' = 0$, $T''' = 2$, $T'''' = 0$, $T^{(5)} = 16$. So

$$\tan x \approx x + \frac{2}{3!}x^3 + \frac{16}{5!}x^5 = x + \frac{1}{3}x^3 + \frac{2}{15}x^5.$$

(Odd function, odd powers only. Doing this with the quotient rule on $\sin / \cos$ is the long way of Oefening 312.) □

Exercise 319

from the dictaat

With the substitution technique, order-4 approximations at 0 of: (a) $\log(2+x)$ (hint: $2+x = 2(1+\frac{x}{2})$); (b) $\log((1+x)^2)$, in a very short and in a longer way.

(a) $\log(2+x) = \log 2 + \log(1+\frac{x}{2})$; substitute $u = \frac{x}{2}$ in $\log(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4}$: $\log(2+x) \approx \log 2 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{24} - \frac{x^4}{64}$.

(b) Short: $\log(1+x)^2 = 2\log(1+x) \approx 2x - x^2 + \frac{2}{3}x^3 - \frac{1}{2}x^4$. Longer: $(1+x)^2 = 1+u$ with $u = 2x+x^2$; then $u^2 = 4x^2+4x^3+x^4$, $u^3 = 8x^3+12x^4+\dots$, $u^4 = 16x^4+\dots$, and $u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4} = 2x+x^2-2x^2-2x^3-\frac{1}{2}x^4+\frac{8}{3}x^3+4x^4-4x^4+\text{h.o.t.} = 2x-x^2+\frac{2}{3}x^3-\frac{1}{2}x^4$. Same answer. ✓ □

Exercise 320

from the dictaat

Order-4 approximations at 0 of (a) $e^{\sin x}$ and (b) $e^{\cos 2x}$; (c) check.

(a) $u = \sin x = x - \frac{1}{6}x^3 + \text{h.o.t.}$ (the x^5 term is beyond order 4). $u^2 = x^2 - \frac{1}{3}x^4 + \dots$, $u^3 = x^3 + \dots$, $u^4 = x^4 + \dots$. Then $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \frac{u^4}{24} = 1 + x - \frac{1}{6}x^3 + \frac{1}{2}x^2 - \frac{1}{6}x^4 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \dots = 1 + x + \frac{1}{2}x^2 - \frac{1}{8}x^4$. (The x^3 terms cancel.)

(b) $\cos 2x = 1 - 2x^2 + \frac{2}{3}x^4 + \dots$ (substitute $2x$ in the cosine series: $\frac{(2x)^2}{2} = 2x^2$, $\frac{(2x)^4}{24} = \frac{2}{3}x^4$). Since $\cos 2x$ does not tend to 0, first split off the constant: $e^{\cos 2x} = e \cdot e^u$ with $u = -2x^2 + \frac{2}{3}x^4 + \dots$; $u^2 = 4x^4 + \dots$; so $e^u = 1 + u + \frac{u^2}{2} + \dots = 1 - 2x^2 + \frac{2}{3}x^4 + 2x^4 = 1 - 2x^2 + \frac{8}{3}x^4$, and $e^{\cos 2x} \approx e(1 - 2x^2 + \frac{8}{3}x^4)$.

(c) Direct differentiation (or Wolfram Alpha) confirms both; the direct route for (a) needs $f' = \cos x e^{\sin x}$, $f'' = (\cos^2 x - \sin x)e^{\sin x}$, ..., much messier. □

NOW YOU 305, 306, 307, 310, 311, 312, 313, 315, 318, 321, 322 (Extra) · solutions in Appendix S

305 P_3 matches f, f', f'', f''' at a . **306** $P_n^{(k)}(a) = f^{(k)}(a)$. **307** Complete the degree-9 expansions of e^x , $\sin x$, $\cos x$, $\log(1+x)$. **310** All Taylor approximations of $2x^3 - 3x^2 + 4$ at 0; is a polynomial equal to its own $P_{n,a}$? **311** Taylor approximations at 0 of even/odd functions. **312** P_5 of $\tan x$ by organised differentiation. **313** k -th derivatives of $\log(1 \pm x)$ at 0; P_5 of $\log(1-x)$; substitution $-x$. **315** From $\sqrt{1-x^2}$: $\frac{1}{2}\sqrt{3} \approx \frac{111}{128}$. **318** No order- ≤ 4 terms were dropped in Voorbeeld 3.6. **321** P_5 of $e^{x-\frac{1}{6}x^3}$. **322 (Extra)** Approximating $\log \frac{1}{2}$, $\log 0$, $\log 1.9$, $\log 2.1$, $\log 2$ by Taylor polynomials.

D
L'HÔPITAL'S RULE AND SOME IMPORTANT LIMITS

Voorbeeld 3.7, Stelling 3.8 · Taylor for limits; l'Hôpital

Taylor: near 0, $\frac{\sin x}{x} = \frac{x + c_3 x^3 + \dots}{x} = 1 + c_3 x^2 + \dots \rightarrow 1$; the exact coefficients do not matter, only that the remaining terms tend to 0. **l'Hôpital:** if $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow a$ (or both $\rightarrow \pm\infty$; $a = \pm\infty$ allowed) and $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$. Voorbeeld 3.9: $\lim_{x \rightarrow 0} \frac{x}{\tan x} = \lim_{x \rightarrow 0} \frac{1}{1/\cos^2 x} = \lim_{x \rightarrow 0} \cos^2 x = 1$. **Check the hypotheses first:** l'Hôpital on a quotient that is not $\frac{0}{0}$ or $\frac{\infty}{\infty}$ gives wrong answers (325d).

Exercise 323

from the dictaat

Repeat the Taylor argument for $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ and $\lim_{x \rightarrow 0} \frac{\sin x}{9x^3}$.

$$\sin 3x = 3x - \frac{(3x)^3}{6} + \dots = 3x - \frac{9}{2}x^3 + \dots, \text{ so } \frac{\sin 3x}{2x} = \frac{3}{2} - \frac{9}{4}x^2 + \dots \rightarrow \frac{3}{2}.$$

$$\frac{\sin x}{9x^3} = \frac{x - \frac{1}{6}x^3 + \dots}{9x^3} = \frac{1}{9x^2} - \frac{1}{54} + \dots; \text{ the first term tends to } +\infty, \text{ so the limit does not exist as a real number } (+\infty). \quad \square$$

Exercise 325

from the dictaat

Determine in two ways (with and without l'Hôpital): (a) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}$; (b) $\lim_{x \rightarrow 1} \frac{\log x}{1-x}$; (c) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x}$; (d) $\lim_{x \rightarrow \pi/2} \frac{1 - \cos x}{\sin x}$; (e) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$.

(a) $\frac{0}{0}$. l'H: $\frac{1/(1+x)}{1} \rightarrow 1$. Taylor: $\frac{x - \frac{1}{2}x^2 + \dots}{x} = 1 - \frac{1}{2}x + \dots \rightarrow 1$. (Also: it is $\log'(1)$.)

(b) $\frac{0}{0}$. l'H: $\frac{1/x}{-1} \rightarrow -1$. Taylor at 1: $\log x = \log(1 + (x-1)) = (x-1) - \frac{1}{2}(x-1)^2 + \dots$, so the quotient is $\frac{(x-1)(1 - \frac{1}{2}(x-1) + \dots)}{-(x-1)} \rightarrow -1$.

(c) $\frac{0}{0}$. l'H: $\frac{\sin x}{\cos x} \rightarrow 0$. Taylor: $\frac{\frac{1}{2}x^2 - \dots}{x - \dots} = \frac{\frac{1}{2}x - \dots}{1 - \dots} \rightarrow 0$.

(d) Not indeterminate: numerator $\rightarrow 1 - 0 = 1$, denominator $\rightarrow 1$; the limit is 1 by the quotient rule. l'Hôpital would wrongly give $\frac{\sin x}{\cos x} \rightarrow \infty$.

(e) $\frac{1}{x} - \frac{1}{\sin x} = \frac{\sin x - x}{x \sin x}$, a $\frac{0}{0}$ form. Taylor: $\frac{-\frac{1}{6}x^3 + \dots}{x^2 - \dots} = \frac{-\frac{1}{6}x + \dots}{1 - \dots} \rightarrow 0$. l'H twice: $\frac{\cos x - 1}{\sin x + x \cos x}$ (still $\frac{0}{0}$) $\rightarrow \frac{-\sin x}{2 \cos x - x \sin x} \rightarrow \frac{0}{2} = 0$. $\square$

Exercise 326

from the dictaat

Prove in order: (a) $\lim_{x \rightarrow \infty} \frac{\log x}{x} = 0$; (b) $\lim_{x \rightarrow \infty} \frac{(\log x)^n}{x} = 0$ for $n \in \mathbb{N}$; (c) $\lim_{t \rightarrow \infty} \frac{t^n}{e^t} = 0$; (d) $\lim_{t \rightarrow \infty} \frac{t^x}{e^t} = 0$ for all real x .

(a) $\frac{\infty}{\infty}$; l'Hôpital: $\frac{1/x}{1} = \frac{1}{x} \rightarrow 0$.

(b) Induction on n ; (a) is $n = 1$. If $\frac{(\log x)^{n-1}}{x} \rightarrow 0$, then l'Hôpital on $\frac{(\log x)^n}{x}$ gives $\frac{n(\log x)^{n-1} \cdot \frac{1}{x}}{1} = n \frac{(\log x)^{n-1}}{x} \rightarrow 0$.

(c) Substitute $t = \log x$, i.e. $x = e^t$; as $t \rightarrow \infty$, $x \rightarrow \infty$, and $\frac{t^n}{e^t} = \frac{(\log x)^n}{x} \rightarrow 0$ by (b).

(d) Choose $n \in \mathbb{N}$ with $n \geq x$. For $t \geq 1$: $0 \leq t^x \leq t^n$, so $0 \leq \frac{t^x}{e^t} \leq \frac{t^n}{e^t} \rightarrow 0$; squeeze.

Meaning: log grows slower than every power, every power grows slower than e^t . □

NOW YOU 324, 327, 328, 329, 330 (Extra) · solutions in Appendix S

324 $\lim_{x \rightarrow 0} \frac{x}{\tan x}$ without l'Hôpital. **327** $\lim_{x \rightarrow 0} x \log x = 0$ from 326(a). **328, 329 (Extra)** The general terms of the series for e^x , $\sin$, $\cos$, $\log(1+x)$; $e^{ix} = \cos x + i \sin x$ from the series. **330 (Extra)** $\frac{1}{1+x} = 1 - x + x^2 - \dots$: meaning for $x = 0, -\frac{1}{2}, -1, 1$; its Taylor series; differentiating the series of $\log(1+x)$.

Watch out

- Always say which steunpunt and which order. Before substituting a known expansion, check that the inner expression tends to 0 (split off e^1 in $e^{\cos 2x}$).
- Keep terms up to the required order and *all* of them: in $\log \cos x$ the u^2 term contributes at order 4 (Oefening 318).
- l'Hôpital only for $\frac{0}{0}$ or $\frac{\infty}{\infty}$; differentiate numerator and denominator separately (no quotient rule).
- Degrees to radians before using a Taylor polynomial (Oefening 309).

Chapter 4

Tapas: four short topics

Contents

- A Pascal's triangle (§4.1)
- B Logarithmic differentiation (§4.2)
- C arcsin, arccos, arctan (§4.3)
- D Hyperbolic functions (§4.4)

Planning

Exercises 401–420 (all worked or solved).

Hoofdpunten: expanding $(a + b)^n$ with Pascal's triangle; arcsin, arccos, arctan with domain, graph and derivative; logarithmic differentiation; simplifying expressions with inverse trig functions. The hyperbolic functions need not be memorised: the skill is deriving an inverse and its derivative from a given formula.

A PASCAL'S TRIANGLE

§4.1

Rows 1; 1 1; 1 2 1; 1 3 3 1; 1 4 6 4 1; ..., each number the sum of the two above it. Row n (counting from 0) gives the coefficients of $(x + y)^n$: $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$.

Exercise 401

from the dictaat

Check this for $n = 0, 1, 2, 3$ and predict $(x + y)^7$.

$(x + y)^0 = 1$; $(x + y)^1 = x + y$; $(x + y)^2 = x^2 + 2xy + y^2$; $(x + y)^3 = (x + y)(x^2 + 2xy + y^2) = x^3 + 3x^2y + 3xy^2 + y^3$: coefficients 1; 1 1; 1 2 1; 1 3 3 1. Continuing the triangle: 1 5 10 10 5 1, 1 6 15 20 15 6 1, 1 7 21 35 35 21 7 1, so

$$(x + y)^7 = x^7 + 7x^6y + 21x^5y^2 + 35x^4y^3 + 35x^3y^4 + 21x^2y^5 + 7xy^6 + y^7.$$

□

NOW YOU 402, 403 · solutions in Appendix S

402 Why is the sum of row n equal to 2^n ? **403** How does Pascal's triangle give the n -th derivative of $f(x)g(x)$? Try $n = 0, 1, 2, \dots$

B LOGARITHMIC DIFFERENTIATION

§4.2 and Voorbeeld 4.1 · take the log first

$\frac{d}{dx} \log f(x) = \frac{f'(x)}{f(x)}$, so $f'(x) = f(x) \cdot (\log f(x))'$. Useful when $\log f$ is easier to differentiate than f : products, quotients, powers and variable exponents become sums, differences and multiples. Voorbeeld

4.1: $f(x) = x^x$, $\log f = x \log x$, $(\log f)' = \log x + 1$, so $f'(x) = (\log x + 1)x^x$. (Alternative for exponents: $a^b = e^{b \log a}$.)

Exercise 404

from the dictaat

Differentiate in two ways (“normally” and logarithmically): (a) $(6x+1)^5(x^4-3)^6$; (b) $\sqrt[5]{\frac{x^2+2}{x^2+3}}$.

(a) *Logarithmic*: $\log f = 5 \log(6x+1) + 6 \log(x^4-3)$, so $\frac{f'}{f} = \frac{30}{6x+1} + \frac{24x^3}{x^4-3}$ and $f' = (6x+1)^5(x^4-3)^6 \left(\frac{30}{6x+1} + \frac{24x^3}{x^4-3} \right)$. *Normal*: product and chain rule, $f' = 30(6x+1)^4(x^4-3)^6 + 24x^3(6x+1)^5(x^4-3)^5$. Both equal $6(6x+1)^4(x^4-3)^5(5(x^4-3) + 4x^3(6x+1)) = 6(6x+1)^4(x^4-3)^5(29x^4 + 4x^3 - 15)$.

(b) *Logarithmic*: $\log g = \frac{1}{5}(\log(x^2+2) - \log(x^2+3))$, $\frac{g'}{g} = \frac{1}{5} \left(\frac{2x}{x^2+2} - \frac{2x}{x^2+3} \right) = \frac{2x}{5(x^2+2)(x^2+3)}$, so $g' = \sqrt[5]{\frac{x^2+2}{x^2+3}} \cdot \frac{2x}{5(x^2+2)(x^2+3)}$. *Normal*: $g' = \frac{1}{5} \left(\frac{x^2+2}{x^2+3} \right)^{-4/5} \cdot \frac{2x(x^2+3) - 2x(x^2+2)}{(x^2+3)^2} = \frac{1}{5} \left(\frac{x^2+2}{x^2+3} \right)^{-4/5} \cdot \frac{2x}{(x^2+3)^2}$, the same after writing $(\cdot)^{-4/5} = (\cdot)^{1/5} \cdot \frac{x^2+3}{x^2+2}$. $\square$

Exercise 405

from the dictaat

Find f' for (a) $f: x \mapsto a^{\sin(bx)}$ ($a > 0, b \neq 0$); (b) $f: x \mapsto (\log(4x+3))^{2x}$.

(a) $\log f = \sin(bx) \log a$, so $f' = f \cdot b \cos(bx) \log a = b \log a \cos(bx) a^{\sin(bx)}$.

(b) $\log f = 2x \log(\log(4x+3))$. Product and chain rule: $(\log f)' = 2 \log(\log(4x+3)) + 2x \cdot \frac{1}{\log(4x+3)} \cdot \frac{4}{4x+3}$, so

$$f'(x) = (\log(4x+3))^{2x} \left(2 \log(\log(4x+3)) + \frac{8x}{(4x+3) \log(4x+3)} \right).$$

 $\square$
Exercise 407

from the dictaat

Find $f'(0)$ if $f(x) = (1-x)(1-2x)^2(1-3x)^3(1-4x)^4(1-5x)^5$.

$\log f = \sum_{k=1}^5 k \log(1-kx)$, so $\frac{f'}{f} = \sum_{k=1}^5 \frac{-k^2}{1-kx}$. At $x=0$: $f(0) = 1$ and $f'(0) = -(1+4+9+16+25) = -55$. (Expanding the product would be madness.) $\square$

NOW YOU 406, 408 · solutions in Appendix S

406 Differentiate $x^{(e^x)}$ and $e^{(x^e)}$. **408** What do the examples have in common that makes logarithmic differentiation work so well, and how would you do them without it?

C CIRCLE FUNCTIONS: ARCSIN, ARCCOS, ARCTAN
Tabel 4.1 · inverse trigonometric functions

$\sin, \cos, \tan$ are periodic, hence not bijective; restrict the domain to make them bijective and invert:

function	restricted domain	inverse	domain of the inverse
$\sin$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$\arcsin$	$[-1, 1]$, values in $[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos$	$[0, \pi]$	$\arccos$	$[-1, 1]$, values in $[0, \pi]$
$\tan$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	$\arctan$	$\mathbb{R}$, values in $(-\frac{\pi}{2}, \frac{\pi}{2})$

Graphs: mirror the restricted graph in the line $y = x$. The American notation $\sin^{-1}$ is avoided (confusion with $1/\sin$). **Derivatives** (from $\sin(\arcsin x) = x$ by the chain rule): $\arcsin' x = \frac{1}{\sqrt{1-x^2}}$, $\arccos' x = \frac{-1}{\sqrt{1-x^2}}$, $\arctan' x = \frac{1}{1+x^2}$ (Oefening 412).

Oefeningen 410, 411 · simplifying

$\arcsin(\sin \alpha) = \alpha$ only for $\alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; otherwise first rewrite $\sin \alpha$ as the sine of an angle in that interval using the unit circle. For $\sin(\arccos x)$ and the like, draw a right triangle with hypotenuse 1 and an angle $\varphi = \arccos x$: adjacent side x , opposite side $\sqrt{1-x^2}$; for $\arctan x$ take legs 1 and x , hypotenuse $\sqrt{1+x^2}$. Then check the sign for negative x .

Exercise 410

from the dictaat

(a) Why is $\arcsin(\sin \alpha) = \alpha$ for $-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$? (b) Simplify $\arcsin(\sin(\pi + \alpha))$ for such α . (c) Largest interval I on which $\arccos(\cos \beta) = \beta$. (d) Simplify $\arccos(\cos(\zeta - \pi))$ for $0 \leq \zeta \leq \pi$. (e) Simplify $\arctan(-\tan(\frac{\pi}{2} + \eta))$ for $\frac{\pi}{2} \leq \eta \leq \pi$. (f) An angle $\lambda \in [-\pi, \pi]$ with $\tan \lambda = -\frac{1}{3}\sqrt{3}$ but $\arctan(\tan \lambda) \neq \lambda$.

(a) On $[-\frac{\pi}{2}, \frac{\pi}{2}]$ the sine is bijective and $\arcsin$ is defined as its inverse there, so $\arcsin$ undoes $\sin$: $\arcsin(\sin \alpha) = \alpha$.

(b) The point at angle $\pi + \alpha$ is the point at angle α rotated by a half turn, so its height is $\sin(\pi + \alpha) = -\sin \alpha = \sin(-\alpha)$. Since $-\alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\arcsin$ returns $-\alpha$: $\arcsin(\sin(\pi + \alpha)) = -\alpha$.

(c) $I = [0, \pi]$, the restricted domain of $\cos$ (for β outside it, $\arccos(\cos \beta)$ is the angle in $[0, \pi]$ with the same cosine, not β).

(d) $\cos(\zeta - \pi) = -\cos \zeta = \cos(\pi - \zeta)$ (reflection in the y-axis), and $\pi - \zeta \in [0, \pi]$. So $\arccos(\cos(\zeta - \pi)) = \pi - \zeta$.

(e) $-\tan(\frac{\pi}{2} + \eta) = \tan(-\frac{\pi}{2} - \eta) = \tan(\frac{\pi}{2} - \eta)$ ($\tan$ is odd and has period π). For $\frac{\pi}{2} \leq \eta < \pi$, $\frac{\pi}{2} - \eta \in (-\frac{\pi}{2}, 0]$, inside the restricted domain, so the expression equals $\frac{\pi}{2} - \eta$. (At $\eta = \pi$, $\tan \frac{3\pi}{2}$ is undefined.)

(f) $\tan \lambda = -\frac{1}{3}\sqrt{3}$ for $\lambda = -\frac{\pi}{6} + k\pi$. In $[-\pi, \pi]$: $\lambda = -\frac{\pi}{6}$ (then $\arctan(\tan \lambda) = \lambda$) or $\lambda = \frac{5\pi}{6}$, for which $\arctan(\tan \frac{5\pi}{6}) = \arctan(-\frac{1}{3}\sqrt{3}) = -\frac{\pi}{6} \neq \frac{5\pi}{6}$. So $\lambda = \frac{5\pi}{6}$. $\square$

Exercise 411

from the dictaat

(a) With $\arccos x = \varphi$ and hypotenuse 1, express the legs in x . (b) Simplify $\sin(\arccos x)$. (c) Check the simplification for obtuse φ ($x < 0$). (d) Simplify $\cos(\arcsin x)$; what changes, and does it hold for $x < 0$? (e) Simplify $\cos(\arctan x)$. (f) Simplify $\tan(\arccos x)$ and check for $x < 0$.

(a) $\cos \varphi = x$, so the adjacent leg is x and the opposite leg is $\sqrt{1-x^2}$ (Pythagoras).

(b) $\sin(\arccos x) = \sin \varphi = \sqrt{1-x^2}$.

- (c) For $\frac{\pi}{2} < \varphi < \pi$, $x = \cos \varphi < 0$; the triangle in the right figure has adjacent leg $|x| = -x$, but its height is still $\sqrt{1-x^2}$, and $\sin \varphi > 0$ for $\varphi \in [0, \pi]$. So $\sin(\arccos x) = \sqrt{1-x^2}$ for all $x \in [-1, 1]$.
- (d) Now $\varphi = \arcsin x$: opposite leg x , adjacent leg $\sqrt{1-x^2}$: $\cos(\arcsin x) = \sqrt{1-x^2}$. For $x < 0$, $\varphi \in [-\frac{\pi}{2}, 0)$ and $\cos \varphi \geq 0$, so the positive root is right.
- (e) $\varphi = \arctan x$: take the adjacent leg 1 and the opposite leg x ; hypotenuse $\sqrt{1+x^2}$; $\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}$ (positive, as $\varphi \in (-\frac{\pi}{2}, \frac{\pi}{2})$).
- (f) $\tan(\arccos x) = \frac{\sin \varphi}{\cos \varphi} = \frac{\sqrt{1-x^2}}{x}$ ($x \neq 0$). For $x < 0$, φ is obtuse and $\tan \varphi < 0$; the formula is negative too. ✓ □

Exercise 412

from the dictaat

Show in the same way that $\arccos' x = \frac{-1}{\sqrt{1-x^2}}$ and $\arctan' x = \frac{1}{1+x^2}$.

Differentiate $\cos(\arccos x) = x$: $-\sin(\arccos x) \cdot \arccos' x = 1$, so $\arccos' x = \frac{-1}{\sin(\arccos x)} = \frac{-1}{\sqrt{1-x^2}}$ by 411(b).

Differentiate $\tan(\arctan x) = x$ with $\tan' = 1 + \tan^2$: $(1 + \tan^2(\arctan x)) \arctan' x = 1$, i.e. $(1 + x^2) \arctan' x = 1$, so $\arctan' x = \frac{1}{1+x^2}$. □

NOW YOU 409, 413, 414 (Extra) · solutions in Appendix S

409 Fill in Tabel 4.2 ($\arcsin t$, $\arccos t$, $\arctan t$ for the standard values t); mark “not nice” versus “not defined”. **413** Check that the derivatives match the shapes of the graphs. **414 (Extra)** Locate $\sec t$, $\csc t$, $\cot t$ in the unit circle.

D HYPERBOLIC FUNCTIONS
§4.4

$\sinh t = \frac{e^t - e^{-t}}{2}$, $\cosh t = \frac{e^t + e^{-t}}{2}$. Properties to *derive*, not memorise: $\sinh$ is odd with only zero 0, $\cosh$ is even and ≥ 1 ; $\sinh' = \cosh$, $\cosh' = \sinh$; $\cosh^2 t - \sinh^2 t = 1$ (Oefeningen 415–418). $\sinh$ is bijective $\mathbb{R} \rightarrow \mathbb{R}$; its inverse $\operatorname{arsinh} x = \log(x + \sqrt{x^2 + 1})$ has derivative $\frac{1}{\sqrt{x^2 + 1}}$ (Oefeningen 419, 420; compare $\arcsin'$). Chapter 7 has the same for $\cosh$ and $\tanh$.

Exercise 419

from the dictaat

Let $\operatorname{arsinh} x = t$. (a) Check $x = \frac{1}{2}(e^t - e^{-t})$. (b) Substitute $e^t = u$ and reduce to an expression in x and u . (c) Solve the resulting quadratic equation for u . (d) Conclude $\operatorname{arsinh} x = \log(x + \sqrt{x^2 + 1})$ and explain why $\log(x - \sqrt{x^2 + 1})$ is not an option.

(a) $t = \operatorname{arsinh} x$ means $x = \sinh t = \frac{1}{2}(e^t - e^{-t})$. (b) With $u = e^t$ ($u > 0$) and $e^{-t} = \frac{1}{u}$: $x = \frac{1}{2}(u - \frac{1}{u})$.

(c) Multiply by $2u$: $2xu = u^2 - 1$, i.e. $u^2 - 2xu - 1 = 0$, so $u = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$.

(d) $u = e^t$ must be positive. Since $\sqrt{x^2 + 1} > |x|$, the root $x - \sqrt{x^2 + 1}$ is negative for every x and log of it

does not exist; the other root $x + \sqrt{x^2 + 1}$ is positive. Hence $t = \log(x + \sqrt{x^2 + 1})$. □

Exercise 420

from the dictaat

Differentiate the expression found in 419; the result should resemble $\arcsin'$.

Chain rule with $(\sqrt{x^2 + 1})' = \frac{x}{\sqrt{x^2 + 1}}$:

$$\operatorname{arsinh}' x = \frac{1 + \frac{x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}}.$$

Compare $\arcsin' x = \frac{1}{\sqrt{1 - x^2}}$: only the sign inside the root differs. □

NOW YOU 415, 416, 417, 418 · solutions in Appendix S

415 Even/odd and zeros of $\sinh$, $\cosh$. **416** Sketch e^x , e^{-x} , $\sinh x$, $\cosh x$ in one figure. **417** Derivatives of $\sinh$, $\cosh$; compare with $\sin$, $\cos$. **418** $\cosh t - \sinh t$, $\cosh t + \sinh t$, and hence $\cosh^2 t - \sinh^2 t = 1$.

Watch out

- $\arcsin(\sin \alpha) = \alpha$ only on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $\arccos(\cos \beta) = \beta$ only on $[0, \pi]$, $\arctan(\tan \lambda) = \lambda$ only on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Draw the circle.
- $\arcsin$ and $\arccos$ take only $t \in [-1, 1]$; $\arctan(\frac{1}{2})$ exists but has no nice value.
- Logarithmic differentiation needs $f > 0$ (or use $\log |f|$); the final answer must be multiplied back by f .

Differential equations

Contents

- A First acquaintance (§5.1)
- B Homogeneous linear equations (§5.2.1)
- C Inhomogeneous equations (§5.2.2)
- D Initial value problems (§5.2.3)

Planning

Exercises 501–524 (all worked or solved).

Hoofdpunten: what a d.v. is; order, (in)homogeneous, initial value, integration constant, homogeneous/particular/general solution; recognising linearity; solving a homogeneous d.v. by separating variables; solving an inhomogeneous first-order linear initial value problem; checking that a function solves a d.v.

A FIRST ACQUAINTANCE

§5.1, Definition 5.3 · what a differential equation is

A **differential equation** (d.v.) relates a function to one or more of its derivatives. Voorbeeld 5.1 (radioactive decay): $y' = ky$, solved by $y(t) = ce^{kt}$ for every c (a whole family; c is fixed by the initial amount). Voorbeeld 5.2 (falling with air resistance): $m \frac{dv}{dt} = mg - kv$; without solving: v increases, ever more slowly, and cannot exceed $\frac{mg}{k}$; the solution is $v(t) = \frac{mg}{k}(1 - e^{-kt/m})$. The **order** is the highest derivative that occurs. Solving a d.v. means finding *all* functions that satisfy it; often impossible in closed form, but **checking** a proposed solution is always possible: substitute it. When the variable is clear one writes $u'' + 6u' + 5u = 10x + 2$ for $u''(x) + \dots$.

Exercise 502

from the dictaat

For $y(t) = ce^{kt}$: (a) find c (grams) if you start at $t = 0$ with 3 g of Radium-226; (b) find k if after 100 years 2.87281 g is left; (c) find the half-life; (d) what do $c = 0$ and $k = 0$ mean physically?

(a) $y(0) = ce^0 = c = 3$.

(b) $3e^{100k} = 2.87281$, so $k = \frac{1}{100} \log \frac{2.87281}{3} \approx -4.33 \cdot 10^{-4}$ per year (negative: decay, Oefening 501).

(c) The half-life T satisfies $e^{kT} = \frac{1}{2}$, so $T = \frac{\log \frac{1}{2}}{k} = \frac{\log 2}{4.33 \cdot 10^{-4}} \approx 1600$ years (independent of the starting amount).

(d) $c = 0$: there is no radium at all ($y \equiv 0$). $k = 0$: y is constant, the substance does not decay (a stable isotope). □

Exercise 506

from the dictaat

Check that the following solve $h' = h^2$: (a) $t \mapsto 0$ on $\mathbb{R}$; (b) $t \mapsto -\frac{1}{t}$ on $(0, \infty)$; (c) $t \mapsto \frac{1}{1-t}$ on $(-\infty, 1)$.

(a) $h' = 0 = 0^2$. (b) $h' = \frac{1}{t^2} = \left(-\frac{1}{t}\right)^2 = h^2$. (c) $h' = \frac{1}{(1-t)^2} = h^2$ (chain rule: $\frac{d}{dt}(1-t)^{-1} = -(1-t)^{-2} \cdot (-1)$). Note that one d.v. has very different solutions, each on its own domain. $\square$

NOW YOU 501, 503, 504, 505, 507 · solutions in Appendix S

501 What do you know about k in $y' = ky$ for decay? **503** Orders of (5.2)–(5.7). **504** Check $v(t) = \frac{mg}{k}(1 - e^{-kt/m})$ in Voorbeeld 5.2. **505** Solutions of $g'^2 + g^2 = 1$: the constant -1 , and more. **507** Check $2x - 2$ and $c_1 e^{-x} + c_2 e^{-5x} + 2(x-1)$ in $u'' + 6u' + 5u = 10x + 2$.

B LINEAR FIRST-ORDER EQUATIONS: THE HOMOGENEOUS CASE
Definitions 5.4, 5.5 · linear, homogeneous

A **linear first-order d.v.** can be written as $y' + a(x)y = f(x)$ with known a, f ; it is **homogeneous** if $f \equiv 0$. (General n -th order linear: $y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_0(x)y = f(x)$.) $g'^2 + g^2 = 1$ and $h' = h^2$ are not linear; $j' - xj = 0$ is linear, homogeneous, first order; $u'' + 6u' + 5u = 10x + 2$ is linear, inhomogeneous, second order. The zero function always solves the homogeneous equation (Oefening 508).

§5.2.1 · separating variables

For $\frac{dy}{dx} + 2(x+1)y = 0$: put everything with y on one side and everything with x on the other, $\frac{dy}{y} = -2(x+1)dx$; integrate both sides, $\log|y| = -x^2 - 2x + C_1$; exponentiate, $y = C_2 e^{-x^2 - 2x}$. Remarks: one integration constant suffices (a constant on each side is just their difference); $C_2 = e^{C_1}$ is renamed to a new constant; the absolute value disappears because C_2 may be negative, and $C_2 = 0$ gives the zero solution. **General rule (Oefening 512):** $y' + a(x)y = 0$ has general solution $y = Ce^{-A(x)}$ with A a primitive of a . Whatever manipulations you use, **justify the result by substituting it into the d.v.**

Exercise 511

from the dictaat

Solve the homogeneous equations: (a) $y' + 3y = 0$; (b) $y' - xy = 0$; (c) $y' = -\frac{2y}{x}$; (d) $4y = x^2 y'$; (e) $y' + \sin(x)y = 0$.

(a) $\frac{dy}{y} = -3dx$, $\log|y| = -3x + C_1$: $y = Ce^{-3x}$. Check: $-3Ce^{-3x} + 3Ce^{-3x} = 0$.

(b) $\frac{dy}{y} = xdx$, $\log|y| = \frac{1}{2}x^2 + C_1$: $y = Ce^{x^2/2}$. Check: $y' = xCe^{x^2/2} = xy$.

(c) $\frac{dy}{y} = -\frac{2dx}{x}$, $\log|y| = -2\log|x| + C_1$: $y = Cx^{-2} = \frac{C}{x^2}$. Check: $y' = -2Cx^{-3} = -\frac{2}{x} \cdot \frac{C}{x^2}$.

(d) $y' = \frac{4y}{x^2}$, $\frac{dy}{y} = \frac{4}{x^2}dx$, $\log|y| = -\frac{4}{x} + C_1$: $y = Ce^{-4/x}$. Check: $y' = \frac{4}{x^2}Ce^{-4/x}$, and $x^2 y' = 4Ce^{-4/x} = 4y$.

(e) $\frac{dy}{y} = -\sin x dx$, $\log|y| = \cos x + C_1$: $y = Ce^{\cos x}$. Check: $y' = -\sin x Ce^{\cos x} = -\sin(x)y$. $\square$

NOW YOU 508, 509, 510, 512, 513 · solutions in Appendix S

508 The zero function solves $y' + a(x)y = 0$. **509** Check the manipulation that separates the variables. **510** Substitute $y = Ce^{-x^2 - 2x}$ (all C , also 0). **512** General rule for $y' + a(x)y = 0$. **513** Check your solutions of 511 (done above).

C THE INHOMOGENEOUS EQUATION

Definition 5.6, Stelling 5.7 · general = homogeneous + particular

A **particular solution** y_p is any one solution of $y' + a(x)y = f(x)$. If y_h is the general solution of the homogeneous equation and y_p a particular solution, then $y = y_h + y_p$ is the general solution of the inhomogeneous equation (Oefening 514: the sum solves it, and two particular solutions differ by a homogeneous solution). So: first solve the homogeneous equation, then find *one* y_p .

§5.2.2 · three ways to a particular solution

Guessing (gokken met voorkennis): for $y' + 3y = x + 2$ try $y_p = ax + b$: $3ax + 3b + a = x + 2$ gives $a = \frac{1}{3}$, $b = \frac{5}{9}$; general solution $y = Ce^{-3x} + \frac{1}{3}x + \frac{5}{9}$. Try the same type as f : polynomial, Ae^{5x} for e^{5x} , constants for constants.

Integrating factor: multiply $y' + ay = f$ by $u = e^{\int a dx}$; since $u' = au$, the left side becomes $(uy)'$, so $uy = \int u f dx$ and $y = \frac{1}{u} \int u f dx$. Voorbeeld 5.8: $y' + \frac{2}{x}y = 2x + 1$, $u = e^{2 \log x} = x^2$, $(x^2 y)' = 2x^3 + x^2$, $y = \frac{1}{2}x^2 + \frac{1}{3}x + \frac{C}{x^2}$.

Variation of constants: replace C in $y_h = Ce^{-A(x)}$ by $C(x)$ and substitute: the $C(x)$ -terms cancel and $C'(x)e^{-A(x)} = f(x)$ remains, so $C(x) = \int f(x)e^{A(x)} dx$ (Oefening 518). Example: $xy' + 4\frac{y}{x} = \frac{1}{x}$ has $y_h = Ce^{4/x}$; $C'(x) = \frac{1}{x^2}e^{-4/x}$, $C(x) = \frac{1}{4}e^{-4/x} + c$, $y = \frac{1}{4} + ce^{4/x}$.

Any method is allowed on the exam; **always check the result in the equation.**

Exercise 516

from the dictaat

Use an integrating factor to solve: (a) $y' + 2y = 3$; (b) $y' + 2xy = 3e^{-x^2}$.

(a) $a = 2$, $u = e^{2x}$: $(e^{2x}y)' = 3e^{2x}$, so $e^{2x}y = \frac{3}{2}e^{2x} + C$ and $y = \frac{3}{2} + Ce^{-2x}$. Check: $y' + 2y = -2Ce^{-2x} + 3 + 2Ce^{-2x} = 3$.

(b) $a = 2x$, $u = e^{x^2}$: $(e^{x^2}y)' = e^{x^2} \cdot 3e^{-x^2} = 3$, so $e^{x^2}y = 3x + C$ and $y = (3x + C)e^{-x^2}$. Check: $y' = 3e^{-x^2} - 2x(3x + C)e^{-x^2}$ and $2xy = 2x(3x + C)e^{-x^2}$; sum $3e^{-x^2}$. $\square$

Exercise 517

from the dictaat

Solve by a method of your choice: (a) $y' + xy = x$; (b) $y' - 3y = e^{5x}$; (c) $y' + \frac{2xy}{1+x^2} = 2x$; (d) $xy' + (1+x^2)y = x$.

(a) Homogeneous: $y_h = Ce^{-x^2/2}$ (511b with the sign changed). Guess a constant: $y_p = 1$ gives $0 + x = x$. General solution $y = 1 + Ce^{-x^2/2}$.

(b) $y_h = Ce^{3x}$. Guess $y_p = Ae^{5x}$: $5Ae^{5x} - 3Ae^{5x} = e^{5x}$ gives $A = \frac{1}{2}$. So $y = Ce^{3x} + \frac{1}{2}e^{5x}$.

(c) $a = \frac{2x}{1+x^2}$ has primitive $\log(1+x^2)$, so the integrating factor is $u = 1+x^2$: $((1+x^2)y)' = 2x(1+x^2) = 2x + 2x^3$, hence $(1+x^2)y = x^2 + \frac{1}{2}x^4 + C$ and $y = \frac{x^2 + \frac{1}{2}x^4 + C}{1+x^2}$.

(d) Divide by x : $y' + (\frac{1}{x} + x)y = \frac{1}{x}$. Primitive of a : $\log|x| + \frac{1}{2}x^2$, so $u = xe^{x^2/2}$ (for $x > 0$): $(xe^{x^2/2}y)' = xe^{x^2/2}$, hence $xe^{x^2/2}y = e^{x^2/2} + C$ and $y = \frac{1}{x} + \frac{C}{x}e^{-x^2/2} = \frac{1 + Ce^{-x^2/2}}{x}$. Check: with $y = \frac{1}{x} + \frac{C}{x}e^{-x^2/2}$ one finds $y' = -\frac{1}{x^2} - \frac{C}{x^2}e^{-x^2/2} - Ce^{-x^2/2}$, so $xy' + (1+x^2)y = (-\frac{1}{x} - \frac{C}{x}e^{-x^2/2} - Cxe^{-x^2/2}) + (\frac{1}{x} + x + \frac{C}{x}e^{-x^2/2} + Cxe^{-x^2/2}) = x$. $\checkmark$ $\square$

NOW YOU 514, 515, 518 · solutions in Appendix S

514 Prove Stelling 5.7: (a) $y_h + y_p$ solves the inhomogeneous equation; (b) $\tilde{y}_p - y_p$ solves the homogeneous one; (c) why does this prove the theorem? **515** Find the integrating factor x^2 of Voorbeeld 5.8 from $e^{\int a}$. **518** Variation of constants for general $y' + a(x)y = f(x)$.

D INITIAL VALUE PROBLEMS
Definition 5.9, Voorbeeld 5.10 · fixing the constant

The general solution is a family (one integration constant for a first-order equation). An **initial value** $y(a) = b$ picks one member: substitute $x = a$ into the *general* solution $y = y_h + y_p$ and solve for C (Oefening 521). Voorbeeld 5.10: $y' + \frac{2}{x}y = 2x + 1$, $y(-1) = \frac{2}{3}$: from $\frac{2}{3} = \frac{1}{2} - \frac{1}{3} + C$ we get $C = \frac{1}{2}$.

Exercise 519

from the dictaat

$x^2y' + 4y = 1$ has $y_h = Ce^{4/x}$ and $y_p = \frac{1}{4}$. Solve the initial value problem with $y(4) = 1$.

General solution $y = \frac{1}{4} + Ce^{4/x}$. At $x = 4$: $1 = \frac{1}{4} + Ce^1$, so $C = \frac{3}{4e}$ and $y = \frac{1}{4} + \frac{3}{4e}e^{4/x} = \frac{1}{4} + \frac{3}{4}e^{4/x-1}$.
Check: $y(4) = \frac{1}{4} + \frac{3}{4} = 1$ ✓; and $x^2y' = x^2 \cdot \frac{3}{4}e^{4/x-1} \cdot (-\frac{4}{x^2}) = -3e^{4/x-1}$, $4y = 1 + 3e^{4/x-1}$, sum 1. ✓ □

Exercise 522

from the dictaat

A pizza leaves the freezer at 8 o'clock at -18°C ; kitchen temperature $T_k = 20^\circ\text{C}$; model $\frac{dT}{dt} = -p(T - T_k)$, $p > 0$. (a) Argue that T approaches T_k quickly at first, then ever more slowly. (b) At 9 o'clock the pizza is -9°C ; solve the d.v. and find p . (c) When is the pizza thawed (0°C)? (d) How warm must the kitchen be for the pizza to be thawed at 9 o'clock?

(a) While $T < T_k$ the right side $-p(T - T_k)$ is positive, so T rises; as T gets closer to T_k the difference $T - T_k$ shrinks, so T' shrinks: slower and slower, never passing T_k .

(b) Linear: $T' + pT = pT_k$; $T_h = Ce^{-pt}$, $T_p = T_k$ (constant guess). So $T(t) = 20 + Ce^{-pt}$ with t in hours after 8. $T(0) = -18$ gives $C = -38$: $T(t) = 20 - 38e^{-pt}$. $T(1) = -9$: $38e^{-p} = 29$, $e^{-p} = \frac{29}{38}$, $p = \log \frac{38}{29} \approx 0.270$ per hour.

(c) $T = 0$: $38e^{-pt} = 20$, $t = \frac{\log(38/20)}{p} = \frac{\log 1.9}{\log(38/29)} \approx \frac{0.642}{0.270} \approx 2.37$ hours: about 10:22.

(d) With kitchen temperature T_k : $T(t) = T_k + (-18 - T_k)e^{-pt}$ (same p , a property of the pizza). $T(1) = 0$: $T_k + (-18 - T_k) \cdot \frac{29}{38} = 0$, so $T_k(1 - \frac{29}{38}) = 18 \cdot \frac{29}{38}$, $T_k = \frac{18 \cdot 29}{9} = 58^\circ\text{C}$. □

Exercise 523

from the dictaat

Solve by separation: $y' = -2y^3$, $y(1) = \frac{1}{3}$.

Not linear, but separable: $\frac{dy}{y^3} = -2dx$, so $-\frac{1}{2y^2} = -2x + C_1$, i.e. $\frac{1}{y^2} = 4x + c$. The initial value: $9 = 4 + c$, $c = 5$, so $y^2 = \frac{1}{4x+5}$ and, since $y(1) > 0$, $y = \frac{1}{\sqrt{4x+5}}$ (for $x > -\frac{5}{4}$). Check: $y' = -\frac{1}{2}(4x+5)^{-3/2} \cdot 4 = -2(4x+5)^{-3/2} = -2y^3$ and $y(1) = \frac{1}{3}$. ✓ □

NOW YOU 520, 521, 524 · solutions in Appendix S

520 Derive the solution of Voorbeeld 5.2 ($mv' = mg - kv$, $v(0) = 0$). **521** Which equation determines C in an initial value problem? **524** Logistic equation $y' = \alpha y(\beta - y)$: constant solutions; $\lim_{t \rightarrow \infty} y(t)$ for $y(0) > 0$ without solving.

Watch out

- Bring the equation to the form $y' + a(x)y = f(x)$ first (divide by the coefficient of y' , as in 517d); the integrating factor is $e^{\int a}$, not $e^{\int f}$.
- The initial condition is applied to the *general* solution $y_h + y_p$, not to y_h alone (Oefening 521).
- Every solution must be checked by substitution; the symbolic manipulations with dy and dx are justified only by that check.
- Keep the integration constant until the end, and keep it even when it turns out to be 0 (the zero solution).

Chapter 6

Finding primitives

Contents

- A** Primitives, improper integrals (§6.1)
- B** Integration by parts (§6.2)
- C** Integration by substitution (§6.3)
- D** Partial fractions (§6.4)
- E** Mixed exercises (§6.5)
- F** (Extra) log and exp (§6.6)

Planning

Exercises 601–635 (all worked or solved).

Hoofdpunten: lots of practice with parts, substitution, partial fractions and rewriting integrands, also in combination; willingness to try something else when an approach fails; the double-angle formulas as a tool; what an improper integral is.

Always check a primitive by differentiating.

A STANDARD PRIMITIVES AND IMPROPER INTEGRALS

§6.1 · know these by heart (constants omitted)

$$\int x^r dx = \frac{x^{r+1}}{r+1} \quad (r \neq -1), \quad \int \frac{dx}{x} = \log|x|, \quad \int e^x dx = e^x, \quad \int \sin x dx = -\cos x, \quad \int \cos x dx = \sin x,$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x, \quad \int \frac{dx}{1+x^2} = \arctan x.$$

Differentiating is automatic; primitiving needs creativity, which you get by trying, failing and learning (“goede keuzes maak je met ervaring, ervaring maak je met foute keuzes”).

§6.1.1 and Voorbeelden 6.1–6.4 · improper integrals

An integral is **improper** if the interval or the integrand is unbounded; evaluate it with a limit: $\int_1^\infty x^{-2} dx = \lim_{b \rightarrow \infty} [-x^{-1}]_1^b = 1$ (**convergent**); $\int_0^1 x^{-2} dx = [-x^{-1}]_0^1 = -1 - \lim_{x \rightarrow 0^+} (-\frac{1}{x})$ does not exist (**divergent**). Two problems (e.g. $\int_0^\infty x^{-2} dx$, or $\int_{-\infty}^\infty$): split the interval and treat each limit separately; the whole integral converges only if every piece does. $\int_{-\infty}^\infty \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$.

Exercise 605

from the dictaat

Convergent? If so, compute: (a) $\int_0^1 \frac{dx}{\sqrt{x}}$; (b) $\int_1^\infty \frac{dx}{\sqrt{x}}$.

A primitive of $x^{-1/2}$ is $2\sqrt{x}$.

(a) Improper at 0 (integrand unbounded): $\int_0^1 x^{-1/2} dx = 2\sqrt{1} - \lim_{a \rightarrow 0^+} 2\sqrt{a} = 2 - 0 = 2$. Convergent, value 2.

(b) Improper at ∞ : $\int_1^b x^{-1/2} dx = 2\sqrt{b} - 2 \rightarrow \infty$ as $b \rightarrow \infty$. Divergent. (Compare x^{-2} : at infinity, a higher power in the denominator is needed; near 0, a lower one.) $\square$

NOW YOU 601, 602, 603, 604, 606 · solutions in Appendix S

601 Why does $\int c dx = cx$ follow from the list? **602** $\frac{1}{x}$ and $\log|x|$ on both sides of 0; $\int_a^b \frac{dx}{x} = \log|b| - \log|a|$ for $a < b < 0$, sign check. **603** $\int \frac{p}{qx+r} dx$. **604** Find $f, g \neq 0$ with $\int f \cdot \int g = \int fg$. **606** $\lim_{a \rightarrow \infty} \int_{-a}^a \sin x dx$; yet $\int_{-\infty}^{\infty} \sin x dx$ diverges.

B INTEGRATION BY PARTS
§6.2 · partiel integreren

From the product rule: $\int f'(x)g(x) dx = f(x)g(x) - \int f(x)g'(x) dx$, also written $\int g df = fg - \int f dg$; with limits, $\int_a^b f'g dx = [fg]_a^b - \int_a^b fg' dx$. Split the integrand sensibly into a factor f' you can primitive and a factor g that becomes simpler when differentiated. Voorbeeld 6.5: $\int xe^x dx$ with $f' = e^x$, $g = x$: $xe^x - \int e^x dx = (x-1)e^x + C$. Voorbeeld 6.6: $\int e^x \sin x dx$: twice by parts brings back the original integral: $I = -e^x \cos x + e^x \sin x - I$, so $I = \frac{1}{2}e^x(\sin x - \cos x) + C$. The wrong choice shows itself by a more complicated new integral (Oefening 609). “Multiply by 1” (Oefening 610): $\int \log x dx = \int 1 \cdot \log x dx$.

Exercise 607

from the dictaat

 Find $\int x \sin x dx$.

$f' = \sin x$ (so $f = -\cos x$), $g = x$ ($g' = 1$): $\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C$. Check: $(-x \cos x + \sin x)' = -\cos x + x \sin x + \cos x = x \sin x$. ✓ □

Exercise 610(c)

from the dictaat

 Determine $\int \log x dx$ and $\int \arctan x dx$ by “multiplying by 1”.

$f' = 1$, $f = x$, $g = \log x$: $\int \log x dx = x \log x - \int x \cdot \frac{1}{x} dx = x \log x - x + C$.

$f' = 1$, $g = \arctan x$: $\int \arctan x dx = x \arctan x - \int \frac{x}{1+x^2} dx = x \arctan x - \frac{1}{2} \log(1+x^2) + C$, since $(\log(1+x^2))' = \frac{2x}{1+x^2}$. □

Exercise 611

from the dictaat

 Compute $\int_1^e x^3 \log^2 x dx$.

Parts with $f' = x^3$, $g = \log^2 x$ ($g' = \frac{2 \log x}{x}$): $\left[\frac{x^4}{4} \log^2 x \right]_1^e - \int_1^e \frac{x^4}{4} \cdot \frac{2 \log x}{x} dx = \frac{e^4}{4} - \frac{1}{2} \int_1^e x^3 \log x dx$. Again by parts: $\int_1^e x^3 \log x dx = \left[\frac{x^4}{4} \log x \right]_1^e - \int_1^e \frac{x^3}{4} dx = \frac{e^4}{4} - \frac{e^4 - 1}{16} = \frac{3e^4 + 1}{16}$. Hence the integral is $\frac{e^4}{4} - \frac{3e^4 + 1}{32} = \frac{5e^4 - 1}{32}$. □

Exercise 612

from the dictaat

The gamma function is $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$ ($x > 0$; assume convergence). (a) $\Gamma(1)$. (b) Show $\Gamma(x+1) =$

$x\Gamma(x)$. (c) $\Gamma(6)$, and the pattern.

(a) $\Gamma(1) = \int_0^\infty e^{-t} dt = \lim_{b \rightarrow \infty} [-e^{-t}]_0^b = 0 + 1 = 1$.

(b) Parts with $f' = e^{-t}$ ($f = -e^{-t}$), $g = t^x$: $\Gamma(x+1) = [-t^x e^{-t}]_0^\infty + x \int_0^\infty t^{x-1} e^{-t} dt$. The boundary term vanishes: at $t = 0$ because $t^x \rightarrow 0$ ($x > 0$), at ∞ because $t^x e^{-t} \rightarrow 0$ (Oefening 326d). So $\Gamma(x+1) = x\Gamma(x)$.

(c) $\Gamma(6) = 5\Gamma(5) = 5 \cdot 4\Gamma(4) = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot \Gamma(1) = 120 = 5!$; likewise $\Gamma(7) = 720 = 6!$, and in general $\Gamma(n) = (n-1)!$: the gamma function extends the factorial to all positive reals. $\square$

NOW YOU 608, 609, 610(a),(b),(d) · solutions in Appendix S

608 Check $\frac{1}{2}e^x(\sin x - \cos x)$ by differentiating. **609** Swap the roles of f' and g in Voorbeelden 6.5 and 6.6: which one goes wrong, and what is the signal? **610** (a) $\int x dx$ and (b) $\int x^2 dx$ by parts with $f' = 1$; (d) try the same for $\int \frac{1}{x} dx$ and explain the paradox.

C INTEGRATION BY SUBSTITUTION

§6.3 · the chain rule backwards

$\int f(u(x)) u'(x) dx = \int f(u) du$. Choose u , compute $du = u'(x) dx$, rewrite the integrand entirely in u , integrate, substitute back. Voorbeeld 6.7: $\int \frac{x}{\sqrt{1+x^2}} dx$, $u = 1 + x^2$, $du = 2x dx$: $\int \frac{1}{\sqrt{u}} \cdot \frac{1}{2} du = \sqrt{u} + C = \sqrt{1+x^2} + C$. Questions to ask: does the integrand remind you of a standard integral? which part would you like to get rid of? is one part the derivative of another? **Definite integrals:** either substitute back and use the original limits, or convert the limits ($u(a)$, $u(b)$) and never substitute back (Voorbeeld 6.8: $\int_0^8 \frac{\cos \sqrt{1+x}}{\sqrt{1+x}} dx = 2 \int_1^3 \cos u du = 2(\sin 3 - \sin 1)$).

§6.3 · trigonometric substitutions

Three variants of $\sin^2 u + \cos^2 u = 1$: for $a^2 - x^2$ put $x = a \sin u$ ($a^2 - x^2 = a^2 \cos^2 u$); for $x^2 - a^2$ put $x = \frac{a}{\cos u}$ ($x^2 - a^2 = a^2 \tan^2 u$); for $a^2 + x^2$ put $x = a \tan u$ ($a^2 + x^2 = \frac{a^2}{\cos^2 u}$). Voorbeeld 6.9: $\int \sqrt{25 - x^2} dx$ with $x = 5 \sin u$ becomes $25 \int \cos^2 u du = \frac{25}{2} \int (\cos 2u + 1) du = \frac{25}{4} \sin 2u + \frac{25}{2} u$, and back: $\frac{5}{2} x \sqrt{1 - (\frac{x}{5})^2} + \frac{25}{2} \arcsin \frac{x}{5} + C$. No absolute value is needed in $\sqrt{1 - \sin^2 u} = \cos u$ because $u \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ (needed anyway for the substitution to be a bijection). Double-angle formulas are essential here.

Exercise 613

from the dictaat

With the given substitution: (a) $\int 16x(3+4x^2)^7 dx$, $u = 3+4x^2$; (b) $\int \sin(2x) \sqrt{2+\sin^2 x} dx$, $u = 2+\sin^2 x$.

(a) $du = 8x dx$, so $16x dx = 2 du$: $\int 2u^7 du = \frac{1}{4} u^8 + C = \frac{1}{4} (3+4x^2)^8 + C$.

(b) $du = 2 \sin x \cos x dx = \sin(2x) dx$: $\int \sqrt{u} du = \frac{2}{3} u^{3/2} + C = \frac{2}{3} (2+\sin^2 x)^{3/2} + C$. $\square$

Exercise 615

from the dictaat

Evaluate $\int \frac{dx}{1 + \sqrt{3x}}$ with $u = \sqrt{3x}$. Hint: be creative with $\frac{du}{dx}$.

$u^2 = 3x$, so $2u du = 3 dx$, $dx = \frac{2}{3} u du$ (differentiate the squared relation rather than the root). Then

$$\begin{aligned} \int \frac{dx}{1 + \sqrt{3x}} &= \int \frac{\frac{2}{3} u}{1 + u} du = \frac{2}{3} \int \left(1 - \frac{1}{1 + u}\right) du \\ &= \frac{2}{3} (u - \log |1 + u|) + C = \frac{2}{3} (\sqrt{3x} - \log(1 + \sqrt{3x})) + C. \end{aligned}$$

□

Exercise 616

from the dictaat

Compute $\int_0^{\pi/2} \sqrt{1 + \cos x} dx$.

Double-angle: $1 + \cos x = 2 \cos^2 \frac{x}{2}$, so $\sqrt{1 + \cos x} = \sqrt{2} |\cos \frac{x}{2}| = \sqrt{2} \cos \frac{x}{2}$ on $[0, \frac{\pi}{2}]$. Then $\int_0^{\pi/2} \sqrt{2} \cos \frac{x}{2} dx = \sqrt{2} [2 \sin \frac{x}{2}]_0^{\pi/2} = 2\sqrt{2} \sin \frac{\pi}{4} = 2\sqrt{2} \cdot \frac{1}{2} \sqrt{2} = 2$. □

Exercise 619

from the dictaat

Compute: (a) $\int \frac{x}{\sqrt{x^2 - 36}} dx$; (b) $\int \frac{dt}{t^2 \sqrt{49 - t^2}}$; (c) $\int_{1/2}^{\frac{1}{2}\sqrt{3}} \frac{dx}{(1 + 4x^2)^2}$.

(a) No trig needed: $u = x^2 - 36$, $du = 2x dx$: $\frac{1}{2} \int u^{-1/2} du = \sqrt{u} = \sqrt{x^2 - 36} + C$.

(b) $t = 7 \sin u$, $dt = 7 \cos u du$, $\sqrt{49 - t^2} = 7 \cos u$: $\int \frac{7 \cos u du}{49 \sin^2 u \cdot 7 \cos u} = \frac{1}{49} \int \frac{du}{\sin^2 u} = -\frac{1}{49} \cot u + C$.

Back: $\cot u = \frac{\cos u}{\sin u} = \frac{\sqrt{1 - (t/7)^2}}{t/7} = \frac{\sqrt{49 - t^2}}{t}$, so the integral is $-\frac{\sqrt{49 - t^2}}{49t} + C$.

(c) $2x = \tan u$, $dx = \frac{1}{2} (1 + \tan^2 u) du = \frac{du}{2 \cos^2 u}$ and $1 + 4x^2 = \frac{1}{\cos^2 u}$: the integrand becomes $\cos^4 u \cdot \frac{1}{2 \cos^2 u} du = \frac{1}{2} \cos^2 u du = \frac{1}{4} (1 + \cos 2u) du$. Limits: $x = \frac{1}{2} \Rightarrow \tan u = 1 \Rightarrow u = \frac{\pi}{4}$; $x = \frac{1}{2}\sqrt{3} \Rightarrow \tan u = \sqrt{3} \Rightarrow u = \frac{\pi}{3}$. So the integral is $\frac{1}{4} [u + \frac{1}{2} \sin 2u]_{\pi/4}^{\pi/3} = \frac{1}{4} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{4} - \frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi}{48} + \frac{\sqrt{3}}{16} - \frac{1}{8} \approx 0.0487$. □

NOW YOU 614, 617, 618 · solutions in Appendix S

614 By substitution: $\int 2 \sin(\pi x) dx$; $\int x e^{-x^2} dx$; $\int \frac{\log^2 x}{x} dx$; $\int \tan x dx$; $\int (1 + \cos x)^3 \sin x dx$. **617** $\int_e^{e^2} \frac{dx}{x \log x}$.

618 Derive $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x$ with $x = \sin u$.

D RATIONAL FUNCTIONS AND PARTIAL FRACTIONS
§6.4 · the standard procedure

1 Degree of numerator $\geq$ degree of denominator: long division first (§0.3). $\frac{(2x-1)^3 + 4}{2x^2 - x} = 4x - 4 + \frac{2x+3}{2x^2 - x}$.

- 2 Denominator of degree 1: $\int \frac{c}{ax+b} dx = \frac{c}{a} \log |ax+b|$.
- 3 Denominator of degree 2, by the number of real zeros (discriminant): **none**: complete the square and reduce to $\int \frac{dx}{1+x^2} = \arctan x$ and $\int \frac{xdx}{1+x^2} = \frac{1}{2} \log(1+x^2)$; **one (double)**: write the numerator in terms of the root, $\frac{3x+5}{(x+1)^2} = \frac{3}{x+1} + \frac{2}{(x+1)^2}$; **two**: partial fractions $\frac{ax+b}{(px+q)(rx+s)} = \frac{A}{px+q} + \frac{B}{rx+s}$, found by putting the right side over one denominator and comparing numerators (or substituting the roots).
- 4 Degree ≥ 3 : factorise the denominator (easy cases only), one term per factor; a repeated factor $(x-c)^k$ or an irreducible quadratic gets a numerator of degree one less than its denominator: $\frac{Ax^2+Bx+C}{(1-x)^3} + \frac{D}{2+x}, \frac{Ax+B}{1+x^2} + \frac{C}{1+2x}$. Sometimes $u = x^2$ lowers the degree instead.

Voorbeeld 6.10: $\frac{x+2}{x^3-x} = \frac{-2}{x} + \frac{3/2}{x-1} + \frac{1/2}{x+1}$. Voorbeeld 6.11: $\frac{4x^2+x+13}{(x-1)^2(x+5)} = \frac{x+2}{(x-1)^2} + \frac{3}{x+5}$ with $\frac{x+2}{(x-1)^2} = \frac{1}{x-1} + \frac{3}{(x-1)^2}$.

Exercise 621

from the dictaat

Evaluate $\int \frac{(2x-1)^3+4}{2x^2-x} dx$.

After the division (page 96): $\int (4x-4) dx + \int \frac{2x+3}{x(2x-1)} dx$. Partial fractions: $\frac{2x+3}{x(2x-1)} = \frac{A}{x} + \frac{B}{2x-1}$ with $A(2x-1) + Bx = 2x+3$; $x=0$ gives $A=-3$, $x=\frac{1}{2}$ gives $\frac{1}{2}B=4$, $B=8$ (this is the dictaat's example $\frac{8}{2x-1} - \frac{3}{x}$). Hence

$$\int \frac{(2x-1)^3+4}{2x^2-x} dx = 2x^2 - 4x - 3 \log |x| + 4 \log |2x-1| + C,$$

using $\int \frac{8}{2x-1} dx = 8 \log |2x-1|$. Check by differentiating: $4x-4 - \frac{3}{x} + \frac{8}{2x-1} \checkmark$. □

Exercise 624(a)–(d)

from the dictaat

Evaluate: (a) $\int \frac{dt}{1-t^2}$; (b) $\int \frac{x^3}{x^2+2x-15} dx$; (c) $\int \frac{dx}{x^2+6x+25}$; (d) $\int \frac{4y+3}{4y^2-9} dy$.

(a) $\frac{1}{1-t^2} = \frac{1}{(1-t)(1+t)} = \frac{1}{2} \left(\frac{1}{1-t} + \frac{1}{1+t} \right)$, so the integral is $\frac{1}{2} (-\log |1-t| + \log |1+t|) + C = \frac{1}{2} \log \left| \frac{1+t}{1-t} \right| + C$.

(b) Divide: $x^3 = (x^2+2x-15)(x-2) + (19x-30)$. Then $\frac{19x-30}{(x+5)(x-3)} = \frac{A}{x+5} + \frac{B}{x-3}$: $A(x-3) + B(x+5) = 19x-30$; $x=3$: $8B=27$; $x=-5$: $-8A=-125$. So the integral is $\frac{x^2}{2} - 2x + \frac{125}{8} \log |x+5| + \frac{27}{8} \log |x-3| + C$.

(c) No real zeros ($36-100 < 0$): complete the square, $x^2+6x+25 = (x+3)^2+16 = 16 \left(1 + \left(\frac{x+3}{4} \right)^2 \right)$; with $u = \frac{x+3}{4}$, $dx = 4du$: $\int \frac{4du}{16(1+u^2)} = \frac{1}{4} \arctan u = \frac{1}{4} \arctan \frac{x+3}{4} + C$.

(d) $4y^2-9 = (2y-3)(2y+3)$: $\frac{4y+3}{(2y-3)(2y+3)} = \frac{A}{2y-3} + \frac{B}{2y+3}$, $A(2y+3) + B(2y-3) = 4y+3$;

$$y = \frac{3}{2}: 6A = 9, A = \frac{3}{2}; y = -\frac{3}{2}: -6B = -3, B = \frac{1}{2}. \text{ Integral: } \frac{3}{2} \cdot \frac{1}{2} \log |2y - 3| + \frac{1}{2} \cdot \frac{1}{2} \log |2y + 3| = \frac{3}{4} \log |2y - 3| + \frac{1}{4} \log |2y + 3| + C. \quad \square$$

Exercise 625

from the dictaat

 Solve the logistic equation $y' = \frac{1}{50}y(100 - y)$, $y(0) = 25$, by separating variables.

$$\frac{dy}{y(100 - y)} = \frac{dt}{50}. \text{ Partial fractions: } \frac{1}{y(100 - y)} = \frac{1}{100} \left(\frac{1}{y} + \frac{1}{100 - y} \right). \text{ Integrating: } \frac{1}{100} (\log y - \log(100 - y)) = \frac{t}{50} + C \text{ (for } 0 < y < 100 \text{), i.e. } \log \frac{y}{100 - y} = 2t + C'. \text{ At } t = 0: \log \frac{25}{75} = \log \frac{1}{3} = C'. \text{ So } \frac{y}{100 - y} = \frac{1}{3} e^{2t}, \text{ hence } 3y = (100 - y)e^{2t}, y(3 + e^{2t}) = 100e^{2t} \text{ and}$$

$$y(t) = \frac{100e^{2t}}{3 + e^{2t}} = \frac{100}{1 + 3e^{-2t}}.$$

 Check: $y(0) = 25$; $y \rightarrow 100$ as $t \rightarrow \infty$ (the carrying capacity $\beta = 100$ of Oefening 524); and $y' = \frac{600e^{-2t}}{(1 + 3e^{-2t})^2} = \frac{1}{50}y(100 - y)$ since $100 - y = \frac{300e^{-2t}}{1 + 3e^{-2t}}. \quad \checkmark \quad \square$
NOW YOU 620, 622, 623, 624(e)–(h), 626 · solutions in Appendix S

620 Recombine $\frac{1}{3} \left(\frac{4}{x-2} - \frac{1}{x+4} \right)$. **622** Try partial fractions on $\frac{3x+5}{x^2+2x+1}$: what goes wrong? **623** Finish Voorbeeld 6.11. **624** (e) $\int \frac{4x^2+12x+9}{4x^2+12x+10} dx$; (f) $\int \frac{x dx}{x^4-x^2-6}$; (g) $\int \frac{dx}{(x^2-1)^2}$; (h) $\int \frac{2x^3-3x^2-2x-5}{x^4-1} dx$. **626** $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi$, hence $\frac{22}{7} > \pi$.

E MIXED EXERCISES
§6.5 · choosing a method

Trig integrands: rewrite with the formulas ($\sin 2x = 2 \sin x \cos x$, $\cos 2x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$, products to sums, $\cos^2 + \sin^2 = 1$), then substitute ($u = \sin x$ when an odd power of $\cos$ is left over, and vice versa) or integrate by parts. One method is usually enough (628); sometimes two or three in a row (629). For $\int \frac{dx}{\cos x}$ multiply by $\frac{\cos x}{\cos x}$ and substitute $u = \sin x$ (632). If an approach is not working, try another.

Exercise 627(a),(c),(d)

from the dictaat

 Evaluate (a) $\int \sin x \sin 2x dx$; (c) $\int_0^{\pi/12} \sin^4 \xi d\xi$; (d) $\int \tan^2 \alpha \sin \alpha d\alpha$.

$$(a) \sin 2x = 2 \sin x \cos x: \int 2 \sin^2 x \cos x dx; u = \sin x: \int 2u^2 du = \frac{2}{3} \sin^3 x + C.$$

$$(c) \text{ Reduce the power with double angles: } \sin^4 \xi = \left(\frac{1 - \cos 2\xi}{2} \right)^2 = \frac{1}{4} (1 - 2 \cos 2\xi + \cos^2 2\xi) = \frac{1}{4} \left(1 - 2 \cos 2\xi + \frac{1 + \cos 4\xi}{2} \right) = \frac{3}{8} - \frac{1}{2} \cos 2\xi + \frac{1}{8} \cos 4\xi. \text{ Integrating: } \left[\frac{3\xi}{8} - \frac{1}{4} \sin 2\xi + \frac{1}{32} \sin 4\xi \right]_0^{\pi/12} = \frac{\pi}{32} - \frac{1}{4} \sin \frac{\pi}{6} + \frac{1}{32} \sin \frac{\pi}{3} = \frac{\pi}{32} - \frac{1}{8} + \frac{\sqrt{3}}{64}.$$

$$(d) \tan^2 \alpha \sin \alpha = \frac{\sin^3 \alpha}{\cos^2 \alpha} = \frac{(1 - \cos^2 \alpha) \sin \alpha}{\cos^2 \alpha}; u = \cos \alpha, du = -\sin \alpha d\alpha: -\int \frac{1-u^2}{u^2} du = -\int (u^{-2} - 1) du = \frac{1}{u} + u + C = \frac{1}{\cos \alpha} + \cos \alpha + C. \quad \square$$

Exercise 628(c),(e)

from the dictaat

 One method each: (c) $\int x^3 e^{-x^2} dx$; (e) $\int (\frac{1}{2}(t^2+t)^9 + t(t^2+t)^9) dt$.

 (c) $u = x^2, du = 2x dx$: $\int x^2 e^{-x^2} \cdot x dx = \frac{1}{2} \int u e^{-u} du = \frac{1}{2} (-u e^{-u} - e^{-u}) + C = -\frac{1}{2}(x^2 + 1)e^{-x^2} + C$ (the inner integral by parts as in Voorbeeld 6.5).

 (e) Factor: $(t^2+t)^9(t+\frac{1}{2}) = \frac{1}{2}(t^2+t)^9(2t+1)$, and $2t+1$ is the derivative of t^2+t : $u = t^2+t$ gives $\frac{1}{2} \int u^9 du = \frac{1}{20}(t^2+t)^{10} + C. \quad \square$
Exercise 629(b),(c),(f),(g)

from the dictaat

$$(b) \int \frac{\sqrt{x+1}}{x+2} dx; (c) \int x^3 \sqrt{1-x^2} dx; (f) \int \frac{x \arctan x}{(1+x^2)^{3/2}} dx; (g) \int \frac{x^2}{\sqrt{2-x^2}} dx.$$

 (b) $u = \sqrt{x+1}, x = u^2 - 1, dx = 2u du$: $\int \frac{u \cdot 2u}{u^2+1} du = 2 \int \left(1 - \frac{1}{u^2+1}\right) du = 2u - 2 \arctan u + C = 2\sqrt{x+1} - 2 \arctan \sqrt{x+1} + C.$

 (c) $u = 1-x^2, x^2 = 1-u, du = -2x dx$: $\int x^2 \sqrt{1-x^2} \cdot x dx = -\frac{1}{2} \int (1-u) \sqrt{u} du = -\frac{1}{2} \left(\frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2}\right) + C = -\frac{1}{3}(1-x^2)^{3/2} + \frac{1}{5}(1-x^2)^{5/2} + C.$

 (f) Parts with $f' = \frac{x}{(1+x^2)^{3/2}}$, whose primitive is $f = -\frac{1}{\sqrt{1+x^2}}$ (substitution $u = 1+x^2$), and $g = \arctan x$: $-\frac{\arctan x}{\sqrt{1+x^2}} + \int \frac{dx}{(1+x^2)^{3/2}}$. The last integral: $x = \tan u, dx = \frac{du}{\cos^2 u}, (1+x^2)^{3/2} = \cos^{-3} u$, so it is $\int \cos u du = \sin u = \frac{x}{\sqrt{1+x^2}}$. Result: $\frac{x - \arctan x}{\sqrt{1+x^2}} + C.$

 (g) $x = \sqrt{2} \sin u, dx = \sqrt{2} \cos u du, \sqrt{2-x^2} = \sqrt{2} \cos u$: $\int \frac{2 \sin^2 u \cdot \sqrt{2} \cos u}{\sqrt{2} \cos u} du = 2 \int \sin^2 u du = u - \sin u \cos u + C$. Back: $u = \arcsin \frac{x}{\sqrt{2}}, \sin u \cos u = \frac{x}{\sqrt{2}} \sqrt{1 - \frac{x^2}{2}} = \frac{1}{2} x \sqrt{2-x^2}$. So $\arcsin \frac{x}{\sqrt{2}} - \frac{1}{2} x \sqrt{2-x^2} + C. \quad \square$
Exercise 630(a),(e),(f)

from the dictaat

 Solve the initial value problems: (a) $xy' - (x+1)y = x^2 - x^3, y(1) = 2$; (e) $y' + y \cos x = 2x e^{-\sin x}, y(\pi) = 0$;

 (f) $y' = \frac{4-2x}{3y^2-5}, y(1) = 3$.

 (a) Divide by x : $y' - (1 + \frac{1}{x})y = x - x^2$. Integrating factor $u = e^{-x-\log x} = \frac{e^{-x}}{x}$: $\left(\frac{e^{-x}}{x} y\right)' = (1-x)e^{-x}$, and $\int (1-x)e^{-x} dx = x e^{-x} + C$ (check: $(x e^{-x})' = e^{-x} - x e^{-x}$). So $\frac{e^{-x}}{x} y = x e^{-x} + C, y = x^2 + C x e^x$. $y(1) = 1 + C e = 2$ gives $C = \frac{1}{e}$: $y = x^2 + x e^{x-1}$.

 (e) Integrating factor $e^{\sin x}$: $(e^{\sin x} y)' = 2x$, so $e^{\sin x} y = x^2 + C, y = (x^2 + C) e^{-\sin x}$. $y(\pi) = (\pi^2 + C) e^0 = 0$ gives $C = -\pi^2$: $y = (x^2 - \pi^2) e^{-\sin x}$.

 (f) Not linear; separate: $(3y^2 - 5) dy = (4 - 2x) dx$, so $y^3 - 5y = 4x - x^2 + C$. At $(1, 3)$: $27 - 15 = 4 - 1 + C$, $C = 9$. The solution is given implicitly by $y^3 - 5y = 4x - x^2 + 9$ (it cannot be solved for y in closed form;

the d.v. holds by implicit differentiation). □

Exercise 632

from the dictaat

$\int \frac{dx}{\cos x}$: (a) multiply numerator and denominator by a suitable factor so that $u = \sin x$ works; (b) integrate by partial fractions; (c) use the log rules and back-substitution to reach $\log \left| \frac{1}{\cos x} + \tan x \right|$.

(a) Multiply by $\cos x$: $\frac{1}{\cos x} = \frac{\cos x}{\cos^2 x} = \frac{\cos x}{1 - \sin^2 x}$. With $u = \sin x$, $du = \cos x dx$: $\int \frac{du}{1 - u^2}$.

(b) $\frac{1}{1 - u^2} = \frac{1}{2} \left(\frac{1}{1 - u} + \frac{1}{1 + u} \right)$ (624a), so $\int \frac{du}{1 - u^2} = \frac{1}{2} \log \left| \frac{1 + u}{1 - u} \right| + C$.

(c) $\frac{1 + u}{1 - u} = \frac{(1 + u)^2}{1 - u^2} = \frac{(1 + \sin x)^2}{\cos^2 x}$, so $\frac{1}{2} \log \frac{(1 + \sin x)^2}{\cos^2 x} = \log \left| \frac{1 + \sin x}{\cos x} \right| = \log \left| \frac{1}{\cos x} + \tan x \right| + C$. □

Exercise 633

from the dictaat

First a substitution, then Oefening 632: (a) $\int \frac{dx}{\sqrt{x^2 - 1}}$; (b) $\int_0^2 \frac{dx}{\sqrt{4 + x^2}}$.

(a) $x = \frac{1}{\cos u}$, $dx = \frac{\sin u}{\cos^2 u} du$, $\sqrt{x^2 - 1} = \tan u$ (for $0 \leq u < \frac{\pi}{2}$): $\int \frac{\sin u / \cos^2 u}{\sin u / \cos u} du = \int \frac{du}{\cos u} = \log \left| \frac{1}{\cos u} + \tan u \right| = \log |x + \sqrt{x^2 - 1}| + C$ (this is $\operatorname{arcosh} x$, Oefening 714).

(b) $x = 2 \tan u$, $dx = \frac{2 du}{\cos^2 u}$, $\sqrt{4 + x^2} = \frac{2}{\cos u}$: the integrand becomes $\frac{du}{\cos u}$; limits $u = 0$ to $u = \frac{\pi}{4}$: $\left[\log \left(\frac{1}{\cos u} + \tan u \right) \right]_0^{\pi/4} = \log(\sqrt{2} + 1) - \log 1 = \log(1 + \sqrt{2})$. □

NOW YOU 627(b),(e),(f), 628(a),(b),(d),(f), 629(a),(d),(e),(h), 630(b),(c),(d),(g), 631 · solutions in Appendix S

627 (b) $\int \cos^5 \psi d\psi$; (e) $\int \tan(\arcsin t) dt$; (f) $\int \sin(\tau + 1) \sin \tau d\tau$. **628** (a) $\frac{x}{\sqrt{x^2 + a^2}}$; (b) $(x^2 + 3x + 2)^{-1}$; (d) $\frac{x^3 + 1}{x\sqrt{x}}$; (f) $x^2 \cos x$. **629** (a) $\log(1 + 1/x^2)$; (d) $\frac{x^3 - 8}{x^2 - 8}$; (e) $e^{\sqrt{1+x}}$; (h) $\sin(\log x)$. **630** (b) $y' = e^{\sqrt{x}}/y$, $y(0) = -2$; (c) $y' + 10y = 1$, $y(\frac{1}{10}) = \frac{2}{10}$; (d) $y' + 3x^2 y = x^2$, $y(0) = 1$; (g) $y' = 2xy^3(2x^2 + 1)$, $y(1) = 1$. **631** $\int_0^2 \sqrt{(x-1)^2} dx$; $\int_0^\infty \frac{dx}{e^x + 1}$.

F (EXTRA) THE NATURAL LOGARITHM AND THE EXPONENTIAL FUNCTION
Definitions 6.12–6.18 · building log and exp from an integral

Define $\log x = \int_1^x \frac{dt}{t}$ for $x > 0$. Then $\log' x = \frac{1}{x}$ (fundamental theorem) and $\log(xy) = \log x + \log y$ (Stelling 6.13: $x \mapsto \log(xy)$ has derivative $\frac{1}{x}$, so it is $\log x + c$, and $x = 1$ gives $c = \log y$): it deserves the name logarithm. Define $\exp = \log_{\text{inv}} : \mathbb{R} \rightarrow \mathbb{R}_{>0}$; then $\exp' = \exp$ (differentiate $\log(\exp x) = x$), $\exp(x + y) = \exp x \exp y$, $e = \exp 1$ (so $\int_1^e \frac{dt}{t} = 1$), and finally $e^x = \exp x$ and $a^x = e^{x \log a}$ for all real x , which at last gives $3^{\sqrt{2}}$ a meaning. $e = \sum_{k \geq 0} \frac{1}{k!}$.

NOW YOU 634 (Extra), 635 (Extra) · solutions in Appendix S

634 How many terms of $\sum \frac{1}{k!}$ give the first 10 decimals of e ? **635** From Definition 6.18: $a^1 = a$, $a^{x+y} = a^x a^y$, $(a^x)^y = a^{xy}$.

 Watch out

- Check every primitive by differentiating; it takes ten seconds and catches most errors.
- In a substitution, convert dx as well (differentiate the relation between u and x ; for $u = \sqrt{\cdot}$ square first). With definite integrals, either convert the limits or substitute back, never half of each.
- Before partial fractions: divide if the numerator's degree is too high, and check the discriminant of a quadratic denominator.
- Improper integrals: one limit per “improperness”; a symmetric cancellation ($\int_{-a}^a \sin x dx = 0$) proves nothing (606).
- Do not forget $+C$, and do not add one to a definite integral.

Chapter 7

More exercises

Contents

- A Mixed exercises (701–724)
- B (Extra) Function investigation (§7.1)

Planning

Exercises 701–730 (all worked or solved).
The dictaat: “use them as extra practice, exam preparation, or for fun”. Each one combines techniques from several chapters; the first question to ask is *which* chapter it belongs to.

A MIXED EXERCISES ON CHAPTERS 0–6

Exercise 701

from the dictaat

f differentiable with inverse f_{inv} , so $f(f_{\text{inv}}(x)) = x$. (a) Differentiate both sides and show $f'_{\text{inv}}(x) = \frac{1}{f'(f_{\text{inv}}(x))}$. (b) For $f(x) = x^3 + x$ find $f'_{\text{inv}}(-10)$.

(a) Chain rule: $f'(f_{\text{inv}}(x)) \cdot f'_{\text{inv}}(x) = 1$, so $f'_{\text{inv}}(x) = \frac{1}{f'(f_{\text{inv}}(x))}$ (where $f' \neq 0$).

(b) $f_{\text{inv}}(-10)$ is the x with $x^3 + x = -10$: $x = -2$ ($-8 - 2 = -10$; f is increasing, so it is the only one).
 $f'(x) = 3x^2 + 1$, $f'(-2) = 13$, so $f'_{\text{inv}}(-10) = \frac{1}{13}$. (No formula for f_{inv} is needed.) $\square$

Exercise 702

from the dictaat

Show with sum and double-angle formulas that $\tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right) = \frac{1}{\cos \varphi} + \tan \varphi$.

Right side: $\frac{1 + \sin \varphi}{\cos \varphi}$. With half angles, $1 + \sin \varphi = \cos^2 \frac{\varphi}{2} + \sin^2 \frac{\varphi}{2} + 2 \sin \frac{\varphi}{2} \cos \frac{\varphi}{2} = (\cos \frac{\varphi}{2} + \sin \frac{\varphi}{2})^2$
and $\cos \varphi = \cos^2 \frac{\varphi}{2} - \sin^2 \frac{\varphi}{2} = (\cos \frac{\varphi}{2} - \sin \frac{\varphi}{2})(\cos \frac{\varphi}{2} + \sin \frac{\varphi}{2})$. Hence

$$\begin{aligned} \frac{1 + \sin \varphi}{\cos \varphi} &= \frac{\cos \frac{\varphi}{2} + \sin \frac{\varphi}{2}}{\cos \frac{\varphi}{2} - \sin \frac{\varphi}{2}} = \frac{1 + \tan \frac{\varphi}{2}}{1 - \tan \frac{\varphi}{2}} \\ &= \frac{\tan \frac{\pi}{4} + \tan \frac{\varphi}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{\varphi}{2}} = \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right) \end{aligned}$$

by the sum formula for $\tan$ (Oefening 26c) with $\tan \frac{\pi}{4} = 1$. $\square$

Exercise 705

from the dictaat

Second derivative of $\sqrt{\frac{1+x}{1-x}}$, simplified.

Logarithmic differentiation: $\log y = \frac{1}{2}(\log(1+x) - \log(1-x))$, so $\frac{y'}{y} = \frac{1}{2}\left(\frac{1}{1+x} + \frac{1}{1-x}\right) = \frac{1}{1-x^2}$ and

$$y' = \frac{(1+x)^{1/2}}{(1-x)^{1/2}(1-x)(1+x)} = (1+x)^{-1/2}(1-x)^{-3/2}. \text{ Then}$$

$$\begin{aligned} y'' &= -\frac{1}{2}(1+x)^{-3/2}(1-x)^{-3/2} + \frac{3}{2}(1+x)^{-1/2}(1-x)^{-5/2} \\ &= (1+x)^{-3/2}(1-x)^{-5/2} \left(-\frac{1}{2}(1-x) + \frac{3}{2}(1+x) \right) = \frac{1+2x}{(1+x)^{3/2}(1-x)^{5/2}}. \end{aligned}$$

□

Exercise 706

from the dictaat

Find all a, b for which $f(x) = ax^2 + b$ ($x < 1$), $f(x) = \frac{1}{a+bx}$ ($x \geq 1$) is differentiable.

Away from $x = 1$ both pieces are differentiable (provided $a + bx \neq 0$ for $x \geq 1$). At $x = 1$ we need continuity and equal one-sided derivatives: $a + b = \frac{1}{a+b}$, so $(a+b)^2 = 1$; and $2a = \frac{-b}{(a+b)^2} = -b$. So $b = -2a$ and $a + b = -a = \pm 1$: $(a, b) = (-1, 2)$ or $(1, -2)$. Check $(1, -2)$: $f(1^-) = 1 - 2 = -1$, $f(1) = \frac{1}{1-2} = -1$; slopes $2a = 2$ and $\frac{-b}{(a+b)^2} = 2$; and $a + bx = 1 - 2x \neq 0$ for $x \geq 1$. Similarly for $(-1, 2)$ ($-1 + 2x \neq 0$ for $x \geq 1$). Two solutions. □

Exercise 710

from the dictaat

$f(x) = x \arctan(2x^3)$. Find the coefficient c_{10} of x^{10} in its Taylor approximation at 0.

$\arctan u = u - \frac{1}{3}u^3 + \frac{1}{5}u^5 - \dots$ (Oefening 709). With $u = 2x^3$: $\arctan(2x^3) = 2x^3 - \frac{8}{3}x^9 + \frac{32}{5}x^{15} - \dots$, so $f(x) = 2x^4 - \frac{8}{3}x^{10} + \dots$ and $c_{10} = -\frac{8}{3}$. (The next term is x^{16} .) □

Exercise 713

from the dictaat

Determine $\lim_{x \rightarrow \frac{1}{2}} \frac{\arcsin x - \frac{\pi}{6}}{x - \frac{1}{2}}$.

Since $\arcsin \frac{1}{2} = \frac{\pi}{6}$, this is $\frac{\arcsin x - \arcsin \frac{1}{2}}{x - \frac{1}{2}}$ with $x \rightarrow \frac{1}{2}$: the derivative of $\arcsin$ at $\frac{1}{2}$ (§2.3.1). $\arcsin' x = \frac{1}{\sqrt{1-x^2}}$, so the limit is $\frac{1}{\sqrt{1-\frac{1}{4}}} = \frac{2}{\sqrt{3}} = \frac{2}{3}\sqrt{3}$. (l'Hôpital gives the same in one step.) □

Exercise 718

from the dictaat

Solve $z'(t) + z(t) = \sin t$ with $z(\frac{\pi}{4}) = 1$.

$z_h = Ce^{-t}$. Guess $z_p = A \sin t + B \cos t$: $z'_p + z_p = (A \cos t - B \sin t) + (A \sin t + B \cos t) = (A - B) \sin t + (A + B) \cos t = \sin t$ gives $A - B = 1$, $A + B = 0$: $A = \frac{1}{2}$, $B = -\frac{1}{2}$. General solution $z = Ce^{-t} + \frac{1}{2}(\sin t - \cos t)$. At $t = \frac{\pi}{4}$: $\sin \frac{\pi}{4} = \cos \frac{\pi}{4}$, so $z(\frac{\pi}{4}) = Ce^{-\pi/4} = 1$, $C = e^{\pi/4}$. Hence $z(t) = e^{\pi/4-t} + \frac{1}{2}(\sin t - \cos t)$. □

Exercise 719

from the dictaat

Primitives of: (a) $\arcsin(2x)$; (b) $\frac{2x+1}{x^4+2x^3+x^2}$; (c) $x \cos(3x)$; (d) $\frac{xe^{2x}}{(1+e^{2x})^2}$.

(a) Parts with $f' = 1$: $x \arcsin(2x) - \int \frac{2x}{\sqrt{1-4x^2}} dx$; with $u = 1-4x^2$, $du = -8x dx$ the last integral is $-\frac{1}{4} \int u^{-1/2} du = -\frac{1}{2} \sqrt{1-4x^2}$. So $x \arcsin(2x) + \frac{1}{2} \sqrt{1-4x^2} + C$.

(b) $x^4 + 2x^3 + x^2 = (x^2 + x)^2$ and $2x + 1 = (x^2 + x)'$: $u = x^2 + x$ gives $\int \frac{du}{u^2} = -\frac{1}{u} = -\frac{1}{x^2 + x} + C$.

(c) Parts with $f' = \cos 3x$, $f = \frac{1}{3} \sin 3x$: $\frac{1}{3} x \sin 3x - \frac{1}{3} \int \sin 3x dx = \frac{1}{3} x \sin 3x + \frac{1}{9} \cos 3x + C$.

(d) Parts with $f' = \frac{e^{2x}}{(1+e^{2x})^2}$, whose primitive is $f = -\frac{1}{2(1+e^{2x})}$ ($u = 1+e^{2x}$), and $g = x$: $-\frac{x}{2(1+e^{2x})} + \frac{1}{2} \int \frac{dx}{1+e^{2x}}$. And $\int \frac{dx}{1+e^{2x}} = \int \frac{e^{-2x}}{e^{-2x}+1} dx = -\frac{1}{2} \log(1+e^{-2x})$ (as in 631b). Result: $-\frac{x}{2(1+e^{2x})} - \frac{1}{4} \log(1+e^{-2x}) + C$, or equivalently $\frac{x}{2} - \frac{x}{2(1+e^{2x})} - \frac{1}{4} \log(1+e^{2x}) + C$. $\square$

Exercise 720

from the dictaat

Determine (or show divergence): (a) $\int_0^\pi \cos t \sqrt{1+\cos^2 t} dt$; (b) $\int_0^1 \frac{\log(1+x)}{\sqrt{x}} dx$.

(a) Write $\cos^2 t = 1 - \sin^2 t$ and substitute $u = \sin t$, $du = \cos t dt$: the limits become $u(0) = 0$ and $u(\pi) = 0$, so the integral is $\int_0^0 \sqrt{2-u^2} du = 0$. (Symmetry: the integrand at $\pi-t$ is the negative of that at t .)

(b) Improper at 0 (the integrand $\sim \frac{x}{\sqrt{x}} = \sqrt{x}$ there, so it is actually bounded: convergent). $x = u^2$, $dx = 2u du$: $\int_0^1 2 \log(1+u^2) du$. Parts: $[2u \log(1+u^2)]_0^1 - \int_0^1 \frac{4u^2}{1+u^2} du = 2 \log 2 - 4 \int_0^1 \left(1 - \frac{1}{1+u^2}\right) du = 2 \log 2 - 4 + \pi$. $\square$

Exercise 722

from the dictaat

Reaction kinetics: $\frac{dx}{dt} = k(a-x)(b-x)$, $x(0) = 0$, $k > 0$. Solve in the form $kt = \dots$ for (a) $a = b$; (b) $a \neq b$.

(a) $\frac{dx}{(a-x)^2} = k dt$ gives $\frac{1}{a-x} = kt + C$; $x(0) = 0$: $C = \frac{1}{a}$. So $kt = \frac{1}{a-x} - \frac{1}{a} = \frac{x}{a(a-x)}$.

(b) Partial fractions: $\frac{1}{(a-x)(b-x)} = \frac{1}{b-a} \left(\frac{1}{a-x} - \frac{1}{b-x} \right)$. Integrating (for $x < a, b$): $\frac{1}{b-a} (-\log(a-x) + \log(b-x)) = kt + C$, and $x(0) = 0$ gives $C = \frac{1}{b-a} \log \frac{b}{a}$. Hence

$$kt = \frac{1}{b-a} \left(\log \frac{b-x}{a-x} - \log \frac{b}{a} \right) = \frac{1}{b-a} \log \frac{a(b-x)}{b(a-x)}.$$

□

Exercise 723

from the dictaat

Determine $\int \tan(1-x) \tan(1+x) dx$. Hint: Oefening 26(c), creatively.

From $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ we get $\tan A \tan B = 1 - \frac{\tan A + \tan B}{\tan(A+B)}$. With $A = 1-x$, $B = 1+x$,

$$A + B = 2:$$

$$\tan(1-x)\tan(1+x) = 1 - \frac{\tan(1-x) + \tan(1+x)}{\tan 2}.$$

Now $\int \tan(1-x)dx = \log|\cos(1-x)|$ (chain rule, inner derivative -1) and $\int \tan(1+x)dx = -\log|\cos(1+x)|$. So

$$\int \tan(1-x)\tan(1+x)dx = x - \frac{1}{\tan 2} \left(\log|\cos(1-x)| - \log|\cos(1+x)| \right) + C.$$

□

Exercise 724

from the dictaat

Euler's beta function $B_{p,q} = \int_0^1 x^{p-1}(1-x)^{q-1}dx$ ($p, q \geq 1$ integers). Prove: (a) $B_{p,q} = B_{q,p}$; (b) $B_{p+1,q} + B_{p,q+1} = B_{p,q}$; (c) $B_{p+1,q} = \frac{p}{p+q}B_{p,q}$; (d) $B_{p+1,q} = \frac{p}{p+q}B_{p,q}$; (e) a closed formula.

(a) Substitute $x = 1 - u$ ($dx = -du$, limits swap): $\int_0^1 (1-u)^{p-1}u^{q-1}du = B_{q,p}$.

(b) $x^p(1-x)^{q-1} + x^{p-1}(1-x)^q = x^{p-1}(1-x)^{q-1}(x + (1-x)) = x^{p-1}(1-x)^{q-1}$; integrate.

(c) Parts with $f' = (1-x)^{q-1}$, $f = -\frac{1}{q}(1-x)^q$, $g = x^p$: $B_{p+1,q} = \left[-\frac{1}{q}x^p(1-x)^q \right]_0^1 + \frac{p}{q} \int_0^1 x^{p-1}(1-x)^q dx = 0 + \frac{p}{q}B_{p,q+1}$.

(d) By (c), $B_{p,q+1} = \frac{q}{p+q}B_{p,q}$; substitute in (b): $B_{p+1,q}(1 + \frac{q}{p}) = B_{p,q}$, so $B_{p+1,q} = \frac{p}{p+q}B_{p,q}$.

(e) $B_{1,q} = \int_0^1 (1-x)^{q-1}dx = \frac{1}{q}$. Applying (d) repeatedly: $B_{p,q} = \frac{p-1}{p+q-1} \cdot \frac{p-2}{p+q-2} \cdots \frac{1}{q+1} \cdot B_{1,q} = \frac{(p-1)!(q-1)!}{(p+q-1)!}$. (Check: $B_{3,4} = \frac{2!3!}{6!} = \frac{1}{60}$.) □

NOW YOU 703, 704, 707, 708, 709, 711, 712, 714, 715, 716, 717, 721 · solutions in Appendix S

703 $\log(49 + \sqrt{49^2 - 1}) + \log(49 - \sqrt{49^2 - 1})$. **704** $f(x) = \log \frac{e^x + 1}{e^x - 1}$: $f \circ f(x) = x$. **707** $\sin$ and $\arctan$ share a tangent line at 0. **708** P_4 of $\frac{1+x^2}{1-x}$. **709** P_5 of $\arctan x$. **711** Build $\frac{-3}{x+2}$ and $\frac{1-x}{2+x}$ by composing $i(x) = \frac{1}{x}$, $p(x) = 1+x$, $m(x) = -x$, $t(x) = 3x$. **712** $\sin(\arctan x)$. **714–716** $\operatorname{arcosh} x = \log(x \pm \sqrt{x^2 - 1})$: derivation, the two branches as mirror images, the derivative in two ways. **717** $\tanh$ and $\operatorname{artanh} x = \frac{1}{2} \log \frac{1+x}{1-x}$. **721** $\int_0^1 \frac{1-x^n}{1-x} dx = \sum_{k=1}^n \frac{1}{k}$.

B (EXTRA) FUNCTION INVESTIGATION
§7.1 · the six steps and the sign table

(1) Domain. (2) Symmetry (even/odd) and a more convenient form of the formula. (3) Intercepts: $f(x) = 0$ and $f(0)$. (4) Behaviour at the edges of the domain, including $\pm\infty$: limits, asymptotes (not every edge gives an asymptote). (5) f' : horizontal tangents where $f' = 0$, corners where f' does not exist; list the critical points. (6) f'' : convex ($f'' > 0$) or concave, inflection points where f'' changes sign. Collect everything in a *tekenschema* (sign table) for f, f', f'' , check for contradictions (an alarm signal means an error somewhere), then sketch, smuggling in no features you did not find. Model (dictaat): $f(x) = \frac{2-x}{2+x} = \frac{4}{2+x} - 1$: domain $x \neq -2$; zero $x = 2$, $f(0) = 1$; asymptotes $y = -1$ and $x = -2$; $f' = -\frac{4}{(2+x)^2} < 0$ everywhere; $f'' = \frac{8}{(2+x)^3}$: concave left of -2 , convex right of it, no inflection point.

Exercise 726(d)

from the dictaat

Investigate $f: x \mapsto \frac{(x+1)^3}{x^2}$ and sketch the graph.

(1) Domain $x \neq 0$. (2) Division gives $f(x) = x + 3 + \frac{3}{x} + \frac{1}{x^2}$. (3) Zero: $x = -1$ (triple zero: the graph crosses the axis with a horizontal tangent); no y-intercept. (4) $x \rightarrow 0^\pm$: numerator $\rightarrow 1$, denominator $\rightarrow 0^\pm$: $f \rightarrow +\infty$ on both sides, vertical asymptote $x = 0$. $x \rightarrow \pm\infty$: $f(x) - (x+3) = \frac{3}{x} + \frac{1}{x^2} \rightarrow 0$: oblique asymptote $y = x + 3$, approached from above for $x > 0$ and from below for $x < 0$ (sign of $\frac{3}{x}$). (5) $f'(x) = \frac{(x+1)^2(x-2)}{x^3}$: zero at $x = -1$ and $x = 2$. Sign: for $x < 0$, $(x-2)/x^3 > 0$: increasing on $(-\infty, 0)$ (with a flat point at -1); on $(0, 2)$ negative: decreasing; on $(2, \infty)$ positive: increasing. Local minimum $f(2) = \frac{27}{4}$. (6) $f''(x) = \frac{6(x+1)}{x^4}$: negative for $x < -1$ (concave), positive for $x > -1$, $x \neq 0$ (convex); inflection point $(-1, 0)$. Sign table (values at $-\infty, -1, 0^-, 0^+, 2, +\infty$): f : $-\infty, 0, +\infty | +\infty, \frac{27}{4}, +\infty$; f' : $+, 0, + | -, 0, +$; f'' : $-, 0, + | +, +, +$.

Sketch: from the lower left along $y = x + 3$ (below it), concave, rising through $(-1, 0)$ with a horizontal tangent, then convex and shooting up to $+\infty$ at $x = 0^-$; on the right, coming down from $+\infty$ at 0^+ to the minimum $(2, \frac{27}{4})$, then rising along $y = x + 3$ (above it). $\square$

Exercise 727

from the dictaat

Four students investigated the same differentiable f on $(0, \infty)$ with the single zero $f(3) = 0$, $\lim_{x \rightarrow \infty} f(x) = -2$ and exactly one inflection point; they claim (a) $\lim_{0^+} f = -\infty$, $f'(3) = 0$, inflection at 6; (b) $\lim_{0^+} f = -4$, $f'(3) > 0$, inflection at 1; (c) $\lim_{0^+} f = +4$, $f'(3) < 0$, inflection at 3; (d) $\lim_{0^+} f = +\infty$, $f'(3) < 0$, inflection at 1. Who can draw a consistent graph?

(b) impossible: $f'(3) > 0$ means f goes from negative to positive at 3; then $f > 0$ just after 3, but $f \rightarrow -2 < 0$, so f would have to cross 0 again, a second zero.

(d) impossible: f decreases from $+\infty$ through $(3, 0)$ to -2 . On $(1, \infty)$ (after the only inflection point) f would be either concave or convex. Concave and decreasing with $f' < 0$ forces $f \rightarrow -\infty$ (the slope keeps getting more negative), contradicting the limit -2 . So f is convex on $(1, \infty)$ and concave on $(0, 1)$; but a concave function on $(0, 1)$ stays below its tangent lines, hence is bounded above near 0, contradicting $f \rightarrow +\infty$.

(a) consistent: f rises from $-\infty$ to a maximum $f(3) = 0$ (touching the axis, $f'(3) = 0$), concave there, then decreases to -2 , which requires it to become convex eventually: one inflection point at some $x > 3$, e.g. 6. $f < 0$ except at 3: one zero. $\checkmark$

(c) consistent: f decreases from 4 through $(3, 0)$ to -2 like a falling S-curve: concave on $(0, 3)$, convex on $(3, \infty)$, inflection at 3; the value at the inflection need not be the midpoint of 4 and -2 . $\checkmark$ $\square$

Exercise 728

from the dictaat

Find all x with $\frac{1}{1-x^2} > \frac{1}{x}$. Check carefully.

Domain: $x \neq 0, \pm 1$. Never multiply by x or $1-x^2$ (unknown signs); bring everything to one side:

$$\frac{1}{1-x^2} - \frac{1}{x} = \frac{x - (1-x^2)}{x(1-x^2)} = \frac{x^2 + x - 1}{x(1-x^2)} > 0.$$

Zeros of the numerator: $x = \frac{-1 \pm \sqrt{5}}{2}$, i.e. $\alpha \approx -1.618$ and $\beta \approx 0.618$. Sign table with the points $\alpha, -1, 0, \beta, 1$: numerator $+$ on $(-\infty, \alpha)$ and (β, ∞) , $-$ between; denominator $x(1-x)(1+x)$: $+$ on

$(-\infty, -1)$, $-$ on $(-1, 0)$, $+$ on $(0, 1)$, $-$ on $(1, \infty)$. Quotient positive on $(-\infty, \alpha)$, $(-1, 0)$ and $(\beta, 1)$.
 Answer: $x < \frac{-1 - \sqrt{5}}{2}$ or $-1 < x < 0$ or $\frac{\sqrt{5} - 1}{2} < x < 1$. Check $x = -2$: $\frac{1}{-3} = -0.33 > -0.5$ ✓;
 $x = -\frac{1}{2}$: $\frac{4}{3} > -2$ ✓; $x = \frac{1}{2}$: $\frac{4}{3} > 2$ ✗; $x = 0.8$: $\frac{2}{0.36} = 2.8 > 1.25$ ✓; $x = 2$: $-\frac{1}{3} > \frac{1}{2}$ ✗. $\square$

Exercise 729

from the dictaat

Prove $\log x \leq x - 1$ for all $x > 0$.

Let $g(x) = x - 1 - \log x$ on $(0, \infty)$. $g(1) = 0$ and $g'(x) = 1 - \frac{1}{x}$, negative on $(0, 1)$ and positive on $(1, \infty)$:
 g decreases to its minimum $g(1) = 0$ and increases afterwards. So $g(x) \geq 0$ with equality only at $x = 1$:
 the graph of g touches the x -axis at 1 without crossing it, i.e. $\log x \leq x - 1$. $\square$

NOW YOU 725, 726(a),(b),(c),(e)–(j), 730 · solutions in Appendix S

725 Investigate $\sqrt{x}$, e^x , $\log x$, $\sin x$, $\cos x$, $\tan x$. **726** (a) $\sqrt{9 - x^2}$; (b) $\frac{x}{\sqrt{1 - x^2}}$; (c) $e^{1 - x^2}$; (e) $\log \frac{2x}{(x - 4)^2}$; (f)
 $(1 + \frac{1}{x})e^{-x}$; (g) $e^{-1/x}$; (h) $\log(x - \frac{1}{x})$; (i) $e^{1 - \frac{1}{1+x}}$; (j) $\arcsin(1 - \sqrt{x})$. **730** Prove $\log x \leq 2(\sqrt{x} - 1)$.

Appendix S

Solutions to the NOW YOU exercises

Contents

- A** Chapter 0
- B** Chapter 1
- C** Chapter 2
- D** Chapter 3
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Planning

Every Oefening of Hoofdstuk 0–7 that is not worked in the main text is solved here, in the order of the dictaat. Compare line by line, and reflect: could you have seen the answer coming? Would another approach work?

A CHAPTER 0

1 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. 61: 1, 61 (prime). 62: 1, 2, 31, 62. 63: 1, 3, 7, 9, 21, 63. 64: 1, 2, 4, 8, 16, 32, 64. 65: 1, 5, 13, 65. Only $64 = 8^2$ is a square (a square has an odd number of divisors; 64 has seven).

3 $\frac{1}{6} - \frac{1}{7} = \frac{7-6}{42} = \frac{1}{42}$. In general $\frac{1}{p} - \frac{1}{q} = \frac{q-p}{pq}$, which equals $\frac{1}{pq}$ exactly when $q - p = 1$: $q = p + 1$ (and $p, q \neq 0$), e.g. $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

6 $x \mapsto x^2$: a parabola through $(-1, 1), (0, 0), (1, 1), (2, 4)$, symmetric in the y-axis, minimum 0 at $x = 0$. $x \mapsto \sqrt{x}$ on $[0, \infty)$: starts at $(0, 0)$ with a vertical tangent, through $(1, 1), (4, 2)$, increasing and bending downwards (it is the mirror image of the right half of the parabola in the line $y = x$).

7 $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x \geq 0$ (the closed dot in the dictaat's graph is at $(0, 1)$; if it were on the lower branch, use $x \leq 0$ and $x > 0$).

8 $y = 3 - \frac{1}{2+x} \Rightarrow \frac{1}{2+x} = 3 - y \Rightarrow 2 + x = \frac{1}{3-y}$ (for $y \neq 3$) $\Rightarrow x = \frac{1}{3-y} - 2$. Swapping letters: $f_{\text{inv}}(x) = \frac{1}{3-x} - 2$; f is bijective from $\mathbb{R} \setminus \{-2\}$ onto $\mathbb{R} \setminus \{3\}$.

12 $1 - \cos^2 x = \sin^2 x$, so $\sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = |\sin x|$. Sanity check $x = -\frac{\pi}{6}$: $\cos^2 = \frac{3}{4}$, $\sqrt{1 - \frac{3}{4}} = \frac{1}{2}$, and $|\sin(-\frac{\pi}{6})| = |-\frac{1}{2}| = \frac{1}{2}$ ✓ (the answer $\sin x = -\frac{1}{2}$ would be wrong).

13 $(p \circ p)(x) = 1 + (1 + x) = 2 + x$; $(p \circ q)(x) = 1 + x^2$; $(q \circ p)(x) = (1 + x)^2$; $(q \circ q)(x) = x^4$.

14 Many: $f(x) = x$; $f(x) = a - x$ for any constant a ; $f(x) = \frac{1}{x}$ and $f(x) = \frac{a}{x}$ ($x \neq 0$); $f(x) = \frac{x+1}{x-1}$ (check: $f(f(x)) = \frac{\frac{x+1}{x-1}+1}{\frac{x+1}{x-1}-1} = \frac{2x}{2} = x$), more generally $f(x) = \frac{ax+b}{cx-a}$. Such functions are their own inverse: their graph is symmetric in the line $y = x$.

16 (a) Even: $x^2, x^4, |x|, \cos x$, constants. Odd: $x, x^3, \frac{1}{x}, \sin x$. (b) $x + 1, e^x, x^2 + x$. (c) If f is both then $f(x) = f(-x) = -f(x)$, so $f(x) = 0$: only the zero function (on any domain symmetric about 0).

18 If 0 is in the domain of an odd f , then $f(0) = -f(-0) = -f(0)$, so $f(0) = 0$: the claim is right. It fails only when 0 is not in the domain: $f(x) = \frac{1}{x}$ is odd and has no value at 0. For even functions nothing of the kind

holds: $f(0)$ can be anything ($\cos 0 = 1$). (What is true: a differentiable even function has $f'(0) = 0$, since f' is odd.)

20(b) Order the divisor: $2x^3 + x$. $6x^4 \div 2x^3 = 3x$; subtract $3x(2x^3 + x) = 6x^4 + 3x^2$: remainder $-4x^3 - 4x^2 + 6x - 12$. $-4x^3 \div 2x^3 = -2$; subtract $-2(2x^3 + x) = -4x^3 - 2x$: remainder $-4x^2 + 8x - 12$, of degree $2 < 3$. So $\frac{6x^4 - 4x^3 - x^2 + 6x - 12}{2x^3 + x} = 3x - 2 + \frac{-4x^2 + 8x - 12}{2x^3 + x}$. Check: $(3x - 2)(2x^3 + x) + (-4x^2 + 8x - 12) = 6x^4 + 3x^2 - 4x^3 - 2x - 4x^2 + 8x - 12 = 6x^4 - 4x^3 - x^2 + 6x - 12 \checkmark$.

21(b) The common factor is $5x^2(x - 14)$: $5x^2(x - 14)(9x^2 - 10x + 6)$. The quadratic has discriminant $100 - 216 < 0$, so it has no real zeros and cannot be factorised further over $\mathbb{R}$. Check: $5x^2(x - 14) \cdot 9x^2 = 45x^4(x - 14)$, etc.

21(c) $x - x^3 + x^5 - x^7 = x(1 - x^2 + x^4 - x^6) = x((1 - x^2) + x^4(1 - x^2)) = x(1 - x^2)(1 + x^4) = x(1 - x)(1 + x)(1 + x^4)$. Check: $(1 - x^2)(1 + x^4) = 1 + x^4 - x^2 - x^6 \checkmark$. ($1 + x^4$ has no real zeros; over $\mathbb{R}$ it still splits as $(x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)$, but that is beyond “simple” factorising.)

22 Square with diagonal 1: the diagonal makes angles $\frac{\pi}{4}$ with the sides; the sides s satisfy $s^2 + s^2 = 1$, so $s = \frac{1}{\sqrt{2}} = \frac{1}{2}\sqrt{2}$. Equilateral triangle with side 1: all angles $\frac{\pi}{3}$; its altitude splits it into two right triangles with angles $\frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$, hypotenuse 1, short side $\frac{1}{2}$ and altitude $\sqrt{1 - \frac{1}{4}} = \frac{1}{2}\sqrt{3}$. These give the table of Oefening 23.

24 (a) The point at angle t is $(\cos t, \sin t)$: $\cos t$ is its horizontal coordinate (the segment on the x -axis), $\sin t$ its vertical coordinate. The vertical tangent line $x = 1$ meets the extended radius at height h ; the triangle $(0, 0), (1, 0), (1, h)$ is similar to $(0, 0), (\cos t, 0), (\cos t, \sin t)$, so $h/1 = \sin t / \cos t = \tan t$. (b) $\sin(\pi + t) = -\sin t$, $\sin(\pi - t) = +\sin t$, $\cos(\pi + t) = -\cos t$, $\cos(\pi - t) = -\cos t$, $\tan(\pi + t) = +\tan t$, $\tan(\pi - t) = -\tan t$ (rotating by π negates both coordinates; reflecting in the y -axis negates only the first). (c) Reflection in the x -axis ($t \mapsto -t$) negates the y -coordinate and keeps the x -coordinate: $\sin(-t) = -\sin t$, $\cos(-t) = \cos t$. Reflection in the line $y = x$ ($t \mapsto \frac{\pi}{2} - t$) swaps the coordinates: $\sin(\frac{\pi}{2} - t) = \cos t$. Pythagoras in the triangle with legs $\cos t, \sin t$ and hypotenuse 1: $\sin^2 t + \cos^2 t = 1$.

27 (a) The velocity of a point moving on a circle is tangent to the circle, and the tangent is perpendicular to the radius OP (a chord PQ with $Q \rightarrow P$ becomes perpendicular to OP by symmetry of the isosceles triangle OPQ). (b) A vector of length 1 at P , perpendicular to OP , pointing counterclockwise. (c) Rotating $OP = (\cos t, \sin t)$ over $+\frac{\pi}{2}$ gives $(-\sin t, \cos t)$: the endpoint is $(-\sin t, \cos t)$. (d) Velocity is the derivative of position: $\frac{d}{dt}(\cos t, \sin t) = (-\sin t, \cos t)$, so $\cos' t = -\sin t$ and $\sin' t = \cos t$.

29 (a) ${}^h\log x = \frac{{}^s\log x}{{}^s\log h}$ for all $x > 0$, so ${}^s\log x = {}^s\log h \cdot {}^h\log x$: the two functions differ by the constant factor ${}^s\log h$. (b) The factor is ${}^s\log h$ (equivalently $1/{}^h\log g$). (c) Taking $x = g$: ${}^h\log g = \frac{{}^s\log g}{{}^s\log h} = \frac{1}{{}^s\log h}$, so ${}^s\log h \cdot {}^h\log g = 1$. True statements in the list: “you can always compute it if you know g and h ”, “it exists provided g and h are positive (and $\neq 1$, so that they can be bases)”, “it is 1”. It does not depend on which base the notation $\log$ refers to, and it is not 0.

B CHAPTER 1

102 $1 + 2i \mapsto (1, 2)$; $3i \mapsto (0, 3)$; $-1 \mapsto (-1, 0)$; $-4 - i \mapsto (-4, -1)$; $2 - i \mapsto (2, -1)$; $4 \mapsto (4, 0)$.

106 $\arg z$ is the angle between the positive real axis and the line through 0 and z ; for $z = 0$ there is no such line (every direction fits). Equivalently $\cos \varphi = a/r$ needs $r = |z| \neq 0$.

107 With $z = a + bi$, $\bar{z} = a - bi$: (a) $\operatorname{Re} \bar{z} = a = \operatorname{Re} z$. (b) $\operatorname{Im} \bar{z} = -b = -\operatorname{Im} z$. (c) $|\bar{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|$. (d) Reflection in the real axis sends the angle φ to $-\varphi$, so $\arg \bar{z} = -\arg z$ (for a negative real z both arguments are π , because of the convention $(-\pi, \pi]$). (e) $\bar{\bar{z}} = \overline{a - bi} = a + bi = z$.

$$108 \quad \overline{1+i} = 1-i; \overline{3i} = -3i; \overline{-20} = -20; \overline{-1-i} = -1+i.$$

$$109 \quad z = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm 2i}{2} = -1 \pm i. \text{ Check: } (-1+i)^2 + 2(-1+i) + 2 = (1-2i-1) - 2 + 2i + 2 = 0 \\ \checkmark.$$

110 Conjugate both sides of $az^2 + bz + c = 0$: by Stelling 1.5, $\overline{az^2 + bz + c} = \bar{a}\bar{z}^2 + \bar{b}\bar{z} + \bar{c} = a\bar{z}^2 + b\bar{z} + c$ since a, b, c are real ($\bar{a} = a$), and $\bar{0} = 0$. So $\bar{z}$ is a solution. (With the abc-formula: the solutions are $\frac{-b \pm \sqrt{4ac-b^2}}{2a}$ when $b^2 < 4ac$, each other's conjugates.)

$$111(\text{b}), (\text{c}) \quad (\text{b}) z+w = i+1-i = 1; zw = i(1-i) = i-i^2 = 1+i. (\text{c}) z+w = -8+12i; zw = (-4+6i)^2 = 16-48i+36i^2 = -20-48i.$$

$$112 \quad (a+bi)(c+di) = ac + adi + bci + bdi^2 = ac + (ad+bc)i - bd = (ac-bd) + (ad+bc)i, \text{ which is Definition 1.4.}$$

$$113 \quad z = a+bi, w = c+di. (\text{a}) \overline{z+w} = \overline{(a+c) + (b+d)i} = (a+c) - (b+d)i = (a-bi) + (c-di) = \bar{z} + \bar{w}. (\text{c}) z\bar{z} = (a+bi)(a-bi) = a^2 - abi + abi - b^2i^2 = a^2 + b^2 = |z|^2. (\text{d}) z + \bar{z} = (a+bi) + (a-bi) = 2a = 2\operatorname{Re} z.$$

$$115 \quad z = a+bi, w = c+di, v = e+fi. (\text{a}) z+w = (a+c) + (b+d)i = (c+a) + (d+b)i = w+z. (\text{b}) zw = (ac-bd) + (ad+bc)i \text{ and } wz = (ca-db) + (cb+da)i: \text{ equal because real multiplication and addition are commutative. } (\text{c}) z+(w+v) = (a+(c+e)) + (b+(d+f))i = ((a+c)+e) + ((b+d)+f)i = (z+w)+v. (\text{d}) wv = (ce-df) + (cf+de)i, \text{ so } z(wv) = (a(ce-df) - b(cf+de)) + (a(cf+de) + b(ce-df))i = (ace - adf - bcf - bde) + (acf + ade + bce - bdf)i; \text{ and } zw = (ac-bd) + (ad+bc)i, \text{ so } (zw)v = ((ac-bd)e - (ad+bc)f) + ((ac-bd)f + (ad+bc)e)i, \text{ the same. } (\text{e}) z(w+v) = (a(c+e) - b(d+f)) + (a(d+f) + b(c+e))i = (ac-bd) + (ad+bc)i + (ae-bf) + (af+be)i = zw + zv.$$

116 $(a+bi) + (c+di) = (a+c) + (b+d)i$ corresponds to the point $(a+c, b+d) = (a, b) + (c, d)$: the sum is the fourth vertex of the parallelogram with vertices 0, z , w .

118 $|iz| = |i||z| = |z|$ and $\arg(iz) = \arg i + \arg z = \arg z + \frac{\pi}{2}$ (Stelling 1.6). So multiplying by i rotates the plane over $\frac{\pi}{2}$ counterclockwise about 0. (Multiplying by i twice is multiplying by -1 : a rotation over π .)

$$119 \quad e^{i\pi} = \cos \pi + i \sin \pi = -1. 4e^{i\pi/3} = 4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right) = 4\left(\frac{1}{2} + \frac{1}{2}\sqrt{3}i\right) = 2 + 2\sqrt{3}i.$$

$$121 \quad |e^{i\varphi}| = \sqrt{\cos^2 \varphi + \sin^2 \varphi} = 1, \text{ so every } e^{i\varphi} \text{ is at distance 1 from 0: on the unit circle, at angle } \varphi.$$

122 The six solutions $e^{ik\pi/3}$ ($k = 0, \dots, 5$) are the vertices of a regular hexagon on the unit circle, starting at 1: $1, \frac{1}{2} + \frac{1}{2}\sqrt{3}i, -\frac{1}{2} + \frac{1}{2}\sqrt{3}i, -1, -\frac{1}{2} - \frac{1}{2}\sqrt{3}i, \frac{1}{2} - \frac{1}{2}\sqrt{3}i$. Conjugation reflects in the real axis: $\overline{e^{ik\pi/3}} = e^{-ik\pi/3} = e^{i(6-k)\pi/3}$, again one of the six (the hexagon is symmetric in the real axis). This is Oefening 110 for the polynomial $z^6 - 1$ with real coefficients.

C CHAPTER 2

202 (a) By the quotient rule, $\lim_{x \rightarrow 0} \frac{e^x}{1+x} = \frac{e^0}{1+0} = 1$ and $\lim_{x \rightarrow 1} \frac{e^{x-1}}{x} = \frac{e^0}{1} = 1$. (b) Substitution rule (Stelling 2.3e) with $g(x) = x - b$: as $x \rightarrow a$, $g(x) \rightarrow a - b$, so $\lim_{x \rightarrow a} f(x - b) = \lim_{x \rightarrow a} f(g(x)) = \lim_{u \rightarrow a-b} f(u)$. Applied to (a): with $b = 1$, $\lim_{x \rightarrow 1} \frac{e^{x-1}}{x} = \lim_{x \rightarrow 0} \frac{e^x}{x+1}$, so the second limit is the first one shifted; both equal 1.

203 (a) The constant function $g(x) = k$ has limit k , so by the product rule $\lim k f(x) = k\varphi$. (b) $\lim_{x \rightarrow a} x = a$ (the identity), so by the product rule $\lim x^i = a^i$, by (a) $\lim a_i x^i = a_i a^i$, and by the sum rule $\lim P(x) = \sum a_i a^i = P(a)$. (c) Write $k = p/q$ with p, q positive integers. $\lim f(x)^p = \varphi^p$ by the product rule applied p times. For the root, the substitution rule with the continuous function $y \mapsto y^{1/q}$ gives $\lim f(x)^{1/q} = \lim_{y \rightarrow \varphi} y^{1/q} = \varphi^{1/q}$ (for even q we need $f \geq 0$). Combining, $\lim f(x)^{p/q} = \lim (f(x)^{1/q})^p = \varphi^{p/q}$. The condition $k > 0$ avoids dividing by φ , which might be 0.

204 (a) For $x > 0$, $h(x) = |x| = x \rightarrow 0$; for $x < 0$, $h(x) = -x \rightarrow 0$. Left and right limits are both 0, so $\lim_{x \rightarrow 0} |x| = 0$. (b) $g = -h$, so by (a) and the rule for constants its limit is $-0 = 0$. (c) For $x \neq 0$: $-1 \leq \sin \frac{1}{x} \leq 1$; multiplying by $|x| \geq 0$: $-|x| \leq |x| \sin \frac{1}{x} \leq |x|$, and $|x| \sin \frac{1}{x} = \pm x \sin \frac{1}{x}$, so in all cases $-|x| \leq x \sin \frac{1}{x} \leq |x|$. By the squeeze theorem $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$. (d) The graph of f oscillates between the lines $y = x$ and $y = -x$, touching them, with oscillations that become infinitely fast near 0; g and h are the two lines (a V and an upside-down V).

206 Put $x = a + h$. As $h \rightarrow 0$, $x \rightarrow a$, and $\frac{f(a+h)-f(a)}{h} = \frac{f(x)-f(a)}{x-a}$. By the substitution rule (Oefening 202b with $b = -a$) the two limits are equal.

207 $\frac{f}{g} = f \cdot g^{-1}$. Product rule: $(f \cdot g^{-1})' = f'g^{-1} + f(g^{-1})'$. Chain rule with outer function $y \mapsto y^{-1}$: $(g^{-1})' = -g^{-2}g'$. So $\left(\frac{f}{g}\right)' = \frac{f'}{g} - \frac{fg'}{g^2} = \frac{f'g - fg'}{g^2}$.

208 $(f \circ g)' = (f' \circ g) \cdot g'$: the derivative of the composition is the derivative of f , composed with g , times the derivative of g .

213 $\tilde{g}(b) - \tilde{g}(a) = (g(b) + c) - (g(a) + c) = g(b) - g(a)$. Whichever primitive you use, the constant cancels; writing “+c” in a definite integral is pointless.

214 (a) $g(x+dx) = \int_0^{x+dx} f(t) dt$. (b) By the cut-up rule, $dg(x) = \int_0^{x+dx} f(t) dt - \int_0^x f(t) dt = \int_x^{x+dx} f(t) dt$: in the figure, one extra strip of width dx at x . (c) On the infinitely thin strip f is constant ($= f(x)$), so $dg(x) = f(x) dx$, i.e. $\frac{dg(x)}{dx} = f(x)$: differentiating the area function gives back f (the fundamental theorem).

216 For instance: (c) reversing the limits reverses the direction of the strips, so the sign flips; (e) shifting the graph c to the right and the limits with it leaves the strips unchanged; (g) for an odd function the strips left of 0 are the mirror images of those right of 0 with opposite sign; (i) a single point has width 0 and contributes no area; (k) taking $|f|$ turns negative strips positive, which can only increase the sum.

217 The dictaat's figure is not reproduced here; the matching rule is: $F(x) = \int_a^x f(t) dt$ starts with $F(a) = 0$, increases exactly where $f > 0$, decreases where $f < 0$, has a horizontal tangent where $f = 0$ (maxima where f changes from + to –, minima the other way), is steepest where $|f|$ is largest, and is linear where f is constant. Conversely f is the slope of F . Check each pair against these points.

220 (a) Socialist: wealth evenly spread, so the poorest fraction x owns fraction x : $v(x) = x$, the diagonal from (0,0) to (1,1). Capitalist: the poorest own almost nothing until near $x = 1$, where v shoots up to 1: $v(0) = 0$, $v(1) = 1$, v increasing (more people own more), $v(x) \leq x$, with slope increasing (convex). (b) $G = 2 \int_0^1 (x - v(x)) dx$ is twice the area between the diagonal and the graph of v (the integrand is ≥ 0); for the socialist graph this area is 0, for the capitalist graph it is almost the whole triangle under the diagonal. (c) $0 \leq G \leq 1$: $G = 0$ when $v(x) = x$ (complete equality); $G \rightarrow 1$ when $v(x) = 0$ for all $x < 1$ (one person owns everything), since $2 \int_0^1 x dx = 1$.

D CHAPTER 3

304 $P_2(x) = P_1(x) + \frac{1}{2}f''(a)(x-a)^2$: the quadratic correction has the sign of $f''(a)$. For e^x ($f'' > 0$) it adds to P_1 : the linear value 1.5 was too small; for $\log(1+x)$ and $\sqrt[3]{x}$ ($f'' < 0$) it subtracts: the linear values were too large. This is the precise form of the hint in 302.

305 $P'_3(x) = f'(a) + f''(a)(x-a) + \frac{1}{2}f'''(a)(x-a)^2$, $P'_3(x) = f''(a) + f'''(a)(x-a)$, $P'''_3(x) = f'''(a)$. At $x = a$ all terms with a factor $(x-a)$ vanish: $P_3(a) = f(a)$, $P'_3(a) = f'(a)$, $P''_3(a) = f''(a)$, $P'''_3(a) = f'''(a)$.

306 The k -th derivative of $(x-a)^m$ is 0 at $x = a$ unless $m = k$, when it is $k!$ (for $m < k$ the term is differentiated away; for $m > k$ a factor $(x-a)^{m-k}$ remains). So $P_n^{(k)}(a) = \frac{f^{(k)}(a)}{k!} \cdot k! = f^{(k)}(a)$ for $0 \leq k \leq n$.

307 All derivatives of e^x at 0 are 1: $P_9 = \sum_{k=0}^9 \frac{x^k}{k!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6}{720} + \frac{x^7}{5040} + \frac{x^8}{40320} + \frac{x^9}{362880}$.

sin: derivatives cycle $\cos, -\sin, -\cos, \sin$, at 0: $1, 0, -1, 0, \dots$: $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!}$. $\cos$: $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!}$.
 $\log(1+x)$: $f^{(k)}(0) = (-1)^{k-1}(k-1)!$ (313a), so $c_k = \frac{(-1)^{k-1}}{k}$: $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \frac{x^6}{6} + \frac{x^7}{7} - \frac{x^8}{8} + \frac{x^9}{9}$.

310 (a) $f(0) = 4$, $f'(x) = 6x^2 - 6x$ so $f'(0) = 0$, $f''(x) = 12x - 6$ so $f''(0) = -6$, $f''' = 12$, higher derivatives 0. $P_0 = P_1 = 4$; $P_2 = 4 - 3x^2$; $P_n = 4 - 3x^2 + 2x^3 = f$ for all $n \geq 3$. (b) Yes. f and $P_{n,a}$ are both polynomials of degree $\leq n$ (expand f in powers of $x - a$: $f(x) = \sum_{k=0}^n c_k(x-a)^k$); by Oefening 306 both have the same derivatives of order $0, \dots, n$ at a , and the derivatives at a determine the coefficients $c_k = f^{(k)}(a)/k!$. So $f = P_{n,a}$ exactly.

311 If f is even then f' is odd, f'' even, etc.; an odd function (defined at 0) vanishes at 0, so all odd-order derivatives of an even f vanish at 0: its Taylor approximations at 0 contain only even powers (cos). For odd f only odd powers occur (sin, tan, $\log \frac{1+x}{1-x}$).

312 With $T = \tan$, $T' = 1/\cos^2$, $T'' = 2\sin/\cos^3$, $T''' = (2\cos^2 + 6\sin^2)/\cos^4 = (2 + 4\sin^2)/\cos^4$, and so on; at 0: $0, 1, 0, 2, 0, 16$. Result $x + \frac{1}{3}x^3 + \frac{2}{15}x^5$; the organised way is Oefening 317 (worked in the text).

313 (a) $\frac{d^k}{dx^k} \log(1+x) = \frac{(-1)^{k-1}(k-1)!}{(1+x)^k}$ (each differentiation multiplies by $-k/(1+x)$), so at 0: $(-1)^{k-1}(k-1)!$, i.e. $0!, -1!, 2!, -3!, \dots$. For $\log(1-x)$ the chain rule gives an extra factor -1 each time: $\frac{d^k}{dx^k} \log(1-x) = -\frac{(k-1)!}{(1-x)^k}$, at 0: $-(k-1)!$, all negative. (b) $c_k = -\frac{1}{k}$: $\log(1-x) \approx -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5}$. (c) Substituting $-x$ in $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$ gives $-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5}$: the same (even powers keep their sign, odd powers flip).

315 Substitute $u = -x^2$ in $\sqrt{1+u} = 1 + \frac{1}{2}u - \frac{1}{8}u^2 + \dots$: $\sqrt{1-x^2} \approx 1 - \frac{1}{2}x^2 - \frac{1}{8}x^4$. At $x = \frac{1}{2}$: $\sqrt{1 - \frac{1}{4}} = \frac{1}{2}\sqrt{3} \approx 1 - \frac{1}{8} - \frac{1}{128} = \frac{128-16-1}{128} = \frac{111}{128}$ (≈ 0.8672 ; true value 0.8660).

318 $u = -\frac{1}{2}x^2 + \frac{1}{24}x^4 + \text{h.o.t.}$ has lowest order 2, so $u^2 = \frac{1}{4}x^4 + \text{h.o.t.}$ has lowest order 4 and $u^3, u^4, \dots$ have order ≥ 6 . In $\log(1+u) = u - \frac{1}{2}u^2 + \frac{1}{3}u^3 - \dots$ everything from u^3 on is beyond order 4, and within u^2 only $\frac{1}{4}x^4$ matters. So the terms kept, $u - \frac{1}{2}u^2$ with u up to x^4 , are exactly all contributions of order ≤ 4 .

321 $u = x - \frac{1}{6}x^3$ tends to 0, so substitute in $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \frac{u^4}{24} + \frac{u^5}{120}$. Up to order 5: $u^2 = x^2 - \frac{1}{3}x^4$, $u^3 = x^3 - \frac{1}{2}x^5$, $u^4 = x^4$, $u^5 = x^5$. Sum: $1 + x - \frac{1}{6}x^3 + \frac{1}{2}x^2 - \frac{1}{6}x^4 + \frac{1}{6}x^3 - \frac{1}{12}x^5 + \frac{1}{24}x^4 + \frac{1}{120}x^5 = 1 + x + \frac{1}{2}x^2 - \frac{1}{8}x^4 - \frac{3}{40}x^5$. (Since $x - \frac{1}{6}x^3$ is $\sin x$ up to order 4, this agrees with Oefening 320a up to x^4 .)

322 (a) $\log \frac{1}{2} = \log(1+x)$ with $x = -\frac{1}{2}$: $P_n(-\frac{1}{2}) = -\sum_{k=1}^n \frac{1}{k2^k}$: $-0.5, -0.625, -0.6667, -0.6823, -0.6885, -0.6911, -0.6922, -0.6927, -0.6929, -0.6931$ (for $n = 1, \dots, 10$) against $\log \frac{1}{2} = -0.693147$: degree 10 is needed for the first three decimals (0.693). (b) $\log 0 = \log(1+(-1))$ would need $x = -1$: $P_n(-1) = -\sum_{k=1}^n \frac{1}{k}$, the harmonic sums, which go to $-\infty$ very slowly; no degree gives a usable number (and $\log 0$ does not exist). (c) $\log 1.9$: $x = 0.9$ is close to 1, so the terms $\frac{0.9^k}{k}$ shrink slowly and many dozens of terms are needed for three decimals. $\log 2.1$: $x = 1.1 > 1$, the terms $\frac{1.1^k}{k}$ grow, the approximations never settle: it does not work at all. (d) $\log 2 = -\log \frac{1}{2} \approx 0.693$. (e) With $\log \frac{1+x}{1-x} = 2(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots)$ and $x = \frac{1}{3}$ (so that $\frac{1+x}{1-x} = 2$): $2(0.33333 + 0.01235 + 0.00082 + 0.00007) = 0.69313$, three decimals right after four terms. Faster because $x = \frac{1}{3}$ is small (each term shrinks by a factor 9) and only odd powers occur.

324 $\frac{x}{\tan x} = \frac{x \cos x}{\sin x} = \cos x \cdot \frac{1}{\sin x/x}$. As $x \rightarrow 0$: $\cos x \rightarrow 1$ and $\frac{\sin x}{x} \rightarrow 1$ (standard limit), so by the product and quotient rules the limit is $1 \cdot \frac{1}{1} = 1$.

327 For $x \rightarrow 0^+$ put $x = \frac{1}{u}$ with $u \rightarrow \infty$: $x \log x = \frac{1}{u} \log \frac{1}{u} = -\frac{\log u}{u} \rightarrow 0$ by 326(a). (Also with l'Hôpital on $\frac{\log x}{1/x}$: $\frac{1/x}{-1/x^2} = -x \rightarrow 0$.)

328 $\frac{x^k}{k!}$ for $k = 0, 1, 2, \dots$ gives $1, x, \frac{x^2}{2}, \frac{x^3}{6}, \dots$; $(-1)^k \frac{x^{2k+1}}{(2k+1)!}$ gives $x, -\frac{x^3}{6}, \frac{x^5}{120}, \dots$; $(-1)^k \frac{x^{2k}}{(2k)!}$ gives $1, -\frac{x^2}{2}, \frac{x^4}{24}, \dots$; $(-1)^{k+1} \frac{x^k}{k}$ gives $x, -\frac{x^2}{2}, \frac{x^3}{3}, \dots$: exactly the Taylor polynomials of Oefening 307.

329 $e^{ix} = \sum_k \frac{(ix)^k}{k!}$ and i^k cycles through $1, i, -1, -i$. The even $k = 2m$ give $i^{2m} = (-1)^m$: $\sum_m (-1)^m \frac{x^{2m}}{(2m)!} = \cos x$; the odd $k = 2m+1$ give $i^{2m+1} = i(-1)^m$: $i \sum_m (-1)^m \frac{x^{2m+1}}{(2m+1)!} = i \sin x$. So $e^{ix} = \cos x + i \sin x$, Definition 1.8.

330 (a) $(1+x)(1-x+x^2-x^3+\dots) = 1-x+x^2-x^3+\dots+x-x^2+x^3-\dots = 1$: all terms cancel except 1. Dividing by $1+x$: $\frac{1}{1+x} = 1-x+x^2-x^3+\dots$ (b) $x=0$: $1=1$, fine. $x=-\frac{1}{2}$: $\frac{1}{1/2} = 2$ and $1+\frac{1}{2}+\frac{1}{4}+\dots = 2$, fine. $x=-1$: $\frac{1}{0}$ against $1+1+1+\dots$: both “infinite”, not a meaningful equation. $x=1$: $\frac{1}{2}$ against $1-1+1-1+\dots$, whose partial sums are $1, 0, 1, 0, \dots$: nonsense. (c) $f(x) = (1+x)^{-1}$ has $f^{(k)}(x) = (-1)^k k! (1+x)^{-k-1}$, so $f^{(k)}(0)/k! = (-1)^k$: the Taylor series is $1-x+x^2-x^3+\dots$, the same as in (a). (d) $\frac{d}{dx}(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\dots) = 1-x+x^2-x^3+\dots$, which is the series of $\frac{1}{1+x} = \frac{d}{dx} \log(1+x)$: differentiating term by term gives the right derivative (for $|x| < 1$, where the series converge).

E CHAPTER 4

402 Put $x=y=1$ in $(x+y)^n = \sum (\text{row } n) x^k y^{n-k}$: the sum of row n is $(1+1)^n = 2^n$. (Also: every entry of row n is added into two entries of row $n+1$, so the row sum doubles each step, starting from 1.)

403 $(fg)' = f'g + fg'$; $(fg)'' = f''g + 2f'g' + fg''$; $(fg)''' = f'''g + 3f''g' + 3f'g'' + fg'''$. The coefficients are the rows of Pascal’s triangle: $(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$ (Leibniz’ rule), because each differentiation of a term $f^{(k)} g^{(n-k)}$ produces $f^{(k+1)} g^{(n-k)} + f^{(k)} g^{(n-k+1)}$, exactly the “sum of the two above” rule.

406 $f(x) = x^{(e^x)}$: $\log f = e^x \log x$, $(\log f)' = e^x \log x + \frac{e^x}{x}$, so $f'(x) = x^{e^x} e^x (\log x + \frac{1}{x})$. $g(x) = e^{(x^e)}$: chain rule with $(x^e)' = ex^{e-1}$: $g'(x) = ex^{e-1} e^{x^e}$ (no logarithm needed: the exponent e is a constant).

408 They are products, quotients or powers of simpler functions, or have the variable in the exponent: the logarithm turns products into sums, powers into multiples and $a^{b(x)}$ into $b(x) \log a$, after which only sums are left to differentiate. Without it: use the product/quotient rule repeatedly (404, 407), or rewrite $u(x)^{v(x)} = e^{v(x) \log u(x)}$ and use the chain rule (405, 406).

409 (a) $t=0$: $\arcsin 0 = 0$, $\arccos 0 = \frac{\pi}{2}$, $\arctan 0 = 0$. $t \rightarrow \infty$: $\arctan t \rightarrow \frac{\pi}{2}$ (the others are undefined). $t = \frac{1}{2}$: $\frac{\pi}{6}, \frac{\pi}{3}$, $\arctan \frac{1}{2}$ not nice. $t = -\frac{1}{2}$: $-\frac{\pi}{6}, \frac{2\pi}{3}$, not nice. $t = \frac{1}{2}\sqrt{2}$: $\frac{\pi}{4}, \frac{\pi}{4}$, not nice; $t = -\frac{1}{2}\sqrt{2}$: $-\frac{\pi}{4}, \frac{3\pi}{4}$, not nice. $t = \frac{1}{2}\sqrt{3}$: $\frac{\pi}{3}, \frac{\pi}{6}$, not nice; $t = -\frac{1}{2}\sqrt{3}$: $-\frac{\pi}{3}, \frac{5\pi}{6}$, not nice. $t=1$: $\frac{\pi}{2}, 0, \frac{\pi}{4}$; $t=-1$: $-\frac{\pi}{2}, \pi, -\frac{\pi}{4}$. $t = \frac{1}{3}\sqrt{3}$: $\arcsin, \arccos$ not nice, $\arctan = \frac{\pi}{6}$; $t = -\frac{1}{3}\sqrt{3}$: not nice, not nice, $-\frac{\pi}{6}$. $t = \sqrt{3}$: $\arcsin, \arccos$ do not exist ($\sqrt{3} > 1$), $\arctan \sqrt{3} = \frac{\pi}{3}$; $t = -\sqrt{3}$: do not exist, $-\frac{\pi}{3}$. (b) “Not nice” (/) for $\arctan$ of $\pm\frac{1}{2}, \pm\frac{1}{2}\sqrt{2}, \pm\frac{1}{2}\sqrt{3}$ and for $\arcsin, \arccos$ of $\pm\frac{1}{3}\sqrt{3}$; “does not exist” for $\arcsin, \arccos$ of $\pm\sqrt{3}$ and of $t \rightarrow \infty$.

413 $\arcsin' x = 1/\sqrt{1-x^2} \geq 1$, equal to 1 at 0 and tending to ∞ at ± 1 : the graph rises with slope 1 through the origin and becomes vertical at the endpoints $(\pm 1, \pm \frac{\pi}{2})$, the mirror image of the flat tops of the sine wave. $\arccos' = -\arcsin'$: the same shape falling from $(-1, \pi)$ to $(1, 0)$. $\arctan' x = 1/(1+x^2)$ is 1 at 0 and tends to 0 for $x \rightarrow \pm\infty$: the graph has slope 1 at the origin and flattens towards the horizontal asymptotes $y = \pm \frac{\pi}{2}$.

414 In the figure $\tan t$ is the vertical segment on the tangent line $x=1$. The ray from O through the point at angle t meets that line at distance $\sqrt{1+\tan^2 t} = \frac{1}{\cos t} = \sec t$ from O (hypotenuse of the triangle with legs 1 and $\tan t$). Similarly, with the horizontal tangent line $y=1$: the ray meets it at the point $(\cot t, 1)$ (the triangle $O, (0, 1), (\cot t, 1)$ is similar to the triangle with legs $\cos t, \sin t$), so $\cot t$ is the horizontal segment on $y=1$ and $\csc t = \sqrt{1+\cot^2 t} = \frac{1}{\sin t}$ is the distance from O to that point.

415 $\sinh(-t) = \frac{e^{-t}-e^t}{2} = -\sinh t$: odd. $\cosh(-t) = \cosh t$: even. $\sinh t = 0$ iff $e^t = e^{-t}$ iff $e^{2t} = 1$ iff $t = 0$. $\cosh t = 0$ would need $e^t = -e^{-t} < 0$: impossible, $\cosh$ has no zeros (in fact $\cosh t \geq 1$).

416 e^x rises through $(0, 1)$, e^{-x} is its mirror image; $\cosh$ is their average: a U-shaped even curve with minimum $(0, 1)$, hugging $\frac{1}{2}e^x$ for large x and $\frac{1}{2}e^{-x}$ for large negative x ; $\sinh$ is half their difference: an odd

increasing curve through the origin with slope 1, close to $\frac{1}{2}e^x$ for large x and to $-\frac{1}{2}e^{-x}$ for large negative x . Both $\sinh$ and $\cosh$ lie below $\frac{1}{2}(e^x + e^{-x})$ -type bounds; $\cosh x > \sinh x$ everywhere (difference $e^{-x} > 0$).

417 $\sinh' t = \frac{e^t + e^{-t}}{2} = \cosh t$ and $\cosh' t = \frac{e^t - e^{-t}}{2} = \sinh t$. Like $\sin' = \cos$, but $\cosh' = +\sinh$ without the minus sign of $\cos' = -\sin$.

418 $\cosh t - \sinh t = e^{-t}$ and $\cosh t + \sinh t = e^t$. Multiplying (difference of squares): $\cosh^2 t - \sinh^2 t = (\cosh t - \sinh t)(\cosh t + \sinh t) = e^{-t}e^t = 1$.

F CHAPTER 5

501 The amount $y > 0$ decreases, so $y' < 0$; from $y' = ky$ it follows that $k < 0$.

503 (5.2) $g'^2 + g^2 = 1$: first order. (5.3) first. (5.4) first. (5.5) first. (5.6) second (u''). (5.7) fourth (f'''').

504 $v(t) = \frac{mg}{k}(1 - e^{-kt/m})$; $v'(t) = \frac{mg}{k} \cdot \frac{k}{m} e^{-kt/m} = ge^{-kt/m}$, so $mv' = mge^{-kt/m}$; and $mg - kv = mg - mg(1 - e^{-kt/m}) = mge^{-kt/m}$. Equal. Also $v(0) = \frac{mg}{k}(1 - 1) = 0$.

505 (a) $g = -1$: $g' = 0$, $0 + (-1)^2 = 1$. (b) Also $g = 1$; and the non-constant solutions $g(t) = \sin t$, $g(t) = \cos t$, more generally $g(t) = \sin(t + c)$ (and $-\sin t$, $-\cos t$): $\cos^2 t + \sin^2 t = 1$.

507 (a) $u = 2x - 2$: $u' = 2$, $u'' = 0$: $0 + 12 + 5(2x - 2) = 10x + 2$. (b) For $u = c_1 e^{-x}$: $u'' + 6u' + 5u = c_1(1 - 6 + 5)e^{-x} = 0$; for $c_2 e^{-5x}$: $c_2(25 - 30 + 5)e^{-5x} = 0$. Adding the solution of (a) (the equation is linear): the sum gives $10x + 2$.

508 $y = 0$: $y' = 0$ and $a(x) \cdot 0 = 0$, so $y' + a(x)y = 0$.

509 From $\frac{dy}{dx} = -2(x + 1)y$: divide by y (assuming $y \neq 0$) and “multiply by dx ”: $\frac{dy}{y} = -2(x + 1) dx$. (The separate symbols dy , dx are a formal shorthand; the result is justified afterwards by substitution, Oefening 510.)

510 $y = Ce^{-x^2-2x}$: $y' = C(-2x - 2)e^{-x^2-2x} = -2(x + 1)y$, so $y' + 2(x + 1)y = 0$ for every C , including $C = 0$ (the zero solution).

512 Let A be a primitive of a . Separating: $\frac{dy}{y} = -a(x) dx$, $\log |y| = -A(x) + C_1$, so the general solution of $y' + a(x)y = 0$ is $y = Ce^{-A(x)}$, $C \in \mathbb{R}$. Check: $y' = -a(x)Ce^{-A(x)} = -a(x)y$.

513 See the checks written after each part of Oefening 511.

514 (a) $(y_h + y_p)' + a(y_h + y_p) = (y_h' + ay_h) + (y_p' + ay_p) = 0 + f = f$. (b) $(\tilde{y}_p - y_p)' + a(\tilde{y}_p - y_p) = (\tilde{y}_p' + a\tilde{y}_p) - (y_p' + ay_p) = f - f = 0$. (c) By (a) every function $y_h + y_p$ is a solution; by (b) every solution $\tilde{y}_p$ is $y_p + (\tilde{y}_p - y_p)$ with $\tilde{y}_p - y_p$ a homogeneous solution, hence of the form $y_h + y_p$. So $\{y_h + y_p\}$ is exactly the set of all solutions.

515 $a(x) = \frac{2}{x}$ has primitive $2\log|x|$, so $u = e^{2\log|x|} = |x|^2 = x^2$.

518 Write $y_h = Ce^{-A(x)}$ and try $y = C(x)e^{-A(x)}$. Then $y' = C'e^{-A} - aCe^{-A}$, so $y' + ay = C'(x)e^{-A(x)}$; the terms with $C(x)$ cancel (they are the homogeneous equation). Requiring this to equal f : $C'(x) = f(x)e^{A(x)}$, so $C(x) = \int f(x)e^{A(x)} dx + c$ and $y = e^{-A(x)} \left(\int f(x)e^{A(x)} dx + c \right)$: the same formula as the integrating-factor method ($u = e^A$).

520 $mv' = mg - kv$ is linear: $v' + \frac{k}{m}v = g$. Homogeneous: $v_h = Ce^{-kt/m}$. Particular: the constant $v_p = \frac{mg}{k}$ (then $v' = 0$ and $\frac{k}{m} \cdot \frac{mg}{k} = g$). General: $v = \frac{mg}{k} + Ce^{-kt/m}$; $v(0) = 0$ gives $C = -\frac{mg}{k}$, so $v(t) = \frac{mg}{k}(1 - e^{-kt/m})$.

521 The third option: substitute $x = a$ into the general solution $y = y_h + y_p$ and solve $y_h(a) + y_p(a) = b$ for C (the constant sits in y_h). Solving $y_h(a) = b$ alone or $y_p(a) = b$ ignores half of the solution; the d.v. itself at $x = a$ contains no C to solve for.

524 (a) $y' = 0$ when $\alpha y(\beta - y) = 0$: $y \equiv 0$ or $y \equiv \beta$. (b) If $0 < y < \beta$ then $y' = \alpha y(\beta - y) > 0$: y increases, but it can never reach β (there the solution would coincide with the constant solution β , which is impossible for two different solutions of the same d.v.), so it increases towards β and $y' \rightarrow 0$. If $y > \beta$ then $y' < 0$: y decreases towards β in the same way. If $y(0) = \beta$ it stays β . In all cases $\lim_{t \rightarrow \infty} y(t) = \beta$: the population tends to the carrying capacity β whatever the (positive) starting size.

G CHAPTER 6

601 $c = c \cdot x^0$, and by linearity (Stelling 2.12f) and the rule for x^r with $r = 0$: $\int c dx = c \int x^0 dx = c \cdot \frac{x^1}{1} = cx$.

602 (a) $\frac{1}{x}$: hyperbola in quadrants I and III; $\log|x|$: the usual log curve for $x > 0$ and its mirror image in the y -axis for $x < 0$ (both branches go to $-\infty$ at 0). (b) For $x < 0$, $\log|x| = \log(-x)$ and the chain rule gives $\frac{1}{-x} \cdot (-1) = \frac{1}{x}$. (c) For $a < b < 0$ the integrand $\frac{1}{x}$ is negative, so the integral is negative; and $\log|b| - \log|a| = \log\left|\frac{b}{a}\right| < 0$ because $|b| < |a|$. Signs agree.

603 For $q \neq 0$: $\int \frac{p}{qx+r} dx = \frac{p}{q} \log|qx+r| + C$ (check: derivative $\frac{p}{q} \cdot \frac{q}{qx+r}$). For $q = 0$ it is $\frac{p}{r}x + C$.

604 Try $f = g$ of the form ce^{cx} : $\int ce^{cx} dx = e^{cx}$, so the left side is e^{2cx} , while $\int c^2 e^{2cx} dx = \frac{c}{2} e^{2cx}$. They agree for $c = 2$: $f(x) = g(x) = 2e^{2x}$ (with the integration constants taken 0).

606 (a) $\sin$ is odd, so $\int_{-a}^a \sin x dx = 0$ for every a , and the limit is 0. (b) Split at 0: $\int_0^b \sin x dx = 1 - \cos b$ has no limit as $b \rightarrow \infty$ (it oscillates between 0 and 2), so $\int_0^\infty \sin x dx$ diverges and therefore so does $\int_{-\infty}^\infty \sin x dx$, even though the symmetric limits in (a) cancel. Each improperness gets its own limit.

608 $\frac{d}{dx} \left(\frac{1}{2} e^x (\sin x - \cos x) \right) = \frac{1}{2} e^x (\sin x - \cos x) + \frac{1}{2} e^x (\cos x + \sin x) = e^x \sin x$. ✓

609 Voorbeeld 6.5 with $f' = x$, $g = e^x$: $\int x e^x dx = \frac{1}{2} x^2 e^x - \int \frac{1}{2} x^2 e^x dx$; the new integral has a higher power of x : worse. That is the signal (the integrand got more complicated). Voorbeeld 6.6 with $f' = e^x$, $g = \sin x$: $\int e^x \sin x dx = e^x \sin x - \int e^x \cos x dx$, and again $= e^x \sin x - (e^x \cos x + \int e^x \sin x dx)$, giving the same result as before. Either choice works there, as long as you make the *same* choice twice (mixing them gives 0 = 0).

610 (a) $f' = 1$, $f = x$, $g = x$: $\int x dx = x \cdot x - \int x \cdot 1 dx$, so $2 \int x dx = x^2$, $\int x dx = \frac{1}{2} x^2 + C$. (b) $\int x^2 dx = x \cdot x^2 - \int x \cdot 2x dx = x^3 - 2 \int x^2 dx$, so $\int x^2 dx = \frac{1}{3} x^3 + C$. (d) $\int \frac{1}{x} dx = x \cdot \frac{1}{x} - \int x \cdot \left(-\frac{1}{x^2}\right) dx = 1 + \int \frac{1}{x} dx$, apparently $0 = 1$. No paradox: both sides are *sets* of primitives, $\log|x| + C_1 = 1 + \log|x| + C_2$, which is fine with $C_1 = C_2 + 1$. An indefinite integral is only determined up to a constant.

614 (a) $u = \pi x$: $-\frac{2}{\pi} \cos(\pi x) + C$. (b) $u = -x^2$, $du = -2x dx$: $-\frac{1}{2} e^{-x^2} + C$. (c) $u = \log x$, $du = \frac{dx}{x}$: $\int u^2 du = \frac{1}{3} \log^3 x + C$. (d) $\tan x = \frac{\sin x}{\cos x}$, $u = \cos x$: $-\int \frac{du}{u} = -\log|\cos x| + C$. (e) $u = 1 + \cos x$, $du = -\sin x dx$: $-\int u^3 du = -\frac{1}{4} (1 + \cos x)^4 + C$.

617 $u = \log x$, $du = \frac{dx}{x}$, limits $u(e) = 1$, $u(e^2) = 2$: $\int_1^2 \frac{du}{u} = \log 2$.

618 $x = \sin u$ with $u \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $dx = \cos u du$, $\sqrt{1-x^2} = \sqrt{\cos^2 u} = \cos u$ (non-negative on that interval): $\int \frac{\cos u du}{\cos u} = u + C = \arcsin x + C$.

620 $\frac{1}{3} \left(\frac{4}{x-2} - \frac{1}{x+4} \right) = \frac{1}{3} \cdot \frac{4(x+4)-(x-2)}{(x-2)(x+4)} = \frac{1}{3} \cdot \frac{3x+18}{x^2+2x-8} = \frac{x+6}{x^2+2x-8}$. ✓

622 $x^2 + 2x + 1 = (x+1)^2$ has a double zero, so $\frac{A}{x+1} + \frac{B}{x+1} = \frac{A+B}{x+1}$ has a constant numerator and can never equal $\frac{3x+5}{(x+1)^2}$: the system for A, B has no solution. Use case 3(b) instead: $\frac{3x+5}{(x+1)^2} = \frac{3}{x+1} + \frac{2}{(x+1)^2}$.

623 From $A + C = 4$, $5A + B - 2C = 1$, $5B + C = 13$: $C = 4 - A$, so $5A + B - 8 + 2A = 1$ gives $B = 9 - 7A$, and $45 - 35A + 4 - A = 13$ gives $A = 1$, $B = 2$, $C = 3$. Then $\int \frac{4x^2+x+13}{x^3+3x^2-9x+5} dx = \int \left(\frac{1}{x-1} + \frac{3}{(x-1)^2} + \frac{3}{x+5} \right) dx = \log|x-1| - \frac{3}{x-1} + 3 \log|x+5| + C$.

624(e)–(h) (e) Numerator = denominator -1 : $\int \left(1 - \frac{1}{(2x+3)^2+1}\right) dx = x - \frac{1}{2} \arctan(2x+3) + C$ (substitute $u = 2x+3$). (f) $u = x^2$, $du = 2x dx$: $\frac{1}{2} \int \frac{du}{(u-3)(u+2)} = \frac{1}{2} \cdot \frac{1}{5} \int \left(\frac{1}{u-3} - \frac{1}{u+2}\right) du = \frac{1}{10} \log \left| \frac{x^2-3}{x^2+2} \right| + C$. (g) $\frac{1}{(x^2-1)^2} = \frac{1}{4} \left(\frac{1}{x-1} - \frac{1}{x+1}\right)^2 = \frac{1}{4} \left(\frac{1}{(x-1)^2} + \frac{1}{(x+1)^2} - \frac{1}{x-1} + \frac{1}{x+1}\right)$ (using $\frac{2}{(x-1)(x+1)} = \frac{1}{x-1} - \frac{1}{x+1}$), so the integral is $\frac{1}{4} \left(-\frac{1}{x-1} - \frac{1}{x+1} + \log \left| \frac{x+1}{x-1} \right| \right) + C = -\frac{x}{2(x^2-1)} + \frac{1}{4} \log \left| \frac{x+1}{x-1} \right| + C$. (h) $x^4 - 1 = (x-1)(x+1)(x^2+1)$ and $\frac{2x^3-3x^2-2x-5}{x^4-1} = \frac{-2}{x-1} + \frac{2}{x+1} + \frac{2x+1}{x^2+1}$ (compare numerators: $-2(x+1)(x^2+1) + 2(x-1)(x^2+1) + (2x+1)(x^2-1) = 2x^3 - 3x^2 - 2x - 5$). Integral: $-2 \log|x-1| + 2 \log|x+1| + \log(1+x^2) + \arctan x + C$.

626 Long division of $x^4(1-x)^4 = x^8 - 4x^7 + 6x^6 - 4x^5 + x^4$ by $x^2 + 1$: quotient $x^6 - 4x^5 + 5x^4 - 4x^2 + 4$, remainder -4 . So $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \left[\frac{x^7}{7} - \frac{2x^6}{3} + x^5 - \frac{4x^3}{3} + 4x \right]_0^1 - 4 \arctan 1 = \frac{1}{7} - \frac{2}{3} + 1 - \frac{4}{3} + 4 - \pi = \frac{22}{7} - \pi$. The integrand is positive on $(0, 1)$, so the integral is positive: $\frac{22}{7} - \pi > 0$.

627(b),(e),(f) (b) $\cos^5 \psi = (1 - \sin^2 \psi)^2 \cos \psi$; $u = \sin \psi$: $\int (1 - u^2)^2 du = u - \frac{2}{3} u^3 + \frac{1}{5} u^5 + C = \sin \psi - \frac{2}{3} \sin^3 \psi + \frac{1}{5} \sin^5 \psi + C$. (e) $\tan(\arcsin t) = \frac{t}{\sqrt{1-t^2}}$ (Oefening 411), and $\int \frac{t dt}{\sqrt{1-t^2}} = -\sqrt{1-t^2} + C$ ($u = 1 - t^2$). (f) $\sin A \sin B = \frac{1}{2}(\cos(A-B) - \cos(A+B))$: $\int \sin(\tau+1) \sin \tau d\tau = \frac{1}{2} \int (\cos 1 - \cos(2\tau+1)) d\tau = \frac{1}{2} \tau \cos 1 - \frac{1}{4} \sin(2\tau+1) + C$.

628(a),(b),(d),(f) (a) $u = x^2 + a^2$: $\sqrt{x^2 + a^2} + C$. (b) $\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$: $\log \left| \frac{x+1}{x+2} \right| + C$. (d) $\frac{x^3+1}{x^{3/2}} = x^{3/2} + x^{-3/2}$: $\frac{2}{5} x^{5/2} - 2x^{-1/2} + C$. (f) Parts twice: $x^2 \sin x - \int 2x \sin x dx = x^2 \sin x + 2x \cos x - 2 \sin x + C$.

629(a),(d),(e),(h) (a) Parts with $f' = 1$: $x \log(1+x^{-2}) - \int x \cdot \frac{-2x^{-3}}{1+x^{-2}} dx = x \log(1 + \frac{1}{x^2}) + \int \frac{2}{x^2+1} dx = x \log(1 + \frac{1}{x^2}) + 2 \arctan x + C$. (d) Divide: $x^3 - 8 = x(x^2 - 8) + 8x - 8$, so $\int \left(x + \frac{8x}{x^2-8} - \frac{8}{x^2-8}\right) dx = \frac{x^2}{2} + 4 \log|x^2 - 8| - 8 \cdot \frac{1}{4\sqrt{2}} \log \left| \frac{x-2\sqrt{2}}{x+2\sqrt{2}} \right| + C$, using $\frac{1}{x^2-8} = \frac{1}{4\sqrt{2}} \left(\frac{1}{x-2\sqrt{2}} - \frac{1}{x+2\sqrt{2}}\right)$; i.e. $\frac{x^2}{2} + 4 \log|x^2 - 8| - \sqrt{2} \log \left| \frac{x-2\sqrt{2}}{x+2\sqrt{2}} \right| + C$. (e) $u = \sqrt{1+x}$, $dx = 2u du$: $\int 2ue^u du = 2(u-1)e^u + C = 2(\sqrt{1+x}-1)e^{\sqrt{1+x}} + C$. (h) $u = \log x$, $x = e^u$, $dx = e^u du$: $\int e^u \sin u du = \frac{1}{2} e^u (\sin u - \cos u)$ (Voorbeeld 6.6) $= \frac{1}{2} x (\sin(\log x) - \cos(\log x)) + C$.

630(b),(c),(d),(g) (b) Separate: $y dy = e^{\sqrt{x}} dx$, so $\frac{1}{2} y^2 = 2(\sqrt{x}-1)e^{\sqrt{x}} + C$ (629e). $y(0) = -2$: $2 = -2 + C$, $C = 4$; so $y^2 = 4(\sqrt{x}-1)e^{\sqrt{x}} + 8$ and, as $y(0) < 0$, $y = -\sqrt{4(\sqrt{x}-1)e^{\sqrt{x}} + 8}$. (c) $y_h = Ce^{-10x}$, $y_p = \frac{1}{10}$: $y(\frac{1}{10}) = \frac{1}{10} + Ce^{-1} = \frac{2}{10}$ gives $C = \frac{e}{10}$: $y = \frac{1}{10}(1 + e^{1-10x})$. (d) $y_h = Ce^{-x^3}$, $y_p = \frac{1}{3}$: $y(0) = \frac{1}{3} + C = 1$: $y = \frac{1}{3} + \frac{2}{3}e^{-x^3}$. (g) $\frac{dy}{y^3} = (4x^3 + 2x) dx$, so $-\frac{1}{2y^2} = x^4 + x^2 + C$; at $(1, 1)$: $-\frac{1}{2} = 2 + C$, $C = -\frac{5}{2}$. Hence $\frac{1}{y^2} = 5 - 2x^2 - 2x^4$ and $y = \frac{1}{\sqrt{5-2x^2-2x^4}}$ (positive root since $y(1) = 1$).

631 (a) $\sqrt{(x-1)^2} = |x-1|$: two triangles, $\int_0^2 |x-1| dx = \frac{1}{2} + \frac{1}{2} = 1$. (b) $\frac{1}{e^x+1} = \frac{e^{-x}}{1+e^{-x}}$, with primitive $-\log(1+e^{-x})$: $\int_0^\infty \frac{dx}{e^x+1} = \lim_{b \rightarrow \infty} (-\log(1+e^{-b})) + \log 2 = \log 2$. Convergent.

634 Partial sums $\sum_{k=0}^n \frac{1}{k!}$: $n = 12$ gives 2.71828182828..., $n = 13$ gives 2.718281828446..., against $e = 2.7182818284590...$. The first ten decimals (7182818284) are correct from $n = 13$ on, i.e. after 14 terms (the error is about $\frac{1}{(n+1)!}$: $\frac{1}{13!} \approx 1.6 \cdot 10^{-10}$ is too big, $\frac{1}{14!} \approx 1.1 \cdot 10^{-11}$ is small enough).

635 (a) $a^1 = e^{1 \cdot \log a} = \exp(\log a) = a$, since $\exp$ is the inverse of $\log$. (b) $a^{x+y} = e^{(x+y) \log a} = e^{x \log a} e^{y \log a} = a^x a^y$ by Stelling 6.16. (c) $\log(a^x) = \log(\exp(x \log a)) = x \log a$, so $(a^x)^y = e^{y \log(a^x)} = e^{yx \log a} = a^{xy}$.

H CHAPTER 7

703 $\log a + \log b = \log(ab)$ with $ab = (49 + \sqrt{49^2 - 1})(49 - \sqrt{49^2 - 1}) = 49^2 - (49^2 - 1) = 1$: the sum is $\log 1 = 0$.

704 Put $r = e^{f(x)} = \frac{e^x + 1}{e^x - 1}$. Then $f(f(x)) = \log \frac{r+1}{r-1}$ with $r+1 = \frac{2e^x}{e^x - 1}$ and $r-1 = \frac{2}{e^x - 1}$, so $\frac{r+1}{r-1} = e^x$ and $f(f(x)) = x$. (Strictly, $f(x)$ is defined when $\frac{e^x+1}{e^x-1} > 0$, i.e. $x > 0$; then $f(x) > 0$ as well, so $f \circ f$ makes sense there. For $x < 0$ one reads $\log |\cdot|$.)

707 Both graphs pass through $(0, 0)$: $\sin 0 = 0 = \arctan 0$. The slopes there: $\cos 0 = 1$ and $\frac{1}{1+0^2} = 1$. Same point, same slope: the common tangent line is $y = x$.

708 $\frac{1+x^2}{1-x} = (1+x^2)(1+x+x^2+x^3+x^4+\dots) = 1+x+2x^2+2x^3+2x^4+\dots$, so $P_4(x) = 1+x+2x^2+2x^3+2x^4$.

709 $f = \arctan$ satisfies $f' = \frac{1}{1+x^2}$, i.e. $(1+x^2)f' = 1$. As in 316, differentiate this relation: $2xf' + (1+x^2)f'' = 0$, $2f' + 4xf'' + (1+x^2)f''' = 0$, $6f'' + 6xf''' + (1+x^2)f^{(4)} = 0$, $12f''' + 8xf^{(4)} + (1+x^2)f^{(5)} = 0$. At 0: $f' = 1$, $f'' = 0$, $f''' = -2$, $f^{(4)} = 0$, $f^{(5)} = 24$. So $P_5(x) = x - \frac{2}{3!}x^3 + \frac{24}{5!}x^5 = x - \frac{1}{3}x^3 + \frac{1}{5}x^5$. (Faster: $f' = 1 - x^2 + x^4 - \dots$, integrate term by term.)

711 (a) $x \xrightarrow{p} 1+x \xrightarrow{p} 2+x \xrightarrow{i} \frac{1}{2+x} \xrightarrow{t} \frac{3}{2+x} \xrightarrow{m} \frac{-3}{2+x}$; $m \circ t \circ i \circ p \circ p$. (b) $\frac{1-x}{2+x} = \frac{3-(2+x)}{2+x} = \frac{3}{2+x} - 1$. From $g = \frac{-3}{2+x}$: $p(g(x)) = 1 - \frac{3}{2+x}$ and $m(p(g(x))) = \frac{3}{2+x} - 1$. So $\frac{1-x}{2+x} = m \circ p \circ m \circ t \circ i \circ p \circ p$.

712 With $\phi = \arctan x$ in a right triangle with legs 1 (adjacent) and x (opposite), hypotenuse $\sqrt{1+x^2}$: $\sin(\arctan x) = \frac{x}{\sqrt{1+x^2}}$ (correct sign for $x < 0$ too, since $\arctan x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\sin$ has the sign of x there).

714 $x = \cosh t = \frac{1}{2}(e^t + e^{-t})$; with $u = e^t$: $2x = u + \frac{1}{u}$, $u^2 - 2xu + 1 = 0$, $u = x \pm \sqrt{x^2 - 1}$. For $x \geq 1$ both roots are positive (their product is 1), so $t = \log(x \pm \sqrt{x^2 - 1})$: two values, because $\cosh$ is not injective.

715 (a) If f is even, then $y = f(x)$ has the solutions x_0 and $-x_0$; the inverse on $x \geq 0$ returns x_0 , the inverse on $x \leq 0$ returns $-x_0$. The two inverse graphs are therefore each other's image under $y \mapsto -y$: reflection in the x -axis. (b) Mirror images means $\log(x - \sqrt{x^2 - 1}) = -\log(x + \sqrt{x^2 - 1}) = \log \frac{1}{x + \sqrt{x^2 - 1}}$, i.e. $\frac{1}{x - \sqrt{x^2 - 1}} = x + \sqrt{x^2 - 1}$. True: $(x - \sqrt{x^2 - 1})(x + \sqrt{x^2 - 1}) = x^2 - (x^2 - 1) = 1$.

716 (a) $\frac{d}{dx} \log(x + \sqrt{x^2 - 1}) = \frac{1 + \frac{x}{\sqrt{x^2 - 1}}}{x + \sqrt{x^2 - 1}} = \frac{1}{\sqrt{x^2 - 1}}$ (the numerator is $\frac{\sqrt{x^2 - 1} + x}{\sqrt{x^2 - 1}}$). For the other branch: $\frac{1 - \frac{x}{\sqrt{x^2 - 1}}}{x - \sqrt{x^2 - 1}} = -\frac{1}{\sqrt{x^2 - 1}}$. (b) Differentiate $\cosh(\operatorname{arcosh} x) = x$: $\sinh(\operatorname{arcosh} x) \cdot \operatorname{arcosh}' x = 1$, and by Oefening

418 $\sinh t = \pm \sqrt{\cosh^2 t - 1} = \pm \sqrt{x^2 - 1}$ (+ for $t \geq 0$, - for $t \leq 0$), so $\operatorname{arcosh}' x = \pm \frac{1}{\sqrt{x^2 - 1}}$, matching (a).

717 $\tanh t = \frac{\sinh t}{\cosh t} = \frac{e^t - e^{-t}}{e^t + e^{-t}}$: odd, $\tanh 0 = 0$, increasing, $\tanh t \rightarrow \pm 1$ as $t \rightarrow \pm \infty$ (divide by e^t): an S-shaped graph between the asymptotes $y = \pm 1$. Derivative (quotient rule and 418): $\tanh' t = \frac{\cosh^2 t - \sinh^2 t}{\cosh^2 t} = \frac{1}{\cosh^2 t} = 1 - \tanh^2 t$ (two forms, like $\tan' = \frac{1}{\cos^2} = 1 + \tan^2$). Inverse: $x = \tanh t = \frac{u - 1/u}{u + 1/u} = \frac{u^2 - 1}{u^2 + 1}$ with $u = e^t$, so $u^2 = \frac{1+x}{1-x}$ and $t = \frac{1}{2} \log \frac{1+x}{1-x}$ for $-1 < x < 1$; its derivative is $\frac{1}{2} \left(\frac{1}{1+x} + \frac{1}{1-x} \right) = \frac{1}{1-x^2}$, consistent with

$$\operatorname{artanh}' x = \frac{1}{1 - \tanh^2(\operatorname{artanh} x)}.$$

721 For $x \neq 1$: $\frac{1-x^n}{1-x} = 1 + x + x^2 + \cdots + x^{n-1}$ (Oefening 330a). Integrating term by term over $[0, 1]$:
 $\sum_{k=0}^{n-1} \frac{1}{k+1} = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$

725 $\sqrt{x}$: domain $[0, \infty)$, through $(0, 0)$ and $(1, 1)$, increasing, concave, vertical tangent at 0, $\rightarrow \infty$. e^x : domain $\mathbb{R}$, positive, $e^0 = 1$, increasing, convex, $\rightarrow 0$ as $x \rightarrow -\infty$ (asymptote $y = 0$), $\rightarrow \infty$ faster than any power. $\log x$: domain $(0, \infty)$, zero at 1, $\rightarrow -\infty$ at 0^+ (vertical asymptote), increasing, concave, $\rightarrow \infty$ slower than any power; mirror image of e^x in $y = x$. $\sin x$: odd, period 2π , zeros $k\pi$, extrema ± 1 at $\frac{\pi}{2} + k\pi$, inflection points at the zeros ($\sin'' = -\sin$). $\cos x$: even, $\cos x = \sin(x + \frac{\pi}{2})$. $\tan x$: odd, period π , domain $x \neq \frac{\pi}{2} + k\pi$ (vertical asymptotes), zeros $k\pi$, increasing on each branch ($\tan' = 1 + \tan^2 > 0$), inflection points at the zeros.

726 Each: domain; symmetry; intercepts; edges; f' ; f'' .

(a) $k(x) = \sqrt{9-x^2}$: domain $[-3, 3]$, even, zeros ± 3 , $k(0) = 3$; $k' = -\frac{x}{\sqrt{9-x^2}}$: maximum 3 at 0, vertical tangents at ± 3 ; $k'' = -\frac{9}{(9-x^2)^{3/2}} < 0$: concave. The upper half of the circle of radius 3.

(b) $h(x) = \frac{x}{\sqrt{1-x^2}}$: domain $(-1, 1)$, odd, zero at 0; $h \rightarrow \pm\infty$ as $x \rightarrow \pm 1$ (vertical asymptotes); $h' = (1-x^2)^{-3/2} > 0$, increasing, $h'(0) = 1$; $h'' = 3x(1-x^2)^{-5/2}$: inflection at 0, convex for $x > 0$.

(c) $g(x) = e^{1-x^2}$: domain $\mathbb{R}$, even, no zeros, $g(0) = e$, $g \rightarrow 0$ at $\pm\infty$ (asymptote $y = 0$); $g' = -2xe^{1-x^2}$: maximum e at 0; $g'' = (4x^2 - 2)e^{1-x^2}$: inflection points at $x = \pm\frac{1}{2}\sqrt{2}$ (value $\sqrt{e}$); bell shape.

(e) $\varphi(x) = \log \frac{2x}{(x-4)^2} = \log 2 + \log x - 2\log|x-4|$: domain $x > 0$, $x \neq 4$; zeros where $2x = (x-4)^2$: $x = 2$ and $x = 8$; $\varphi \rightarrow -\infty$ as $x \rightarrow 0^+$ and as $x \rightarrow \infty$ ($\frac{2x}{(x-4)^2} \rightarrow 0$), $\varphi \rightarrow +\infty$ as $x \rightarrow 4$ (vertical asymptote); $\varphi' = \frac{1}{x} - \frac{2}{x-4} = -\frac{x+4}{x(x-4)}$: positive on $(0, 4)$, negative on $(4, \infty)$, no critical points; $\varphi'' = \frac{x^2+8x-16}{x^2(x-4)^2}$: zero at $x = 4\sqrt{2} - 4 \approx 1.66$, concave before it, convex after (on both sides of 4); inflection point at $x = 4\sqrt{2} - 4$.

(f) $\psi(x) = (1 + \frac{1}{x})e^{-x}$: domain $x \neq 0$; zero at -1 ; $\psi \rightarrow +\infty$ as $x \rightarrow 0^+$, $-\infty$ as $x \rightarrow 0^-$, 0^+ as $x \rightarrow \infty$, $+\infty$ as $x \rightarrow -\infty$; $\psi' = -\frac{x^2+x+1}{x^2}e^{-x} < 0$ everywhere: decreasing on each branch; $\psi'' = \frac{(x+1)(x^2+2)}{x^3}e^{-x}$: convex for $x < -1$ and for $x > 0$, concave on $(-1, 0)$; inflection point $(-1, 0)$.

(g) $z(x) = e^{-1/x}$: domain $x \neq 0$, positive; $z \rightarrow 0$ as $x \rightarrow 0^+$, $\rightarrow +\infty$ as $x \rightarrow 0^-$, $\rightarrow 1$ as $x \rightarrow \pm\infty$ (asymptote $y = 1$, approached from below on the right and from above on the left); $z' = \frac{1}{x^2}e^{-1/x} > 0$, increasing on each branch; $z'' = \frac{1-2x}{x^4}e^{-1/x}$: convex for $x < \frac{1}{2}$ ($x \neq 0$), concave for $x > \frac{1}{2}$, inflection at $(\frac{1}{2}, e^{-2})$.

(h) $L(x) = \log(x - \frac{1}{x})$: domain where $\frac{x^2-1}{x} > 0$: $(-1, 0) \cup (1, \infty)$; zeros where $x^2 - x - 1 = 0$: $x = \frac{1 \pm \sqrt{5}}{2}$ (one in each interval); $L \rightarrow -\infty$ at -1^+ and 1^+ , $+\infty$ at 0^- and $+\infty$; $L' = \frac{x^2+1}{x(x^2-1)} > 0$ on the domain: increasing; $L'' = -\frac{x^4+4x^2-1}{x^2(x^2-1)^2}$: zero at $x = \pm\sqrt{\sqrt{5}-2} \approx \pm 0.486$; on $(-1, -0.486)$ convex, on $(-0.486, 0)$ concave (inflection at -0.486); on $(1, \infty)$ concave.

(i) $e^{1-\frac{1}{1+x}}$: domain $x \neq -1$, positive; $\rightarrow 0$ as $x \rightarrow -1^+$, $\rightarrow +\infty$ as $x \rightarrow -1^-$, $\rightarrow e$ as $x \rightarrow \pm\infty$ (asymptote $y = e$); derivative $\frac{1}{(1+x)^2}e^{1-\frac{1}{1+x}} > 0$: increasing on each branch; second derivative $-\frac{2x+1}{(1+x)^4}e^{1-\frac{1}{1+x}}$: convex for $x < -\frac{1}{2}$ ($x \neq -1$), concave for $x > -\frac{1}{2}$, inflection at $x = -\frac{1}{2}$ (value e^{-1}).

(j) $\arcsin(1 - \sqrt{x})$: domain $0 \leq x \leq 4$ ($-1 \leq 1 - \sqrt{x} \leq 1$); values from $\frac{\pi}{2}$ at 0 through 0 at $x = 1$ to $-\frac{\pi}{2}$ at 4; derivative $-\frac{1}{2\sqrt{x}\sqrt{2\sqrt{x}-x}} < 0$: decreasing, vertical tangents at 0 and 4; second derivative changes sign at $\sqrt{x} = \frac{3}{2}$, i.e. $x = \frac{9}{4}$: convex on $(0, \frac{9}{4})$, concave on $(\frac{9}{4}, 4)$.

730 Let $h(x) = 2(\sqrt{x} - 1) - \log x$ on $(0, \infty)$. $h(1) = 0$ and $h'(x) = \frac{1}{\sqrt{x}} - \frac{1}{x} = \frac{\sqrt{x}-1}{x}$, negative on $(0, 1)$, positive on $(1, \infty)$. So h has its minimum 0 at $x = 1$ and $h \geq 0$: $\log x \leq 2(\sqrt{x} - 1)$. (This is sharper than 729 for x near 1, since $2(\sqrt{x} - 1) \leq x - 1$ by $(\sqrt{x} - 1)^2 \geq 0$.)